



## PGA11x Zero-Drift Programmable Gain Amplifier With Mux

### 1 Features

- Rail-to-Rail Input and Output
- Offset: 25  $\mu\text{V}$  (Typical), 100  $\mu\text{V}$  (Maximum)
- Zero Drift: 0.35  $\mu\text{V}/^\circ\text{C}$  (Typical), 1.2  $\mu\text{V}/^\circ\text{C}$  (Maximum)
- Low Noise: 12  $\text{nV}/\sqrt{\text{Hz}}$
- Input Offset Current:  $\pm 5$  nA Maximum ( $25^\circ\text{C}$ )
- Gain Error: 0.1% Maximum ( $G \leq 32$ ), 0.3% Maximum ( $G > 32$ )
- Binary Gains: 1, 2, 4, 8, 16, 32, 64, 128 (PGA112, PGA116)
- Scope Gains: 1, 2, 5, 10, 20, 50, 100, 200 (PGA113, PGA117)
- Gain Switching Time: 200 ns
- 2 Channel MUX: PGA112, PGA113  
10 Channel MUX: PGA116, PGA117
- Four Internal Calibration Channels
- Amplifier Optimized for Driving CDAC ADCs
- Output Swing: 50 mV to Supply Rails
- $\text{AV}_{\text{DD}}$  and  $\text{DV}_{\text{DD}}$  for Mixed Voltage Systems
- $\text{I}_\text{Q} = 1.1$  mA (Typical)
- Software and Hardware Shutdown:  $\text{I}_\text{Q} \leq 4$   $\mu\text{A}$  (Typical)
- Temperature Range:  $-40^\circ\text{C}$  to  $125^\circ\text{C}$
- SPI™ Interface (10 MHz) With Daisy-Chain Capability

### 2 Applications

- Remote e-Meter Reading
- Automatic Gain Control
- Portable Data Acquisition
- PC-Based Signal Acquisition Systems
- Test and Measurement
- Programmable Logic Controllers
- Battery-Powered Instruments
- Handheld Test Equipment

### 3 Description

The PGA112 and PGA113 devices (binary and scope gains) offer two analog inputs, a three-pin SPI interface, and software shutdown in a 10-pin, VSSOP package. The PGA116 and PGA117 (binary and scope gains) offer 10 analog inputs, a SPI interface with daisy-chain capability, and hardware and software shutdown in a 20-pin TSSOP package.

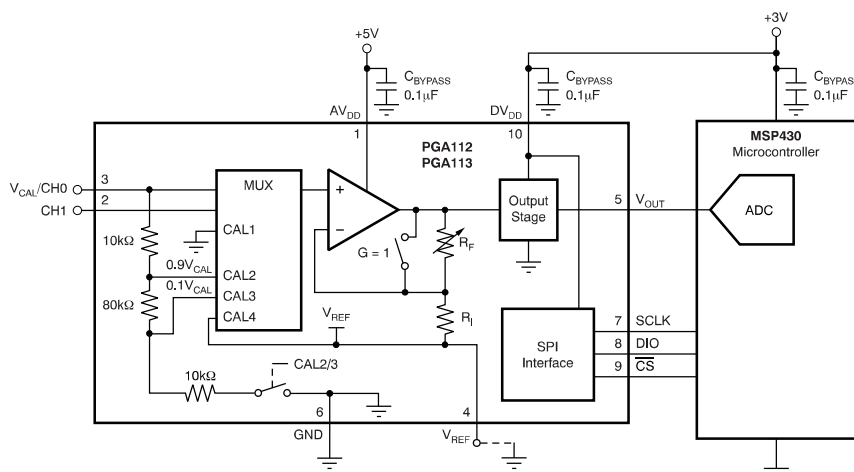
All versions provide internal calibration channels for system-level calibration. The channels are tied to GND,  $0.9 V_{\text{CAL}}$ ,  $0.1 V_{\text{CAL}}$ , and  $V_{\text{REF}}$ , respectively.  $V_{\text{CAL}}$ , an external voltage connected to Channel 0, is used as the system calibration reference. Binary gains are: 1, 2, 4, 8, 16, 32, 64, and 128; scope gains are: 1, 2, 5, 10, 20, 50, 100, and 200.

#### Device Information<sup>(1)</sup>

| PART NUMBER    | PACKAGE    | BODY SIZE (NOM)   |
|----------------|------------|-------------------|
| PGA112, PGA113 | VSSOP (10) | 3.00 mm × 3.00 mm |
| PGA116, PGA117 | TSSOP (20) | 6.50 mm × 4.40 mm |

(1) For all available packages, see the orderable addendum at the end of the data sheet.

#### Simplified Schematic



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## 4 Revision History

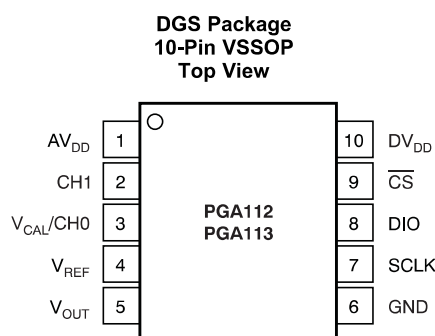
NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

| Changes from Revision B (September 2008) to Revision C                                                                                                                                                                                                                                                                                                                 | Page |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <ul style="list-style-type: none"> <li>Added ESD Ratings table, Thermal Information table, Feature Description section, Device Functional Modes, Application and Implementation section, Power Supply Recommendations section, Layout section, Device and Documentation Support section, and Mechanical, Packaging, and Orderable Information section .....</li> </ul> | 1    |

## 5 Device Comparison

| DEVICE | NO. OF MUX INPUTS | GAINS (EIGHT EACH) | SPI DAISY-CHAIN | SHUTDOWN |          | PACKAGE  |
|--------|-------------------|--------------------|-----------------|----------|----------|----------|
|        |                   |                    |                 | HARDWARE | SOFTWARE |          |
| PGA112 | 2                 | Binary             | No              | No       | ✓        | VSSOP-10 |
| PGA113 | 2                 | Scope              | No              | No       | ✓        | VSSOP-10 |
| PGA116 | 10                | Binary             | ✓               | ✓        | ✓        | TSSOP-20 |
| PGA117 | 10                | Scope              | ✓               | ✓        | ✓        | TSSOP-20 |

## 6 Pin Configuration and Functions



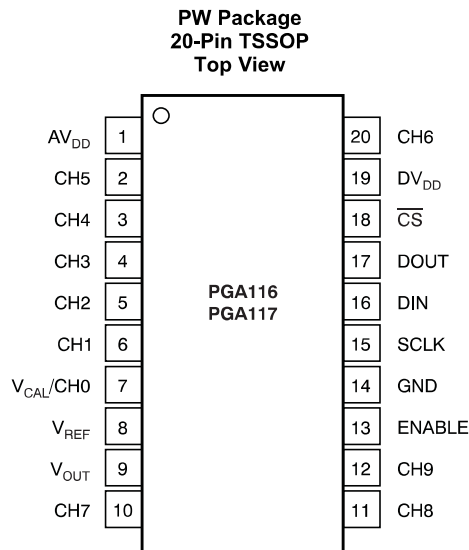
**Pin Functions: PGA112, PGA113**

| PIN |                       | I/O | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----|-----------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NO. | NAME                  |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 1   | AV <sub>DD</sub>      | I   | Analog supply voltage (2.2 V to 5.5 V)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2   | CH1                   | I   | Input MUX channel 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 3   | V <sub>CAL</sub> /CH0 | I   | Input MUX channel 0 and V <sub>CAL</sub> input. For system calibration purposes, connect this pin to a low-impedance external reference voltage to use internal calibration channels. The four internal calibration channels are connected to GND, 0.9 V <sub>CAL</sub> , 0.1 V <sub>CAL</sub> , and V <sub>REF</sub> , respectively. V <sub>CAL</sub> is loaded with 100 kΩ (typical) when internal calibration channels CAL2 or CAL3 are selected. Otherwise, V <sub>CAL</sub> /CH0 appears as high impedance.                                |
| 4   | V <sub>REF</sub>      | I   | Reference input pin. Connect external reference for V <sub>OUT</sub> offset shift or to midsupply for midsupply referenced systems. V <sub>REF</sub> must be connected to a low-impedance reference capable of sourcing and sinking at least 2 mA or V <sub>REF</sub> must be connected to GND.                                                                                                                                                                                                                                                 |
| 5   | V <sub>OUT</sub>      | O   | Analog voltage output. When AV <sub>DD</sub> < DV <sub>DD</sub> , V <sub>OUT</sub> is clamped to AV <sub>DD</sub> + 300 mV.                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 6   | GND                   | —   | Ground pin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 7   | SCLK                  | I   | Clock input for SPI serial interface                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 8   | DIO                   | I   | Data input/output for SPI serial interface. DIO contains a weak, 10-μA internal pulldown current source.                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 9   | CS                    | I   | Chip select line for SPI serial interface                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 10  | DV <sub>DD</sub>      | I   | Digital and op amp output stage supply voltage (2.2 V to 5.5 V). Useful in multi-supply systems to prevent overvoltage and lockup condition on an analog-to-digital (ADC) input (for example, a microcontroller with an ADC running on 3 V and the PGA powered from 5 V). Digital I/O levels to be relative to DV <sub>DD</sub> . DV <sub>DD</sub> should be bypassed with a 0.1-μF ceramic capacitor, and DV <sub>DD</sub> must supply the current for the digital portion of the PGA as well as the load current for the op amp output stage. |

**PGA112, PGA113, PGA116, PGA117**

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**Pin Functions: PGA116, PGA117**

| PIN |                       | I/O | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----|-----------------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NO. | NAME                  |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 1   | AV <sub>DD</sub>      | I   | Analog supply voltage (2.2 V to 5.5 V)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 2   | CH5                   | I   | Input MUX channel 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 3   | CH4                   | I   | Input MUX channel 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 4   | CH3                   | I   | Input MUX channel 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5   | CH2                   | I   | Input MUX channel 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 6   | CH1                   | I   | Input MUX channel 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 7   | V <sub>CAL</sub> /CH0 | I   | Input MUX channel 0 and V <sub>CAL</sub> input. For system calibration purposes, connect this pin to a low-impedance external reference voltage to use internal calibration channels. The four internal calibration channels are connected to GND, 0.9 V <sub>CAL</sub> , 0.1 V <sub>CAL</sub> , and V <sub>REF</sub> , respectively. V <sub>CAL</sub> is loaded with 100 kΩ (typical) when internal calibration channels CAL2 or CAL3 are selected. Otherwise, V <sub>CAL</sub> /CH0 appears as high impedance.            |
| 8   | V <sub>REF</sub>      | I   | Reference input pin. Connect external reference for V <sub>OUT</sub> offset shift or to midsupply for midsupply referenced systems. V <sub>REF</sub> must be connected to a low-impedance reference capable of sourcing and sinking at least 2 mA or to GND.                                                                                                                                                                                                                                                                |
| 9   | V <sub>OUT</sub>      | O   | Analog voltage output. When AV <sub>DD</sub> < DV <sub>DD</sub> , V <sub>OUT</sub> is clamped to AV <sub>DD</sub> + 300 mV.                                                                                                                                                                                                                                                                                                                                                                                                 |
| 10  | CH7                   | I   | Input MUX channel 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 11  | CH8                   | I   | Input MUX channel 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 12  | CH9                   | I   | Input MUX channel 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 13  | ENABLE                | I   | Hardware enable pin. Logic low puts the part into Shutdown mode (I <sub>Q</sub> < 1 μA).                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 14  | GND                   | —   | Ground pin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 15  | SCLK                  | I   | Clock input for SPI serial interface                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 16  | DIN                   | I   | Data input for SPI serial interface. DIN contains a weak, 10-μA internal pulldown current source to allow for ease of daisy-chain configurations.                                                                                                                                                                                                                                                                                                                                                                           |
| 17  | DOUT                  | O   | Data output for SPI serial interface. DOUT goes to high-Z state when CS goes high for standard SPI interface.                                                                                                                                                                                                                                                                                                                                                                                                               |
| 18  | CS                    | I   | Chip select line for SPI serial interface                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 19  | DV <sub>DD</sub>      | I   | Digital and op amp output stage supply voltage (2.2 V to 5.5 V). Useful in multi-supply systems to prevent overvoltage and lockup condition on an ADC input (for example, a microcontroller with an ADC running on 3 V and the PGA powered from 5 V). Digital I/O levels to be relative to DV <sub>DD</sub> . DV <sub>DD</sub> should be bypassed with a 0.1-μF ceramic capacitor, and DV <sub>DD</sub> must supply the current for the digital portion of the PGA as well as the load current for the op amp output stage. |
| 20  | CH6                   | I   | Input MUX channel 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

## 7 Specifications

### 7.1 Absolute Maximum Ratings

Over operating free-air temperature range, unless otherwise noted.<sup>(1)</sup>

|                                                | MIN       | MAX                    | UNIT |
|------------------------------------------------|-----------|------------------------|------|
| Supply voltage                                 |           | 7                      | V    |
| Signal input terminals, voltage <sup>(2)</sup> | GND – 0.5 | AV <sub>DD</sub> + 0.5 | V    |
| Signal input terminals, current <sup>(2)</sup> |           | ±10                    | mA   |
| Output short circuit                           |           | Continuous             |      |
| Operating temperature                          | –40       | 125                    | °C   |
| Junction temperature                           |           | 150                    | °C   |
| Storage temperature                            | –65       | 150                    | °C   |

- (1) Stresses above these ratings may cause permanent damage. Exposure to absolute maximum conditions for extended periods may degrade device reliability. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those specified is not implied.
- (2) Input terminals are diode-clamped to the power-supply rails. Input signals that can swing more than 0.5 V beyond the supply rails should be current-limited to 10 mA or less.

### 7.2 ESD Ratings

|                                            |                                                                                | VALUE | UNIT |
|--------------------------------------------|--------------------------------------------------------------------------------|-------|------|
| V <sub>(ESD)</sub> Electrostatic discharge | Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001 <sup>(1)</sup>              | ±3000 | V    |
|                                            | Charged-device model (CDM), per JEDEC specification JESD22-C101 <sup>(2)</sup> | ±1000 |      |
|                                            | Machine Model (MM)                                                             | ±300  |      |

(1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.

(2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

### 7.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)

|                       | MIN | NOM | MAX | UNIT |
|-----------------------|-----|-----|-----|------|
| AV <sub>DD</sub>      | 2.2 | 5   | 5.5 | V    |
| DV <sub>DD</sub>      | 2.2 | 5   | 5.5 | V    |
| Operating temperature | –40 | 25  | 125 | °C   |

### 7.4 Thermal Information

| THERMAL METRIC <sup>(1)</sup> |                                              | PGA112,<br>PGA113 | PGA116,<br>PGA117 | UNIT |
|-------------------------------|----------------------------------------------|-------------------|-------------------|------|
|                               |                                              | DGS (VSSOP)       | PW (TSSOP)        |      |
|                               |                                              | 10 PINS           | 20 PINS           |      |
| R <sub>θJA</sub>              | Junction-to-ambient thermal resistance       | 98.3              | 100.3             | °C/W |
| R <sub>θJC(top)</sub>         | Junction-to-case (top) thermal resistance    | 57                | 36.9              | °C/W |
| R <sub>θJB</sub>              | Junction-to-board thermal resistance         | 51.2              | 50.6              | °C/W |
| ψ <sub>JT</sub>               | Junction-to-top characterization parameter   | 1.3               | 2.6               | °C/W |
| ψ <sub>JB</sub>               | Junction-to-board characterization parameter | 36.9              | 50.2              | °C/W |
| R <sub>θJC(bot)</sub>         | Junction-to-case (bottom) thermal resistance | 4.8               | N/A               | °C/W |

(1) For more information about traditional and new thermal metrics, see the *IC Package Thermal Metrics* application report, [SPRA953](#).

**PGA112, PGA113, PGA116, PGA117**

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**7.5 Electrical Characteristics:  $V_S = AV_{DD} = DV_{DD} = 5\text{ V}$** 

at  $T_A = 25^\circ\text{C}$ ,  $R_L = 10\text{ k}\Omega/C_L = 100\text{ pF}$  connected to  $DV_{DD}/2$ , and  $V_{REF} = \text{GND}$  (unless otherwise noted)

| PARAMETER                                    |                      | TEST CONDITIONS                                                                                                                         |  | MIN                                | TYP                    | MAX  | UNIT   |
|----------------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|--|------------------------------------|------------------------|------|--------|
| OFFSET VOLTAGE                               |                      |                                                                                                                                         |  |                                    |                        |      |        |
| Input offset voltage                         | V <sub>OS</sub>      | AV <sub>DD</sub> = DV <sub>DD</sub> = 5 V, V <sub>REF</sub> = V <sub>IN</sub> = AV <sub>DD</sub> /2, V <sub>CM</sub> = 2.5 V            |  |                                    | ±25                    | ±100 | μV     |
|                                              |                      | AV <sub>DD</sub> = DV <sub>DD</sub> = 5 V, V <sub>REF</sub> = V <sub>IN</sub> = AV <sub>DD</sub> /2, V <sub>CM</sub> = 4.5 V            |  |                                    | ±75                    | ±325 | μV     |
| vs temperature, −40°C to 125°C               | dV <sub>OS</sub> /dT | AV <sub>DD</sub> = DV <sub>DD</sub> = 5 V, V <sub>CM</sub> = 2.5 V                                                                      |  | T <sub>A</sub> = −40°C to 125°C    | 0.35                   | 1.2  | μV/°C  |
| vs temperature, −40°C to 85°C                |                      | AV <sub>DD</sub> = DV <sub>DD</sub> = 5 V, V <sub>CM</sub> = 2.5 V                                                                      |  |                                    | 0.15                   | 0.9  | μV/°C  |
| vs temperature, −40°C to 125°C               |                      | AV <sub>DD</sub> = DV <sub>DD</sub> = 5 V, V <sub>CM</sub> = 4.5 V                                                                      |  | T <sub>A</sub> = −40°C to 125°C    | 0.6                    | 1.8  | μV/°C  |
| vs temperature, −40°C to 85°C                |                      | AV <sub>DD</sub> = DV <sub>DD</sub> = 5 V, V <sub>CM</sub> = 4.5 V                                                                      |  |                                    | 0.3                    | 1.3  | μV/°C  |
| vs power supply                              | PSRR                 | AV <sub>DD</sub> = DV <sub>DD</sub> = 2.2 V to 5.5 V, V <sub>CM</sub> = 0.5 V, V <sub>REF</sub> = V <sub>IN</sub> = AV <sub>DD</sub> /2 |  |                                    | 5                      | 20   | μV/V   |
| Over temperature, −40°C to 125°C             |                      | AV <sub>DD</sub> = DV <sub>DD</sub> = 2.2 V to 5.5 V, V <sub>CM</sub> = 0.5 V, V <sub>REF</sub> = V <sub>IN</sub> = AV <sub>DD</sub> /2 |  | T <sub>A</sub> = −40°C to 125°C    | 5                      | 40   | μV/V   |
| INPUT ON-CHANNEL CURRENT                     |                      |                                                                                                                                         |  |                                    |                        |      |        |
| Input on-channel current (Ch0, Ch1)          | I <sub>IN</sub>      | V <sub>REF</sub> = V <sub>IN</sub> = AV <sub>DD</sub> /2                                                                                |  |                                    | ±1.5                   | ±5   | nA     |
| Over temperature, −40°C to 125°C             |                      | V <sub>REF</sub> = V <sub>IN</sub> = AV <sub>DD</sub> /2                                                                                |  | See <i>Typical Characteristics</i> |                        |      | nA     |
| INPUT VOLTAGE RANGE                          |                      |                                                                                                                                         |  |                                    |                        |      |        |
| Input voltage range <sup>(1)</sup>           | I <sub>VR</sub>      |                                                                                                                                         |  | GND − 0.1                          | AV <sub>DD</sub> + 0.1 |      | V      |
| Overvoltage input range                      |                      | No output phase reversal <sup>(2)</sup>                                                                                                 |  | GND − 0.3                          | AV <sub>DD</sub> + 0.3 |      | V      |
| INPUT IMPEDANCE (Channel On) <sup>(3)</sup>  |                      |                                                                                                                                         |  |                                    |                        |      |        |
| Channel input capacitance                    | C <sub>CH</sub>      |                                                                                                                                         |  |                                    | 2                      |      | pF     |
| Channel switch resistance                    | R <sub>SW</sub>      |                                                                                                                                         |  |                                    | 150                    |      | Ω      |
| Amplifier input capacitance                  | C <sub>AMP</sub>     |                                                                                                                                         |  |                                    | 3                      |      | pF     |
| Amplifier input resistance                   | R <sub>AMP</sub>     | Input resistance to GND                                                                                                                 |  |                                    | 10                     |      | GΩ     |
| V <sub>CAL</sub> /CH0                        | R <sub>IN</sub>      | CAL1 or CAL2 selected                                                                                                                   |  |                                    | 100                    |      | kΩ     |
| GAIN SELECTIONS                              |                      |                                                                                                                                         |  |                                    |                        |      |        |
| Nominal gains                                |                      | Binary gains: 1, 2, 4, 8, 16, 32, 64, 128                                                                                               |  | 1                                  |                        | 128  |        |
|                                              |                      | Scope gains: 1, 2, 5, 10, 20, 50, 100, 200                                                                                              |  | 1                                  |                        | 200  |        |
| DC gain error                                | G = 1                | V <sub>OUT</sub> = GND + 85 mV to DV <sub>DD</sub> − 85 mV                                                                              |  |                                    | 0.006%                 | 0.1% |        |
|                                              | 1 < G ≤ 32           | V <sub>OUT</sub> = GND + 85 mV to DV <sub>DD</sub> − 85 mV                                                                              |  |                                    |                        | 0.1% |        |
|                                              | G ≥ 50               | V <sub>OUT</sub> = GND + 85 mV to DV <sub>DD</sub> − 85 mV                                                                              |  |                                    |                        | 0.3% |        |
| DC gain drift                                | G = 1                | V <sub>OUT</sub> = GND + 85 mV to DV <sub>DD</sub> − 85 mV                                                                              |  | T <sub>A</sub> = −40°C to 125°C    | 0.5                    |      | ppm/°C |
|                                              | 1 < G ≤ 32           | V <sub>OUT</sub> = GND + 85 mV to DV <sub>DD</sub> − 85 mV                                                                              |  | T <sub>A</sub> = −40°C to 125°C    | 2                      |      | ppm/°C |
|                                              | G ≥ 50               | V <sub>OUT</sub> = GND + 85 mV to DV <sub>DD</sub> − 85 mV                                                                              |  | T <sub>A</sub> = −40°C to 125°C    | 6                      |      | ppm/°C |
| CAL2 DC gain error <sup>(4)</sup>            |                      | Op Amp + Input = 0.9 V <sub>CAL</sub> , V <sub>REF</sub> = V <sub>CAL</sub> = AV <sub>DD</sub> /2, G = 1                                |  |                                    | 0.02%                  |      |        |
| CAL2 DC gain drift <sup>(4)</sup>            |                      | Op Amp + Input = 0.9 V <sub>CAL</sub> , V <sub>REF</sub> = V <sub>CAL</sub> = AV <sub>DD</sub> /2, G = 1                                |  | T <sub>A</sub> = −40°C to 125°C    | 2                      |      | ppm/°C |
| CAL3 DC gain error <sup>(4)</sup>            |                      | Op Amp + Input = 0.1 V <sub>CAL</sub> , V <sub>REF</sub> = V <sub>CAL</sub> = AV <sub>DD</sub> /2, G = 1                                |  |                                    | 0.02%                  |      |        |
| CAL3 DC gain drift <sup>(4)</sup>            |                      | Op Amp + Input = 0.1 V <sub>CAL</sub> , V <sub>REF</sub> = V <sub>CAL</sub> = AV <sub>DD</sub> /2, G = 1                                |  | T <sub>A</sub> = −40°C to 125°C    | 2                      |      | ppm/°C |
| INPUT IMPEDANCE (CHANNEL OFF) <sup>(3)</sup> |                      |                                                                                                                                         |  |                                    |                        |      |        |
| Input impedance                              | C <sub>CH</sub>      | See <a href="#">Figure 55</a>                                                                                                           |  |                                    | 2                      |      | pF     |

- (1) Gain error is a function of the input voltage. Gain error outside of the range ( $\text{GND} + 85\text{ mV} \leq V_{OUT} \leq DV_{DD} - 85\text{ mV}$ ) increases to 0.5% (typical).
- (2) Input voltages beyond this range must be current-limited to  $< 10\text{ mA}$  through the input protection diodes on each channel to prevent permanent destruction of the device.
- (3) See [Figure 55](#).
- (4) Total  $V_{OUT}$  error must be computed using input offset voltage error multiplied by gain. Includes op amp  $G = 1$  error.

**Electrical Characteristics:  $V_S = AV_{DD} = DV_{DD} = 5\text{ V}$  (continued)**

at  $T_A = 25^\circ\text{C}$ ,  $R_L = 10\text{ k}\Omega/C_L = 100\text{ pF}$  connected to  $DV_{DD}/2$ , and  $V_{REF} = \text{GND}$  (unless otherwise noted)

| PARAMETER                                              |                                     | TEST CONDITIONS                                                                                                               |  | MIN                                         | TYP                     | MAX  | UNIT             |
|--------------------------------------------------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|--|---------------------------------------------|-------------------------|------|------------------|
| INPUT OFF-CHANNEL CURRENT                              |                                     |                                                                                                                               |  |                                             |                         |      |                  |
| Input Off-Channel Current (Ch0, Ch1) <sup>(6)</sup>    | I <sub>LKG</sub>                    | V <sub>REF</sub> = GND, V <sub>OFF-CHANNEL</sub> = AV <sub>DD</sub> /2, V <sub>ON-CHANNEL</sub> = AV <sub>DD</sub> /2 – 0.1 V |  | ±0.05                                       |                         | ±1   | nA               |
| Over temperature, –40°C to 125°C                       |                                     | V <sub>REF</sub> = GND, V <sub>OFF-CHANNEL</sub> = AV <sub>DD</sub> /2, V <sub>ON-CHANNEL</sub> = AV <sub>DD</sub> /2 – 0.1 V |  | See <a href="#">Typical Characteristics</a> |                         |      |                  |
| Channel-to-Channel Crosstalk                           |                                     |                                                                                                                               |  | 130                                         |                         |      | dB               |
| OUTPUT                                                 |                                     |                                                                                                                               |  |                                             |                         |      |                  |
| Voltage output swing from rail                         |                                     | I <sub>OUT</sub> = ±0.25 mA, AV <sub>DD</sub> ≥ DV <sub>DD</sub> <sup>(6)</sup>                                               |  | GND + 0.05                                  | DV <sub>DD</sub> – 0.05 |      | V                |
|                                                        |                                     | I <sub>OUT</sub> = ±5 mA, AV <sub>DD</sub> ≥ DV <sub>DD</sub> <sup>(6)</sup>                                                  |  | GND + 0.25                                  | DV <sub>DD</sub> – 0.25 |      | V                |
| DC output nonlinearity                                 |                                     | V <sub>OUT</sub> = GND + 85 mV to DV <sub>DD</sub> – 85 mV <sup>(7)</sup>                                                     |  | 0.0015                                      |                         |      | %FSR             |
| Short circuit current                                  | I <sub>SC</sub>                     |                                                                                                                               |  | –30/+60                                     |                         |      | mA               |
| Capacitive load drive                                  | C <sub>LOAD</sub>                   |                                                                                                                               |  | See <a href="#">Typical Characteristics</a> |                         |      |                  |
| NOISE                                                  |                                     |                                                                                                                               |  |                                             |                         |      |                  |
| Input voltage noise density                            | e <sub>n</sub>                      | f > 10 kHz, C <sub>L</sub> = 100 pF, V <sub>S</sub> = 5 V                                                                     |  | 12                                          |                         |      | nV/√Hz           |
|                                                        |                                     | f > 10 kHz, C <sub>L</sub> = 100 pF, V <sub>S</sub> = 2.2 V                                                                   |  | 22                                          |                         |      | nV/√Hz           |
| Input voltage noise                                    | e <sub>n</sub>                      | f = 0.1 Hz to 10 Hz, C <sub>L</sub> = 100 pF, V <sub>S</sub> = 5 V                                                            |  | 0.362                                       |                         |      | μV <sub>PP</sub> |
|                                                        |                                     | f = 0.1 Hz to 10 Hz, C <sub>L</sub> = 100 pF, V <sub>S</sub> = 2.2 V                                                          |  | 0.736                                       |                         |      | μV <sub>PP</sub> |
| Input current density                                  | i <sub>n</sub>                      | f = 10 kHz, C <sub>L</sub> = 100 pF                                                                                           |  | 400                                         |                         |      | fA/√Hz           |
| SLEW RATE                                              |                                     |                                                                                                                               |  |                                             |                         |      |                  |
| Slew rate                                              | SR                                  |                                                                                                                               |  | See <a href="#">Table 1</a>                 |                         |      | V/μs             |
| SETTLING TIME                                          |                                     |                                                                                                                               |  |                                             |                         |      |                  |
| Settling time                                          | t <sub>s</sub>                      |                                                                                                                               |  | See <a href="#">Table 1</a>                 |                         |      | μs               |
| FREQUENCY RESPONSE                                     |                                     |                                                                                                                               |  |                                             |                         |      |                  |
| Frequency response                                     |                                     |                                                                                                                               |  | See <a href="#">Table 1</a>                 |                         |      | MHz              |
| THD + NOISE                                            |                                     |                                                                                                                               |  |                                             |                         |      |                  |
|                                                        |                                     | G = 1, f = 1 kHz, V <sub>OUT</sub> = 4 V <sub>PP</sub> at 2.5V <sub>DC</sub> , C <sub>L</sub> = 100 pF                        |  | 0.003%                                      |                         |      |                  |
|                                                        |                                     | G = 10, f = 1 kHz, V <sub>OUT</sub> = 4 V <sub>PP</sub> at 2.5V <sub>DC</sub> , C <sub>L</sub> = 100 pF                       |  | 0.005%                                      |                         |      |                  |
|                                                        |                                     | G = 50, f = 1 kHz, V <sub>OUT</sub> = 4 V <sub>PP</sub> at 2.5V <sub>DC</sub> , C <sub>L</sub> = 100 pF                       |  | 0.03%                                       |                         |      |                  |
|                                                        |                                     | G = 128, f = 1 kHz, V <sub>OUT</sub> = 4 V <sub>PP</sub> at 2.5V <sub>DC</sub> , C <sub>L</sub> = 100 pF                      |  | 0.08%                                       |                         |      |                  |
|                                                        |                                     | G = 200, f = 1 kHz, V <sub>OUT</sub> = 4 V <sub>PP</sub> at 2.5V <sub>DC</sub> , C <sub>L</sub> = 100 pF                      |  | 0.1%                                        |                         |      |                  |
|                                                        |                                     | G = 1, f = 20 kHz, V <sub>OUT</sub> = 4 V <sub>PP</sub> at 2.5V <sub>DC</sub> , C <sub>L</sub> = 100 pF                       |  | 0.02%                                       |                         |      |                  |
|                                                        |                                     | G = 10, f = 20 kHz, V <sub>OUT</sub> = 4 V <sub>PP</sub> at 2.5V <sub>DC</sub> , C <sub>L</sub> = 100 pF                      |  | 0.01%                                       |                         |      |                  |
|                                                        |                                     | G = 50, f = 20 kHz, V <sub>OUT</sub> = 4 V <sub>PP</sub> at 2.5V <sub>DC</sub> , C <sub>L</sub> = 100 pF                      |  | 0.03%                                       |                         |      |                  |
|                                                        |                                     | G = 128, f = 20 kHz, V <sub>OUT</sub> = 4 V <sub>PP</sub> at 2.5V <sub>DC</sub> , C <sub>L</sub> = 100 pF                     |  | 0.08%                                       |                         |      |                  |
|                                                        |                                     | G = 200, f = 20 kHz, V <sub>OUT</sub> = 4 V <sub>PP</sub> at 2.5V <sub>DC</sub> , C <sub>L</sub> = 100 pF                     |  | 0.11%                                       |                         |      |                  |
| POWER SUPPLY                                           |                                     |                                                                                                                               |  |                                             |                         |      |                  |
| Operating voltage range <sup>(6)</sup>                 | AV <sub>DD</sub>                    |                                                                                                                               |  | 2.2                                         |                         | 5.5  | V                |
|                                                        | DV <sub>DD</sub>                    |                                                                                                                               |  | 2.2                                         |                         | 5.5  | V                |
| Quiescent current analog                               | I <sub>OA</sub>                     | I <sub>O</sub> = 0, G = 1, V <sub>OUT</sub> = V <sub>REF</sub>                                                                |  | 0.33                                        |                         | 0.45 | mA               |
| Over temperature, –40°C to 125°C                       |                                     | T <sub>A</sub> = –40°C to 125°C                                                                                               |  |                                             |                         | 0.45 | mA               |
| Quiescent current digital <sup>(8)(9)(10)</sup>        | I <sub>OD</sub>                     | I <sub>O</sub> = 0, G = 1, V <sub>OUT</sub> = V <sub>REF</sub> , SCLK at 10 MHz, CS = Logic 0, DIO or DIN = Logic 0           |  | 0.75                                        |                         | 1.2  | mA               |
| Over temperature, –40°C to 125°C <sup>(8)(9)(10)</sup> |                                     | T <sub>A</sub> = –40°C to 125°C                                                                                               |  |                                             |                         | 1.2  | mA               |
| Shutdown current analog + digital <sup>(8)(9)</sup>    | I <sub>SDA</sub> + I <sub>SDD</sub> | I <sub>O</sub> = 0, V <sub>OUT</sub> = V <sub>REF</sub> , G = 1, SCLK Idle                                                    |  | 4                                           |                         |      | μA               |
|                                                        |                                     | I <sub>O</sub> = 0, V <sub>OUT</sub> = 0, G = 1, SCLK at 10MHz, CS = Logic 0, DIO or DIN = Logic 0                            |  | 245                                         |                         |      | μA               |
| POWER-ON RESET (POR)                                   |                                     |                                                                                                                               |  |                                             |                         |      |                  |
| POR trip voltage                                       |                                     | Digital interface disabled and Command Register set to POR values for DV <sub>DD</sub> < POR Trip Voltage                     |  | 1.6                                         |                         |      | V                |

(5) Maximum specification limitation limited by final test time and capability.

(6) When  $AV_{DD}$  is less than  $DV_{DD}$ , the output is clamped to  $AV_{DD} + 300\text{ mV}$ .

(7) Measurement limited by noise in test equipment and test time.

(8) Does not include current into or out of the  $V_{REF}$  pin. Internal  $R_F$  and  $R_I$  are always connected between  $V_{OUT}$  and  $V_{REF}$ .

(9) Digital logic levels: DIO or DIN = logic 0.  $10\text{-}\mu\text{A}$  internal pulldown current source.

(10) Includes current from op amp output structure.

**PGA112, PGA113, PGA116, PGA117**

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**Electrical Characteristics:  $V_S = AV_{DD} = DV_{DD} = 5\text{ V}$  (continued)**

at  $T_A = 25^\circ\text{C}$ ,  $R_L = 10\text{ k}\Omega$ / $C_L = 100\text{ pF}$  connected to  $DV_{DD}/2$ , and  $V_{REF} = \text{GND}$  (unless otherwise noted)

| PARAMETER                                                                 |                                    | TEST CONDITIONS                                                                                 | MIN             | TYP | MAX                | UNIT               |
|---------------------------------------------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------|-----------------|-----|--------------------|--------------------|
| <b>TEMPERATURE RANGE</b>                                                  |                                    |                                                                                                 |                 |     |                    |                    |
| Specified range                                                           |                                    |                                                                                                 | –40             |     | 125                | $^\circ\text{C}$   |
| Operating range                                                           |                                    |                                                                                                 | –40             |     | 125                | $^\circ\text{C}$   |
| Thermal resistance                                                        | $\theta_{JA}$                      |                                                                                                 |                 |     |                    |                    |
|                                                                           | VSSOP-10                           |                                                                                                 |                 | 164 |                    | $^\circ\text{C/W}$ |
| <b>DIGITAL INPUTS (SCLK, <math>\overline{\text{CS}}</math>, DIO, DIN)</b> |                                    |                                                                                                 |                 |     |                    |                    |
| Logic low                                                                 |                                    |                                                                                                 | 0               |     | $0.3DV_{DD}$       | V                  |
| Input leakage current (SCLK and $\overline{\text{CS}}$ only)              |                                    |                                                                                                 | –1              |     | 1                  | $\mu\text{A}$      |
| Weak pulldown current (DIO, DIN only)                                     |                                    |                                                                                                 |                 | 10  |                    | $\mu\text{A}$      |
| Logic high                                                                |                                    |                                                                                                 | $0.7DV_{DD}$    |     | $DV_{DD}$          | V                  |
| Hysteresis                                                                |                                    |                                                                                                 |                 | 700 |                    | mV                 |
| <b>DIGITAL OUTPUT (DIO, DOUT)</b>                                         |                                    |                                                                                                 |                 |     |                    |                    |
| Logic high                                                                | $I_{OH} = -3\text{ mA}$ (sourcing) |                                                                                                 | $DV_{DD} - 0.4$ |     | $DV_{DD}$          | V                  |
| Logic low                                                                 | $I_{OL} = 3\text{ mA}$ (sinking)   |                                                                                                 | GND             |     | $\text{GND} + 0.4$ | V                  |
| <b>CHANNEL AND GAIN TIMING</b>                                            |                                    |                                                                                                 |                 |     |                    |                    |
| Channel select time                                                       |                                    |                                                                                                 |                 | 0.2 |                    | $\mu\text{s}$      |
| Gain select time                                                          |                                    |                                                                                                 |                 | 0.2 |                    | $\mu\text{s}$      |
| <b>SHUTDOWN MODE TIMING</b>                                               |                                    |                                                                                                 |                 |     |                    |                    |
| Enable time                                                               |                                    |                                                                                                 |                 | 4   |                    | $\mu\text{s}$      |
| Disable time                                                              |                                    | $V_{OUT}$ goes high-impedance, $R_F$ and $R_I$ remain connected between $V_{OUT}$ and $V_{REF}$ |                 | 2   |                    | $\mu\text{s}$      |
| <b>POWER-ON-RESET (POR) TIMING</b>                                        |                                    |                                                                                                 |                 |     |                    |                    |
| POR power-up time                                                         |                                    | $DV_{DD} \geq 2\text{ V}$                                                                       |                 | 40  |                    | $\mu\text{s}$      |
| POR power-down time                                                       |                                    | $DV_{DD} \leq 1.5\text{ V}$                                                                     |                 | 5   |                    | $\mu\text{s}$      |

## 7.6 SPI Timing: $V_S = AV_{DD} = DV_{DD} = 2.2\text{ V to }5\text{ V}$

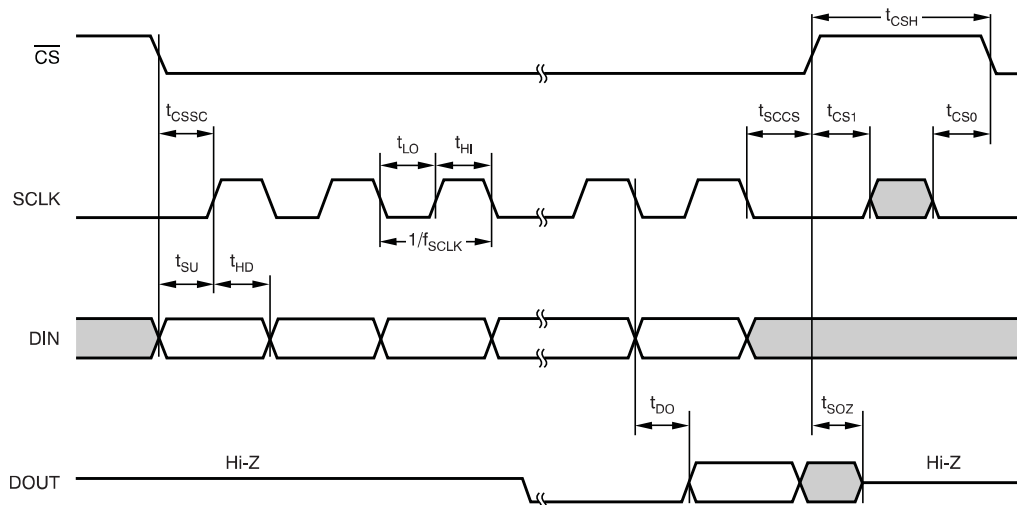
At  $T_A = +25^\circ\text{C}$ ,  $R_L = 10\text{ k}\Omega$ / $C_L = 100\text{ pF}$  connected to  $DV_{DD}/2$ , and  $V_{REF} = \text{GND}$ , unless otherwise noted.

|            |                                                                                           | MIN | NOM | MAX | UNIT          |
|------------|-------------------------------------------------------------------------------------------|-----|-----|-----|---------------|
|            | Input capacitance (SCLK, $\overline{\text{CS}}$ , and DIO pins)                           |     | 1   |     | pF            |
| $t_{RFI}$  | Input rise and fall time <sup>(1)</sup><br>( $\overline{\text{CS}}$ , SCLK, and DIO pins) |     |     | 2   | $\mu\text{s}$ |
| $t_{RFO}$  | Output rise and fall time (DIO pin) <sup>(1)</sup>                                        |     |     | 10  | ns            |
|            | $C_{LOAD} = 60\text{ pF}$                                                                 |     |     |     |               |
| $t_{CSH}$  | $\overline{\text{CS}}$ high time ( $\overline{\text{CS}}$ pin) <sup>(1)</sup>             | 40  |     |     | ns            |
| $t_{CSO}$  | SCLK edge to $\overline{\text{CS}}$ fall setup time <sup>(1)</sup>                        | 10  |     |     | ns            |
| $t_{CSSC}$ | $\overline{\text{CS}}$ fall to first SCLK edge setup time                                 | 10  |     |     | ns            |
| $f_{SCLK}$ | SCLK Frequency <sup>(2)</sup>                                                             |     |     | 10  | MHz           |
| $t_{HI}$   | SCLK high time <sup>(3)</sup>                                                             | 40  |     |     | ns            |
| $t_{LO}$   | SCLK low time <sup>(3)</sup>                                                              | 40  |     |     | ns            |
| $t_{SCCS}$ | SCLK last edge to $\overline{\text{CS}}$ rise setup time <sup>(1)</sup>                   | 10  |     |     | ns            |
| $t_{CS1}$  | $\overline{\text{CS}}$ rise to SCLK edge setup time <sup>(1)</sup>                        | 10  |     |     | ns            |
| $t_{SU}$   | DIN setup time                                                                            | 10  |     |     | ns            |
| $t_{HD}$   | DIN hold time                                                                             | 10  |     |     | ns            |
| $t_{DO}$   | SCLK to DOUT valid propagation delay <sup>(1)</sup>                                       |     |     | 25  | ns            |
| $t_{SOZ}$  | $\overline{\text{CS}}$ rise to DOUT forced to Hi-Z <sup>(1)</sup>                         |     |     | 20  | ns            |

(1) Ensured by design; not production tested.

(2) When using devices in daisy-chain mode, the maximum clock frequency for SCLK is limited by SCLK rise and fall time, DIN setup time, and DOUT propagation delay. See Figure 61. Based on this limitation, the maximum SCLK frequency for daisy-chain mode is 9.09 MHz.

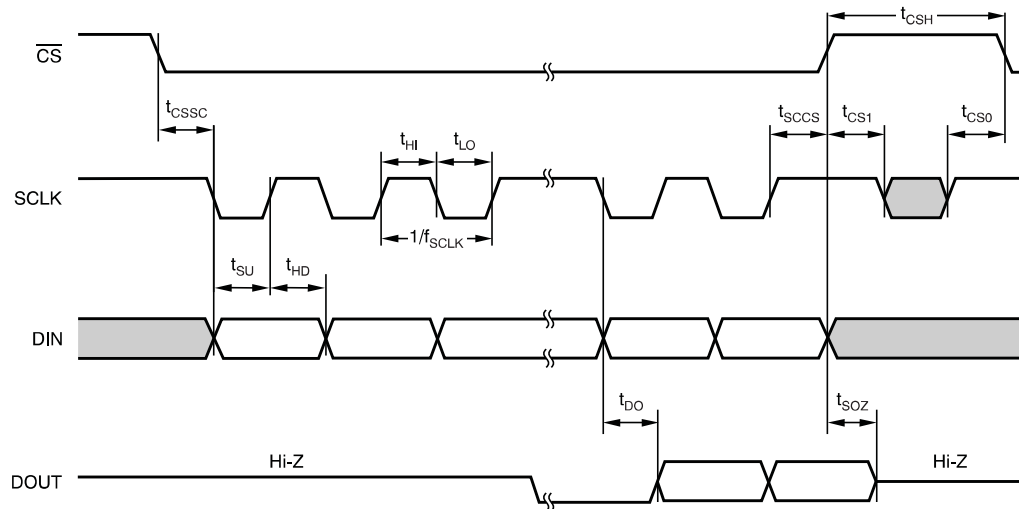
(3)  $t_{HI}$  and  $t_{LO}$  must not be less than  $1/f_{SCLK}$  (maximum).



**Figure 1. SPI Mode 0, 0**

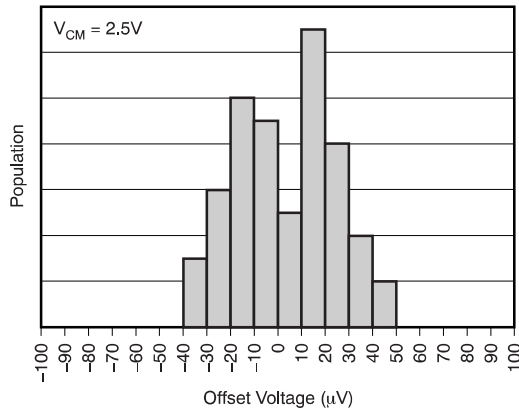
**PGA112, PGA113, PGA116, PGA117**

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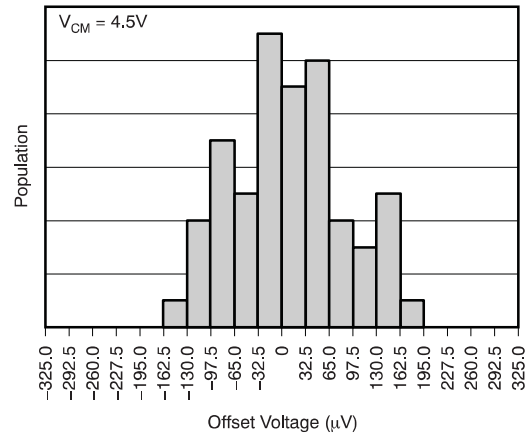
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**Figure 2. SPI Mode 1, 1**

## 7.7 Typical Characteristics

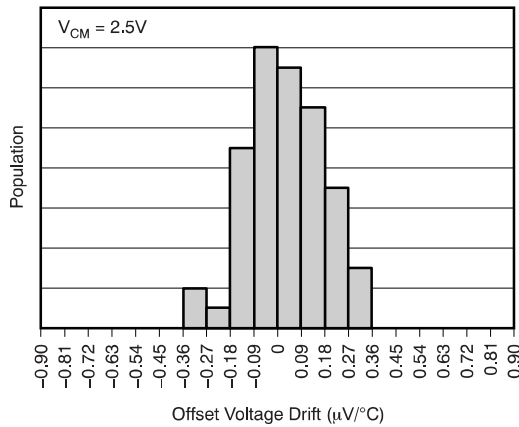
at  $T_A = 25^\circ\text{C}$ ,  $A_{V_{DD}} = DV_{DD} = 5\text{ V}$ ,  $R_L = 10\text{ k}\Omega$  connected to  $DV_{DD}/2$ ,  $V_{REF} = \text{GND}$ , and  $C_L = 100\text{ pF}$ , unless otherwise noted.



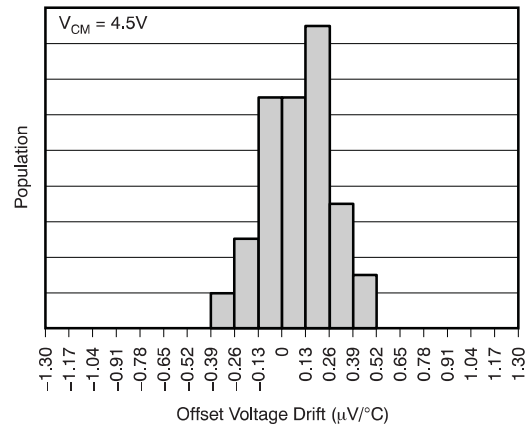
**Figure 3. Offset Voltage**



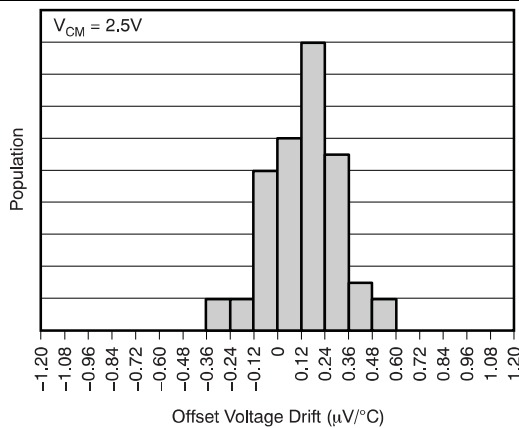
**Figure 4. Offset Voltage**



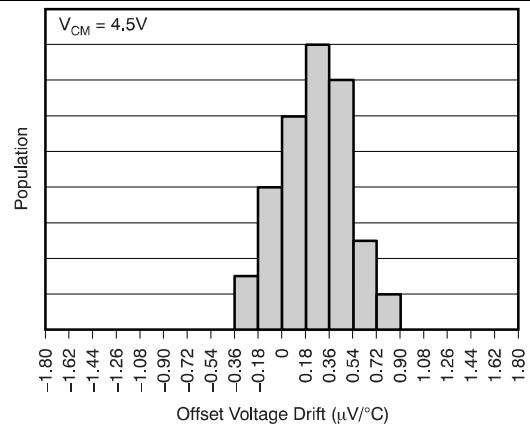
**Figure 5. Offset Voltage Drift (–40°C to 85°C)**



**Figure 6. Offset Voltage Drift (–40°C to 85°C)**



**Figure 7. Offset Voltage Drift (–40°C to 125°C)**

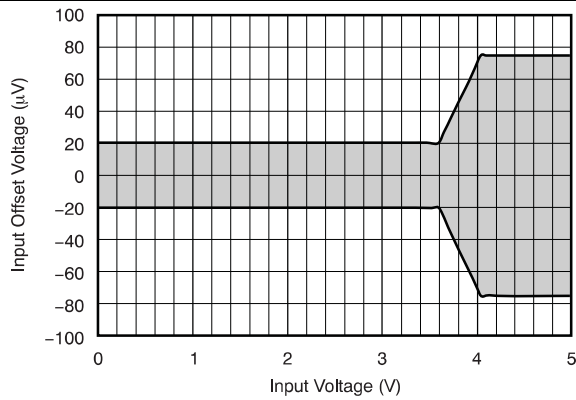
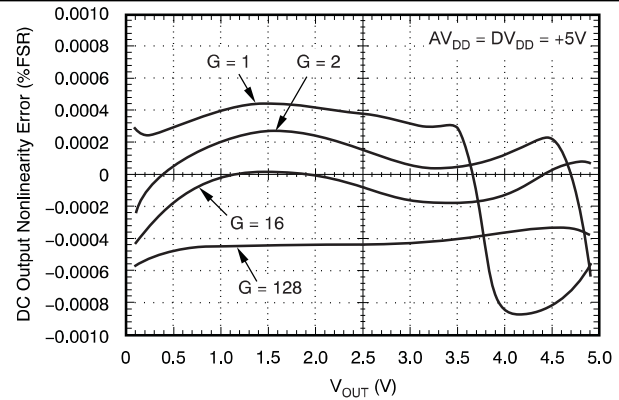
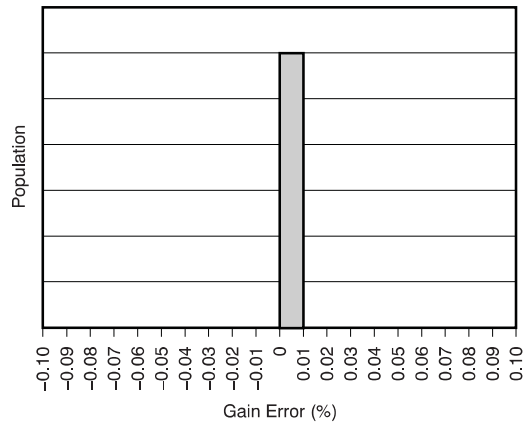
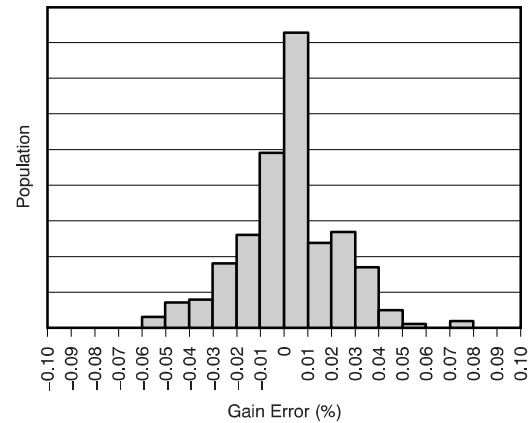
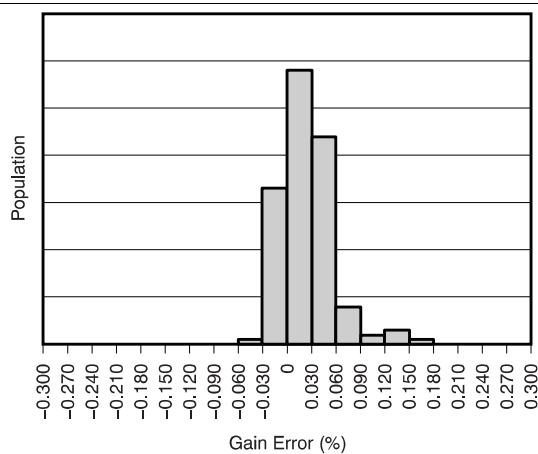
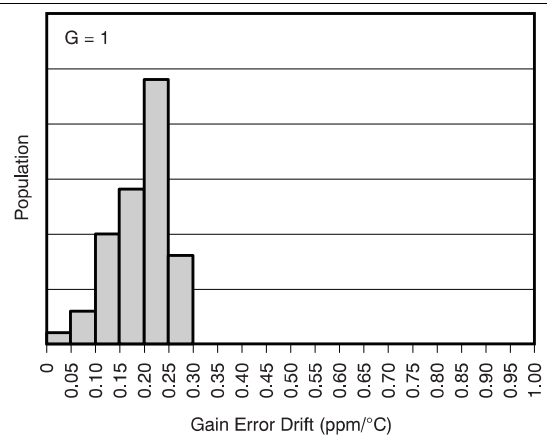


**Figure 8. Offset Voltage Drift (–40°C to 125°C)**

**PGA112, PGA113, PGA116, PGA117**

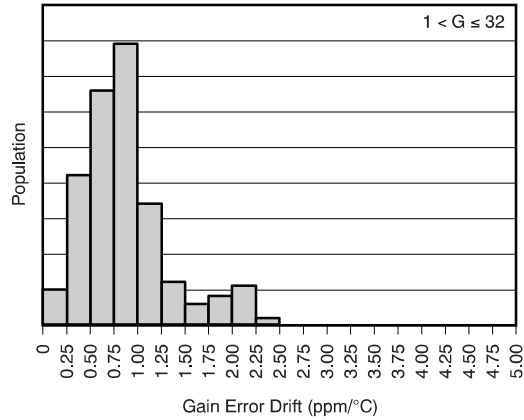
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**Typical Characteristics (continued)**

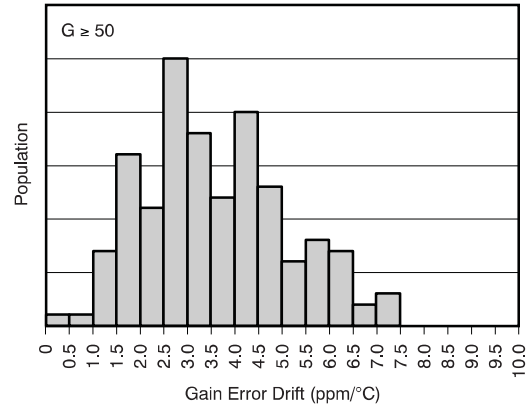
 at  $T_A = 25^\circ\text{C}$ ,  $AV_{DD} = DV_{DD} = 5\text{ V}$ ,  $R_L = 10\text{ k}\Omega$  connected to  $DV_{DD}/2$ ,  $V_{REF} = \text{GND}$ , and  $C_L = 100\text{ pF}$ , unless otherwise noted.

**Figure 9. Input Offset Voltage vs Input Voltage**

**Figure 10. PGA112 and PGA116 Nonlinearity**

**Figure 11. Gain Error (G = 1)**

**Figure 12. Gain Error (1 < G ≤ 32)**

**Figure 13. Gain Error (G ≥ 50)**

**Figure 14. Gain Error Drift (-40°C to 125°C)**

## Typical Characteristics (continued)

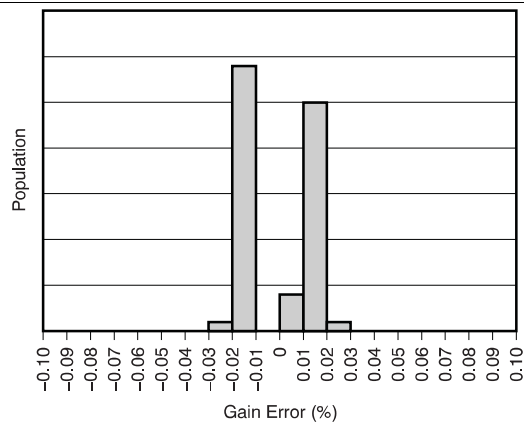
at  $T_A = 25^\circ\text{C}$ ,  $AV_{DD} = DV_{DD} = 5\text{ V}$ ,  $R_L = 10\text{ k}\Omega$  connected to  $DV_{DD}/2$ ,  $V_{REF} = \text{GND}$ , and  $C_L = 100\text{ pF}$ , unless otherwise noted.



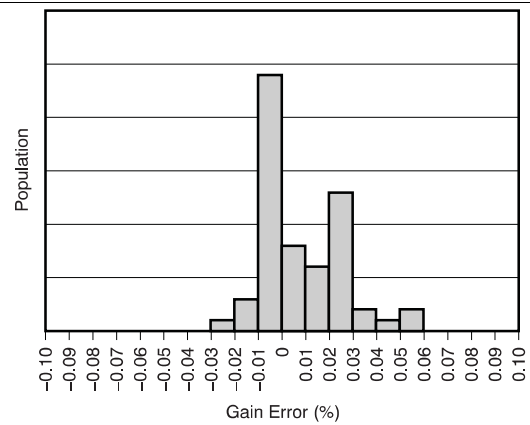
**Figure 15. Gain Error Drift ( $-40^\circ\text{C}$  to  $125^\circ\text{C}$ )**



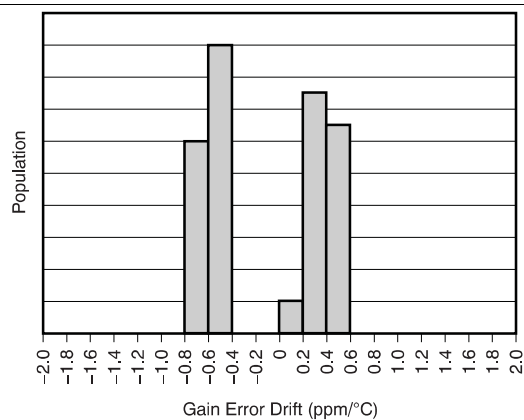
**Figure 16. Gain Error Drift ( $-40^\circ\text{C}$  to  $125^\circ\text{C}$ )**



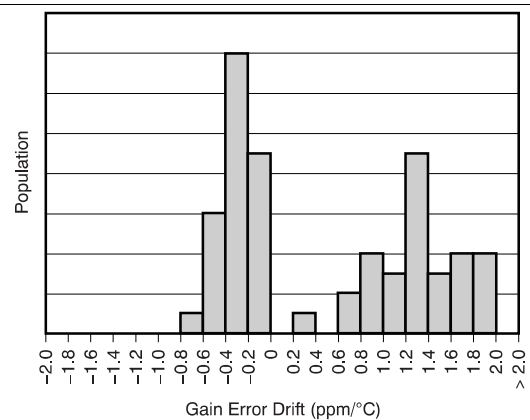
**Figure 17. CAL2 Gain Error**



**Figure 18. CAL3 Gain Error**



**Figure 19. CAL2 Gain Error Drift ( $-40^\circ\text{C}$  to  $125^\circ\text{C}$ )**



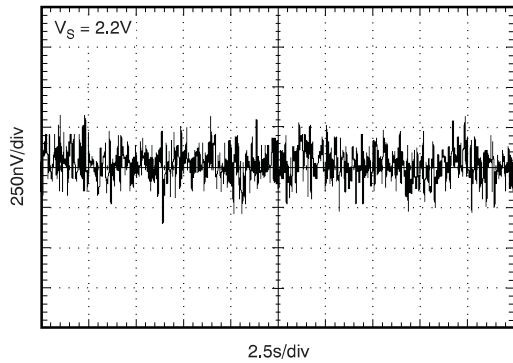
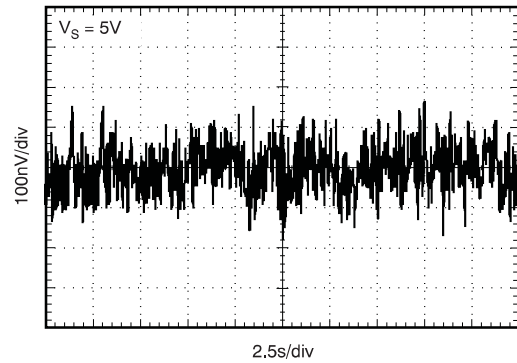
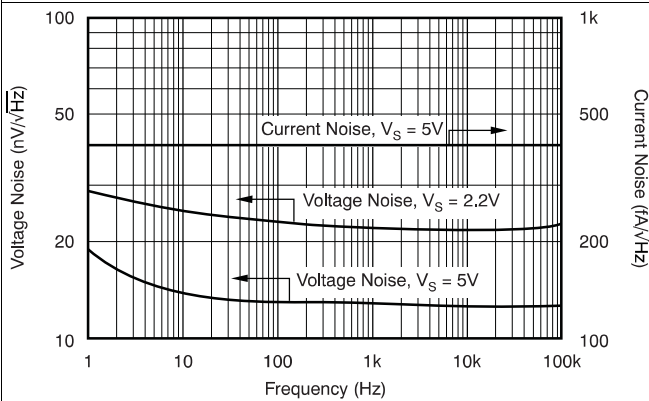
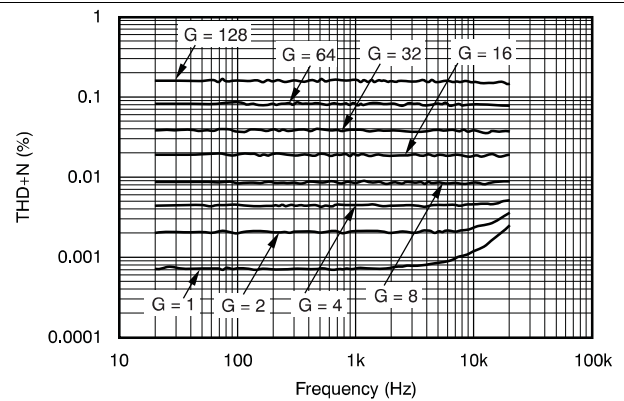
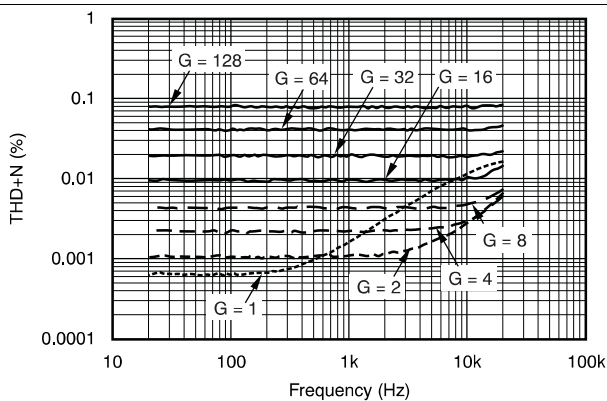
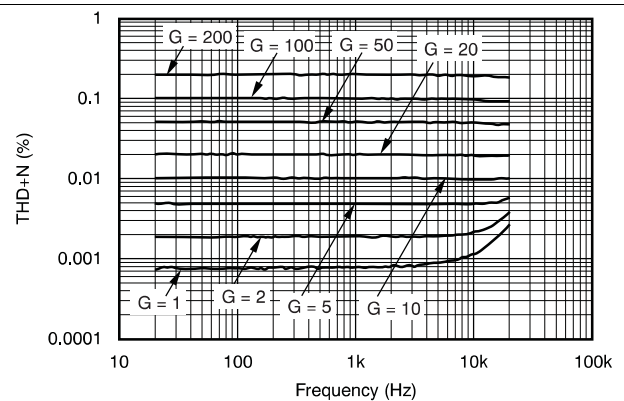
**Figure 20. CAL3 Gain Error Drift ( $-40^\circ\text{C}$  to  $125^\circ\text{C}$ )**

**PGA112, PGA113, PGA116, PGA117**

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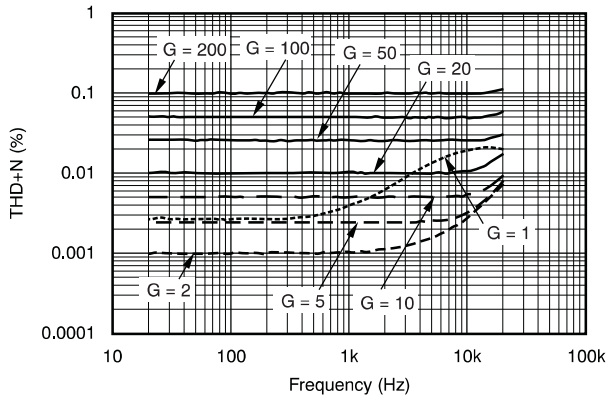
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**Typical Characteristics (continued)**

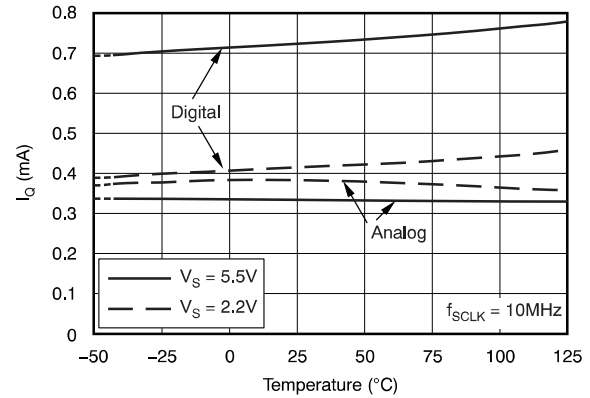
at  $T_A = 25^\circ\text{C}$ ,  $AV_{DD} = DV_{DD} = 5\text{ V}$ ,  $R_L = 10\text{ k}\Omega$  connected to  $DV_{DD}/2$ ,  $V_{REF} = \text{GND}$ , and  $C_L = 100\text{ pF}$ , unless otherwise noted.

**Figure 21. 0.1 Hz To 10 Hz NOISE**

**Figure 22. 0.1 Hz to 10 Hz NOISE**

**Figure 23. Spectral NOISE Density**

**Figure 24. PGA112, PGA116 THD + Noise vs Frequency ( $V_{OUT} = 2\text{ V}_{PP}$ )**

**Figure 25. PGA112, PGA116 THD + NOISE vs Frequency ( $V_{OUT} = 4\text{ V}_{PP}$ )**

**Figure 26. PGA113, PGA117 THD + Noise vs Frequency ( $V_{OUT} = 2\text{ V}_{PP}$ )**

## Typical Characteristics (continued)

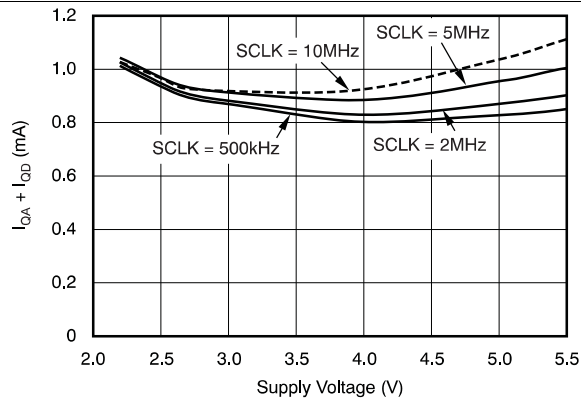
at  $T_A = 25^\circ\text{C}$ ,  $AV_{DD} = DV_{DD} = 5\text{ V}$ ,  $R_L = 10\text{ k}\Omega$  connected to  $DV_{DD}/2$ ,  $V_{REF} = \text{GND}$ , and  $C_L = 100\text{ pF}$ , unless otherwise noted.



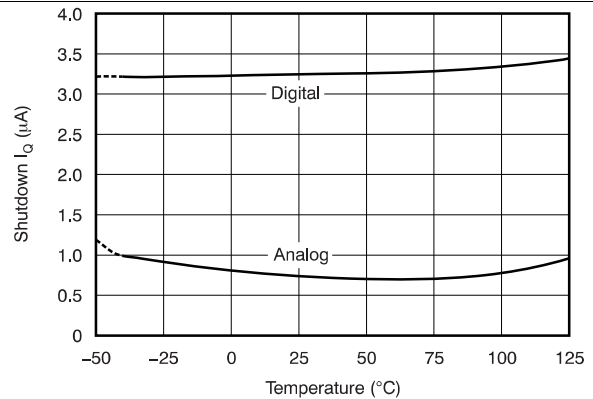
**Figure 27. PGA113, PGA117 THD + Noise vs Frequency**  
( $V_{OUT} = 4\text{ V}_{PP}$ )



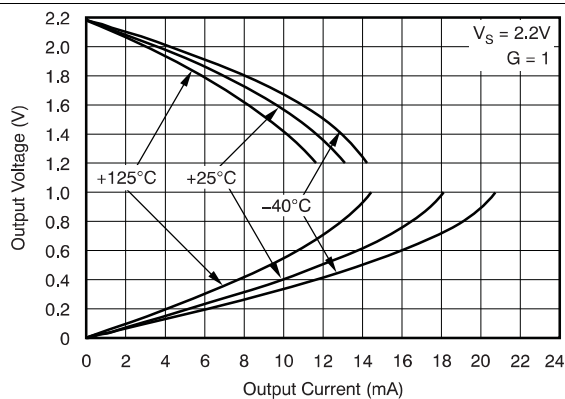
**Figure 28. Quiescent Current vs Temperature**



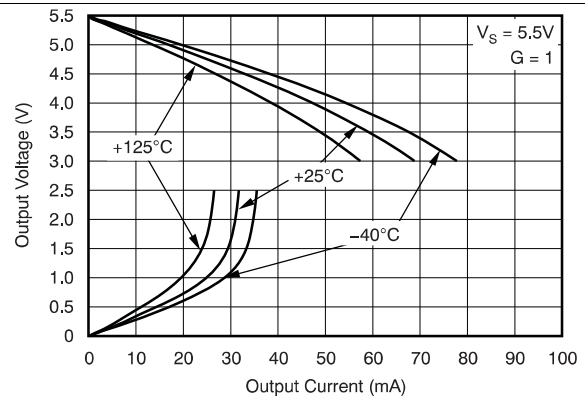
**Figure 29. Total Quiescent Current vs Supply Voltage**



**Figure 30. Shutdown Quiescent Current vs Temperature**



**Figure 31. Output Voltage vs Output Current**



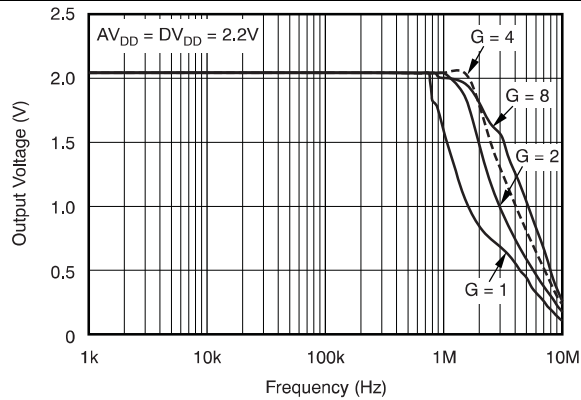
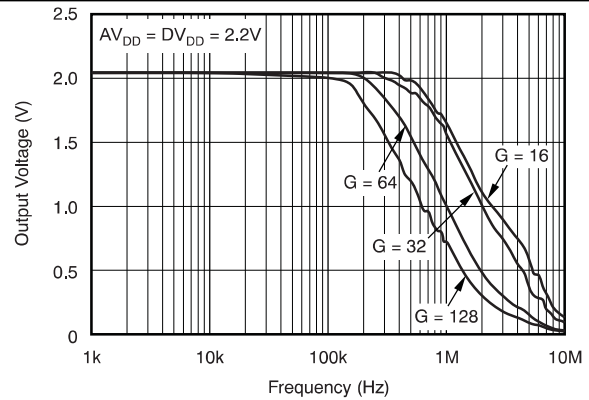
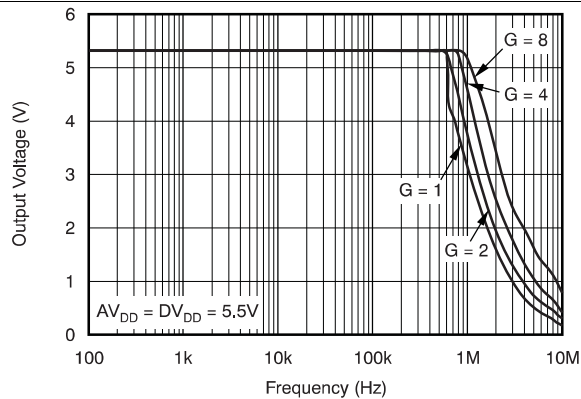
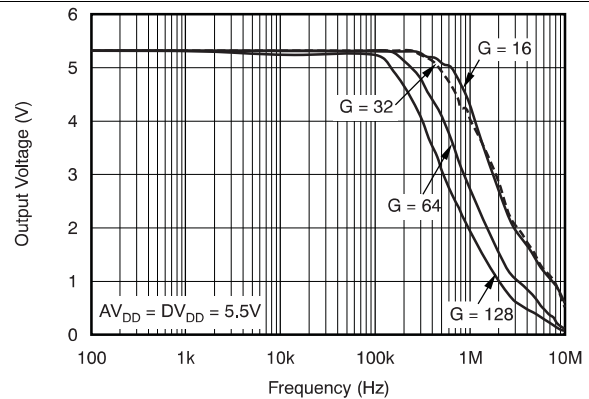
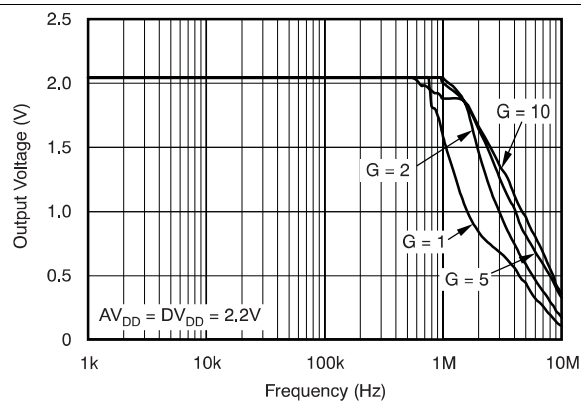
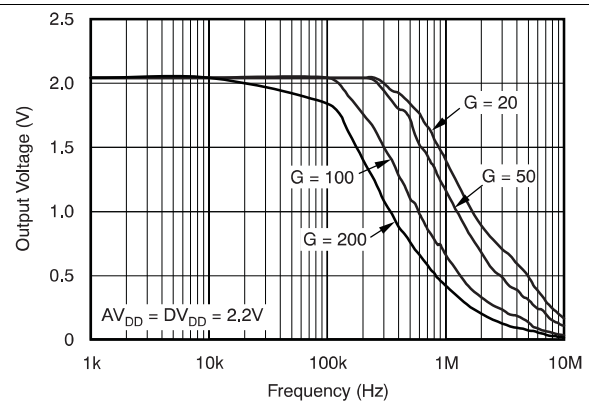
**Figure 32. Output Voltage vs Output Current**

**PGA112, PGA113, PGA116, PGA117**

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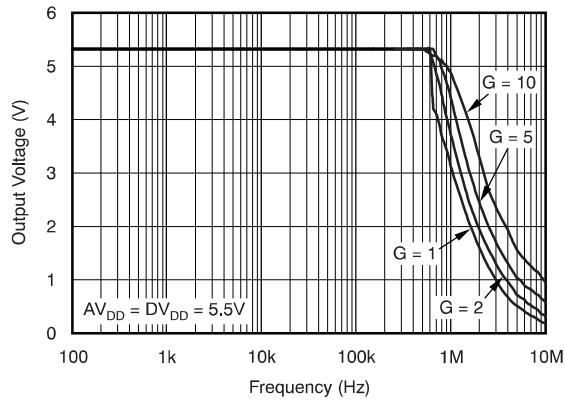
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**Typical Characteristics (continued)**

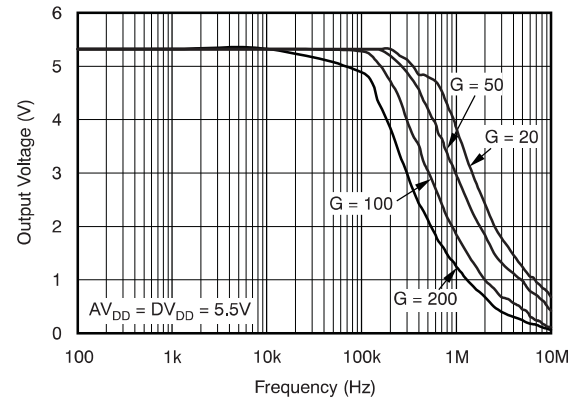
at  $T_A = 25^\circ\text{C}$ ,  $AV_{DD} = DV_{DD} = 5\text{ V}$ ,  $R_L = 10\text{ k}\Omega$  connected to  $DV_{DD}/2$ ,  $V_{REF} = \text{GND}$ , and  $C_L = 100\text{ pF}$ , unless otherwise noted.

**Figure 33. PGA112, PGA116 Output Voltage Swing vs Frequency**

**Figure 34. PGA112, PGA116 Output Voltage Swing vs Frequency**

**Figure 35. PGA112, PGA116 Output Voltage Swing vs Frequency**

**Figure 36. PGA112, PGA116 Output Voltage Swing vs Frequency**

**Figure 37. PGA113, PGA117 Output Voltage Swing vs Frequency**

**Figure 38. PGA113, PGA117 Output Voltage Swing vs Frequency**

## Typical Characteristics (continued)

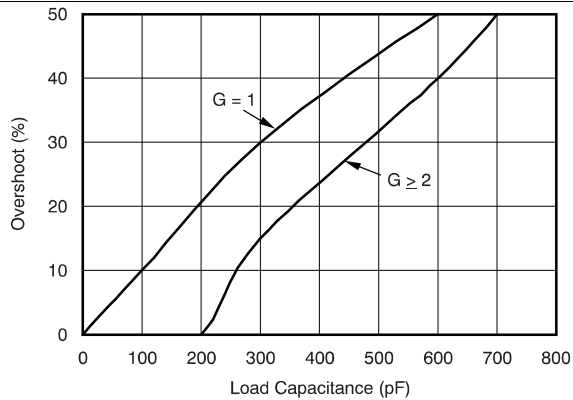
at  $T_A = 25^\circ\text{C}$ ,  $AV_{DD} = DV_{DD} = 5\text{ V}$ ,  $R_L = 10\text{ k}\Omega$  connected to  $DV_{DD}/2$ ,  $V_{REF} = \text{GND}$ , and  $C_L = 100\text{ pF}$ , unless otherwise noted.



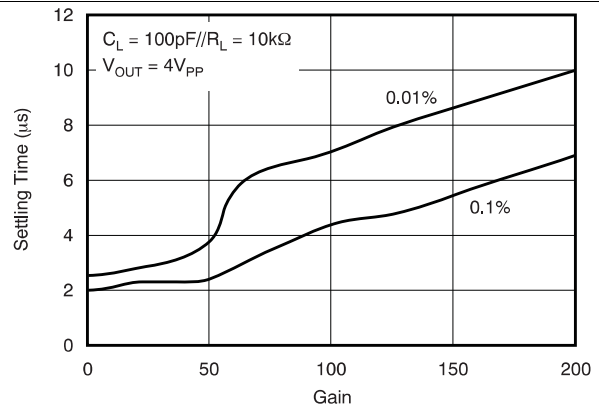
**Figure 39. PGA113, PGA117 Output Voltage Swing vs Frequency**



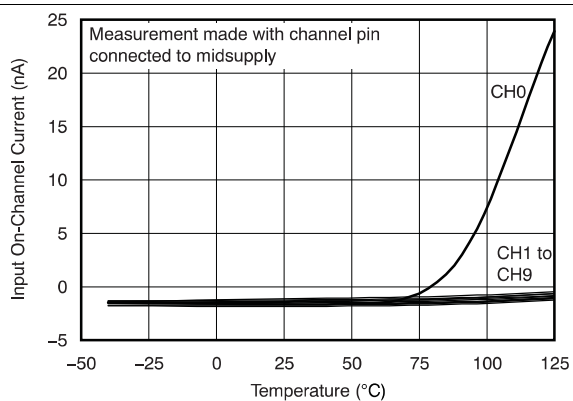
**Figure 40. PGA113, PGA117 Output Voltage Swing vs Frequency**



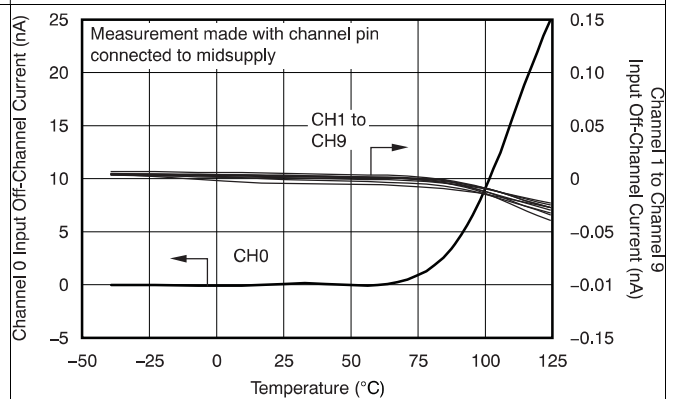
**Figure 41. Small-Signal Overshoot vs Load Capacitance**



**Figure 42. Gain vs Settling Time**



**Figure 43. Input ON-Channel Current vs Temperature**

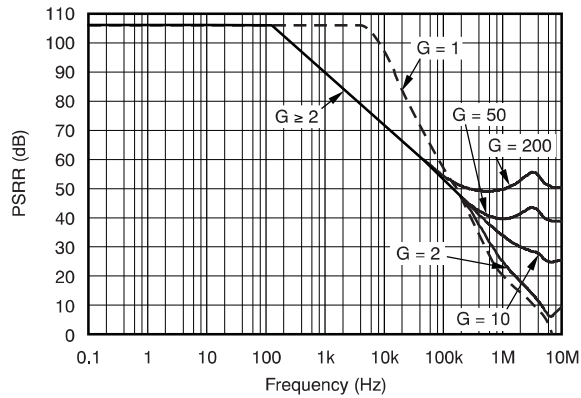
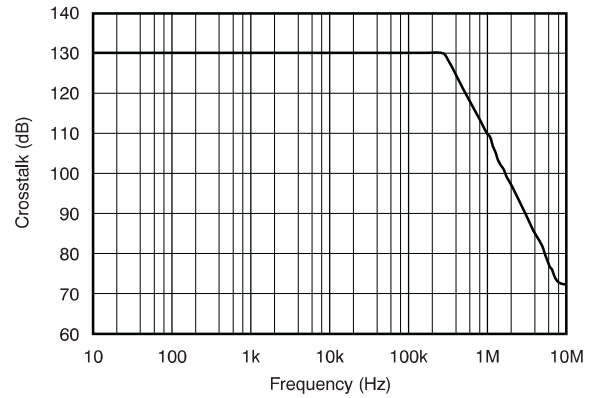
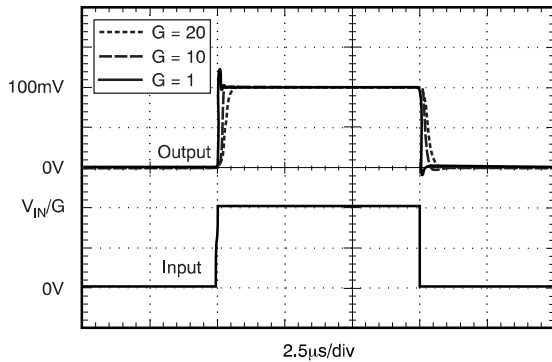
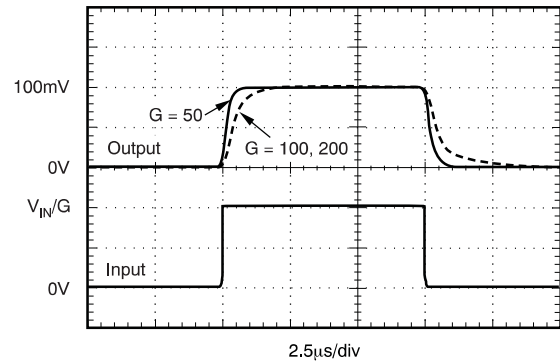
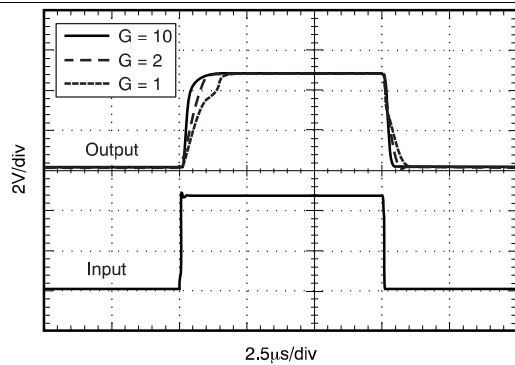
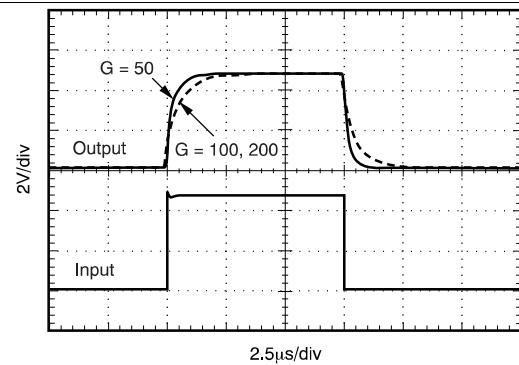


**Figure 44. Input OFF-Channel Leakage Current vs Temperature**

**PGA112, PGA113, PGA116, PGA117**

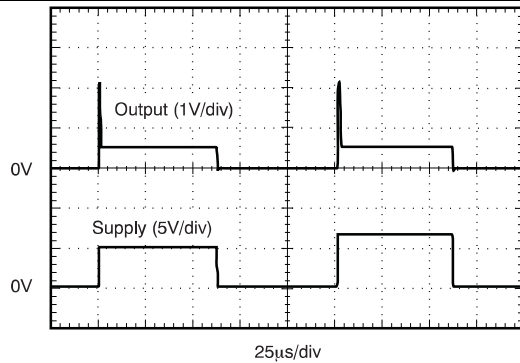
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**Typical Characteristics (continued)**

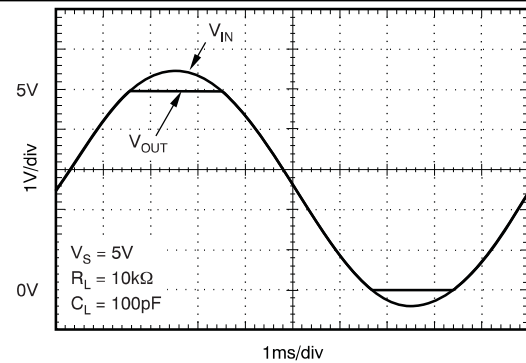
 at  $T_A = 25^\circ\text{C}$ ,  $AV_{DD} = DV_{DD} = 5\text{ V}$ ,  $R_L = 10\text{ k}\Omega$  connected to  $DV_{DD}/2$ ,  $V_{REF} = \text{GND}$ , and  $C_L = 100\text{ pF}$ , unless otherwise noted.

**Figure 45. Power-Supply Rejection Ratio vs Frequency**

**Figure 46. Crosstalk vs Frequency**

**Figure 47. Small-Signal Pulse Response**

**Figure 48. Small-Signal Pulse Response**

**Figure 49. Large-Signal Pulse Response**

**Figure 50. Large-Signal Pulse Response**

## Typical Characteristics (continued)

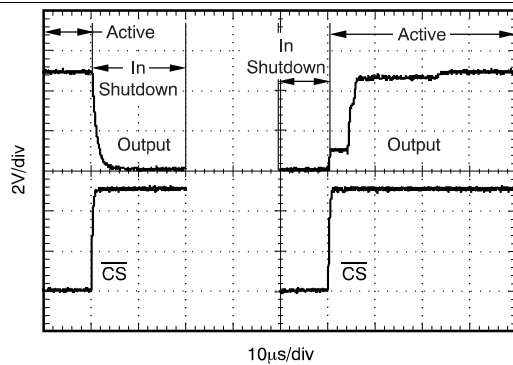
at  $T_A = 25^\circ\text{C}$ ,  $AV_{DD} = DV_{DD} = 5\text{ V}$ ,  $R_L = 10\text{ k}\Omega$  connected to  $DV_{DD}/2$ ,  $V_{REF} = \text{GND}$ , and  $C_L = 100\text{ pF}$ , unless otherwise noted.



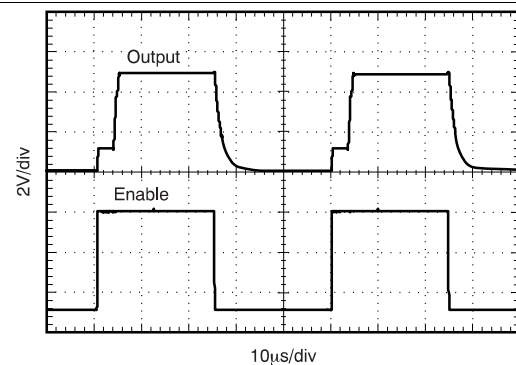
**Figure 51. Power-Up and Power-Down Timing**



**Figure 52. Output Overdrive Performance**



**Figure 53. Output Voltage vs Shutdown Mode**



**Figure 54. PGA116, PGA117 Hardware Shutdown Mode**

## 8 Detailed Description

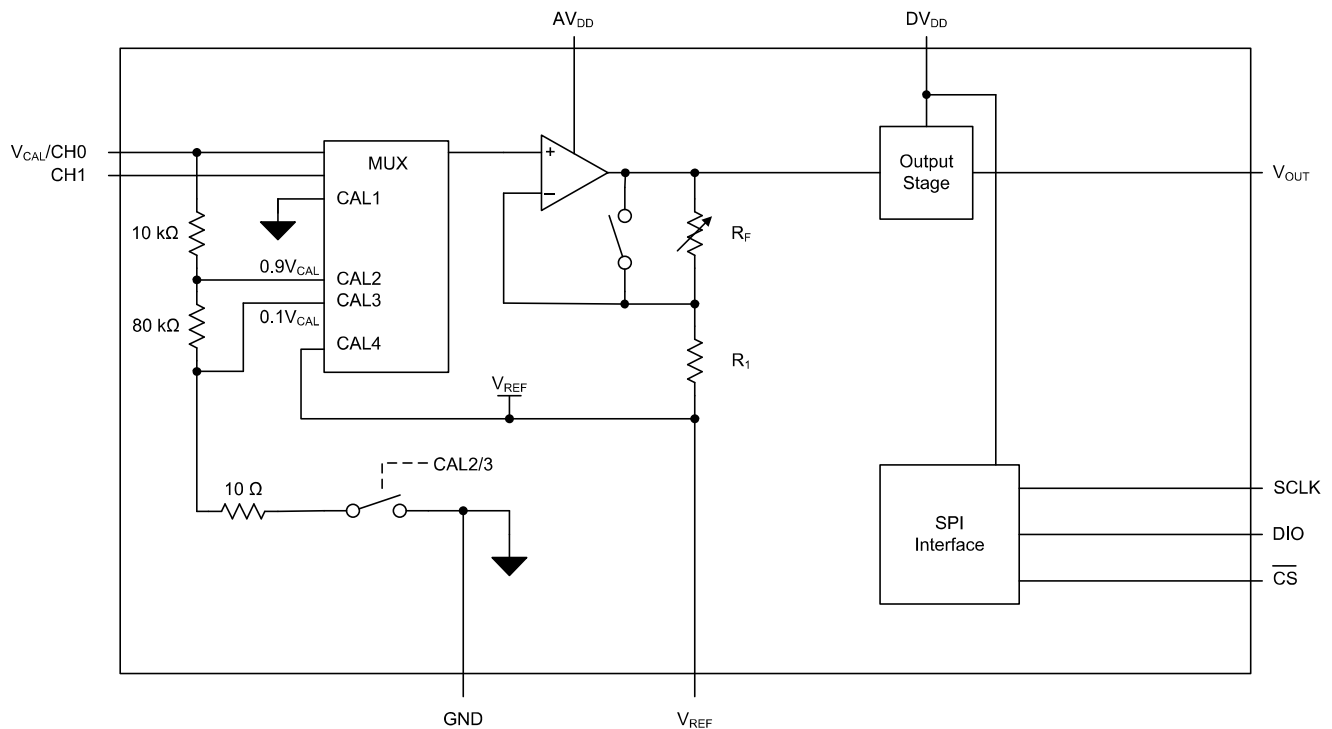
### 8.1 Overview

The PGA112 and PGA113 devices (binary and scope gains) offer two analog inputs, a three-pin SPI interface, and software shutdown in an 10-pin VSSOP package. The PGA116 and PGA117 (binary and scope gains) offer 10 analog inputs, and hardware and software shutdown in a 20-pin TSSOP package.

All versions provide internal calibration channels for system-level calibration. The channels are tied to GND,  $0.9 V_{CAL}$ ,  $0.1 V_{CAL}$ , and  $V_{REF}$ , respectively.  $V_{CAL}$ , an external voltage connected to Channel 0, is used as the system calibration reference. Binary gains are: 1, 2, 4, 8, 16, 32, 64, and 128; scope gains are: 1, 2, 5, 10, 20, 50, 100, and 200.

The PGA uses a SPI interface with daisy-chain capability, a standard serial peripheral interface (SPI). Both SPI Mode 0,0 and Mode 1,1 are supported, as shown in [Figure 56](#) and described in [Table 2](#).

### 8.2 Functional Block Diagram



### 8.3 Feature Description

Featuring low offset, low offset drift and low noise, the PGA11x series provides a flexible analog building block for a variety of applications. The PGA112 and PGA116 offer binary gains of 1, 2, 4, 8, 16, 32, 64, 128 and a 2 channel MUX while the PGA113 and PGA117 offer scope gains of 1, 2, 5, 10, 20, 50, 100, 200 and a 10 channel MUX.

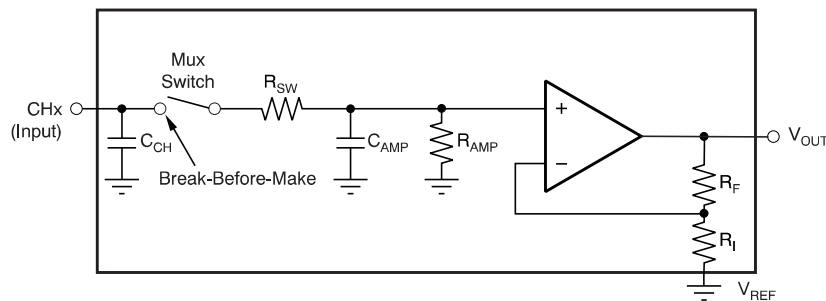
### 8.4 Device Functional Modes

The PGA112 and PGA113 devices have a software shutdown mode, and the PGA116 and PGA117 devices offer both a hardware and software shutdown mode, see [Shutdown and Power-On-Reset \(POR\)](#) for additional information. The PGA uses a standard serial peripheral interface (SPI). Both SPI Mode 0,0 and Mode 1,1 are supported. More information regarding serial communications, including daisy chaining can be found in [Serial Interface Information](#).

## 8.5 Programming

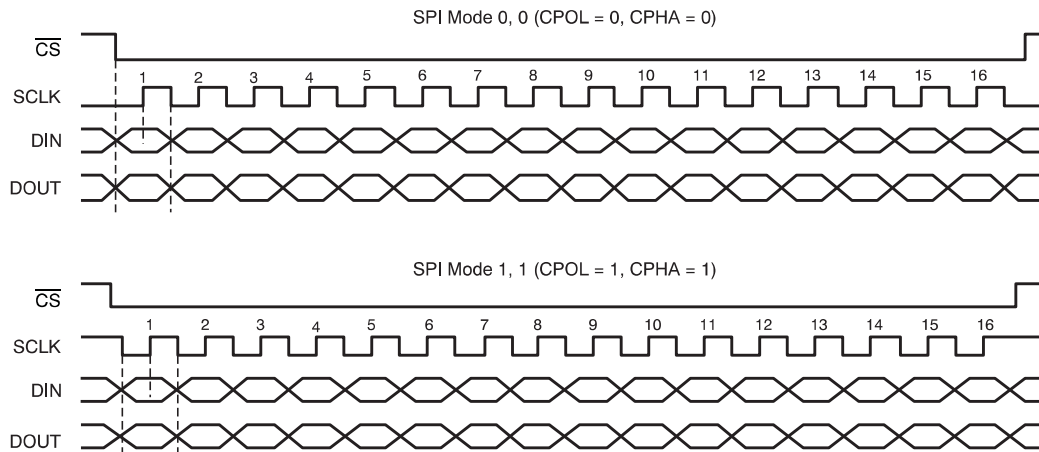
**Table 1. Frequency Response Versus Gain ( $C_L = 100$  pf,  $R_L = 10$  k $\omega$ )**

| BINARY GAIN (V/V) | TYPICAL -3-dB FREQUENCY (MHz) | SLEW RATE-FALL (V/ $\mu$ s) | SLEW RATE-RISE (V/ $\mu$ s) | 0.1% SETTling TIME: 4 V <sub>PP</sub> ( $\mu$ s) | 0.01% SETTling TIME: 4 V <sub>PP</sub> ( $\mu$ s) | SCOPE GAIN (V/V) | TYPICAL -3-dB FREQUENCY (MHz) | SLEW RATE-FALL (V/ $\mu$ s) | SLEW RATE-RISE (V/ $\mu$ s) | 0.1% SETTling TIME: 4 V <sub>PP</sub> ( $\mu$ s) | 0.01% SETTling TIME: 4 V <sub>PP</sub> ( $\mu$ s) |
|-------------------|-------------------------------|-----------------------------|-----------------------------|--------------------------------------------------|---------------------------------------------------|------------------|-------------------------------|-----------------------------|-----------------------------|--------------------------------------------------|---------------------------------------------------|
| 1                 | 10                            | 8                           | 3                           | 2                                                | 2.55                                              | 1                | 10                            | 8                           | 3                           | 2                                                | 2.55                                              |
| 2                 | 3.8                           | 9                           | 6.4                         | 2                                                | 2.6                                               | 2                | 3.8                           | 9                           | 6.4                         | 2                                                | 2.6                                               |
| 4                 | 2                             | 12.8                        | 10.6                        | 2                                                | 2.6                                               | 5                | 1.8                           | 12.8                        | 10.6                        | 2                                                | 2.6                                               |
| 8                 | 1.8                           | 12.8                        | 10.6                        | 2                                                | 2.6                                               | 10               | 1.8                           | 12.8                        | 10.6                        | 2.2                                              | 2.6                                               |
| 16                | 1.6                           | 12.8                        | 12.8                        | 2.3                                              | 2.6                                               | 20               | 1.3                           | 12.8                        | 9.1                         | 2.3                                              | 2.8                                               |
| 32                | 1.8                           | 12.8                        | 13.3                        | 2.3                                              | 3                                                 | 50               | 0.9                           | 9.1                         | 7.1                         | 2.4                                              | 3.8                                               |
| 64                | 0.6                           | 4                           | 3.5                         | 3                                                | 6                                                 | 100              | 0.38                          | 4                           | 3.5                         | 4.4                                              | 7                                                 |
| 128               | 0.35                          | 2.5                         | 2.5                         | 4.8                                              | 8                                                 | 200              | 0.23                          | 2.3                         | 2                           | 6.9                                              | 10                                                |



**Figure 55. Equivalent Input Circuit**

## 8.6 Serial Interface Information



**Figure 56. SPI Mode 0,0 And Mode 1,1**

## Serial Interface Information (continued)

**Table 2. SPI Mode Setting Description**

| MODE | CPOL | CPHA             | CPOL DESCRIPTION | CPHA DESCRIPTION                                                                     |
|------|------|------------------|------------------|--------------------------------------------------------------------------------------|
| 0, 0 | 0    | 0 <sup>(1)</sup> | Clock idles low  | Data are read on the rising edge of clock. Data change on the falling edge of clock. |
| 1, 1 | 1    | 1 <sup>(2)</sup> | Clock idles high | Data are read on the rising edge of clock. Data change on the falling edge of clock. |

(1) CPHA = 0 means sample on first clock edge (rising or falling) after a valid  $\overline{CS}$ .

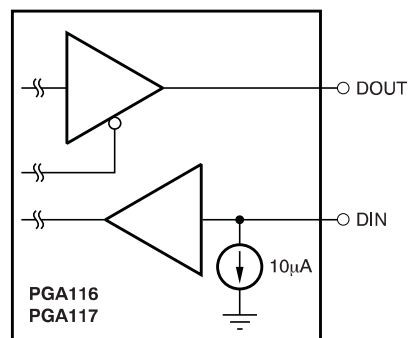
(2) CPHA = 1 means sample on second clock edge (rising or falling) after a valid  $\overline{CS}$ .

### 8.6.1 Serial Digital Interface: SPI Modes

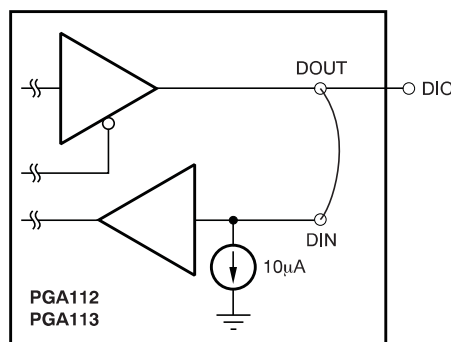
The PGA uses a standard serial peripheral interface (SPI). Both SPI Mode 0,0 and Mode 1,1 are supported, as shown in [Figure 56](#) and described in [Table 2](#).

If there are not even-numbered increments of 16 clocks (that is, 16, 32, 64, and so forth) between  $\overline{CS}$  going low (falling edge) and  $\overline{CS}$  going high (rising edge), the device takes no action. This condition provides reliable serial communication. Furthermore, this condition also provides a way to quickly reset the SPI interface to a known starting condition for data synchronization. Transmitted data are latched internally on the rising edge of  $\overline{CS}$ .

On the PGA116 and PGA117 devices,  $\overline{CS}$ , DIN, and SCLK are Schmitt-triggered CMOS logic inputs. DIN has a weak internal pulldown to support daisy-chain communications on the PGA116 and PGA117 devices. DOUT is a CMOS logic output. When  $\overline{CS}$  is high, the state of DOUT is high-impedance. When  $\overline{CS}$  is low, DOUT is driven as illustrated in [Figure 57](#).


**Figure 57. Digital I/O Structure—PGA116 and PGA117**

On the PGA112 and PGA113 devices, there are digital output and digital input gates both internally connected to the DIO pin. DIN is an input-only gate and DOUT is a digital output that can give a 3-state output. The DIO pin has a weak 10-µA pulldown current source to prevent the pin from floating in systems with a high-impedance SPI DOUT line. When  $\overline{CS}$  is high, the state of the internal DOUT gate is high-impedance. When  $\overline{CS}$  is low, the state of DIO depends on the previous valid SPI communication; either DIO becomes an output to clock out data or it remains an input to receive data. This structure is shown in [Figure 58](#).


**Figure 58. Digital I/O Structure—PGA112 and PGA113**

## 8.6.2 Serial Digital Interface: SPI Daisy-Chain Communications

To reduce the number of I/O port pins used on a microcontroller, the PGA116 and PGA117 support SPI daisy-chain communications with full read and write capability. A two-device daisy-chain configuration is shown in Figure 59, although any number of devices can be daisy-chained. The SPI daisy-chain communication uses a common SCLK and CS line for all devices in the daisy chain, rather than each device requiring a separate CS line. The daisy-chain mode of communication routes data serially through each device in the chain by using its respective DIN and DOUT pins as shown. Special commands are used (see Table 4) to ensure that data are written or read in the proper sequence. There is a special daisy-chain NOP command (No Operation) which, when presented to the desired device in the daisy-chain, causes no changes in that respective device. Detailed timing diagrams for daisy-chain operation are shown in Figure 63 through Figure 65.

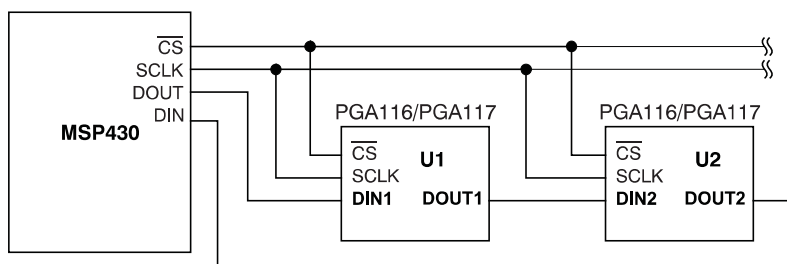


Figure 59. Daisy-Chain Read and Write Configuration

The PGA112 and PGA113 devices can be used as the last device in a daisy-chain as shown in Figure 60 if *write-only* communication is acceptable, because the PGA112 and PGA113 devices have no separate DOUT pin to connect back to the microcontroller DIN pin to read back data in this configuration.

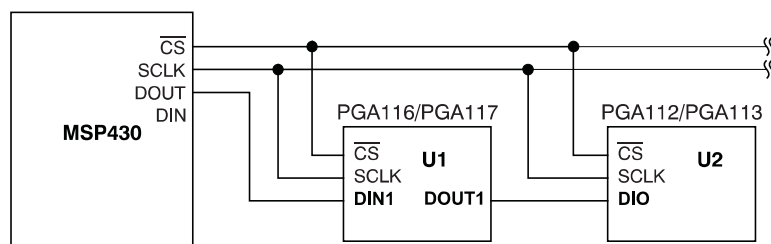


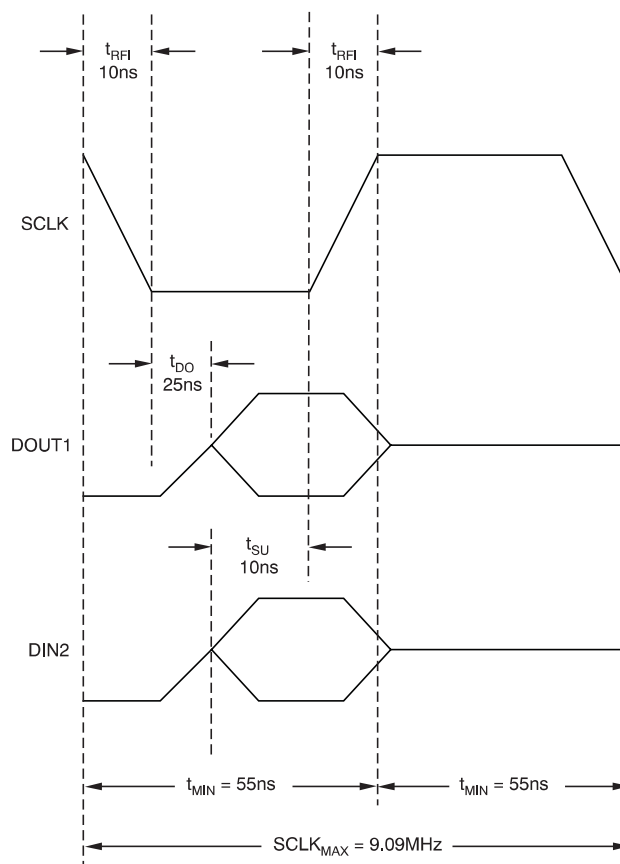
Figure 60. Daisy-Chain Write-Only Configuration

The maximum SCLK frequency that can be used in daisy-chain operation is directly related to SCLK rise and fall times, DIN setup time, and DOUT propagation delay. Any number of two or more devices have the same limitations because it is the timing considerations between adjacent devices that limit the clock speed.

Figure 61 analyzes the maximum SCLK frequency for daisy-chain mode based on the circuit of Figure 59. A clock rise and fall time of 10 ns is assumed to allow for extra bus capacitance that could occur as a result of multiple devices in the daisy-chain.

**PGA112, PGA113, PGA116, PGA117**

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**Figure 61. Daisy-Chain Maximum SCLK Frequency**

### 8.6.3 SPI Serial Interface

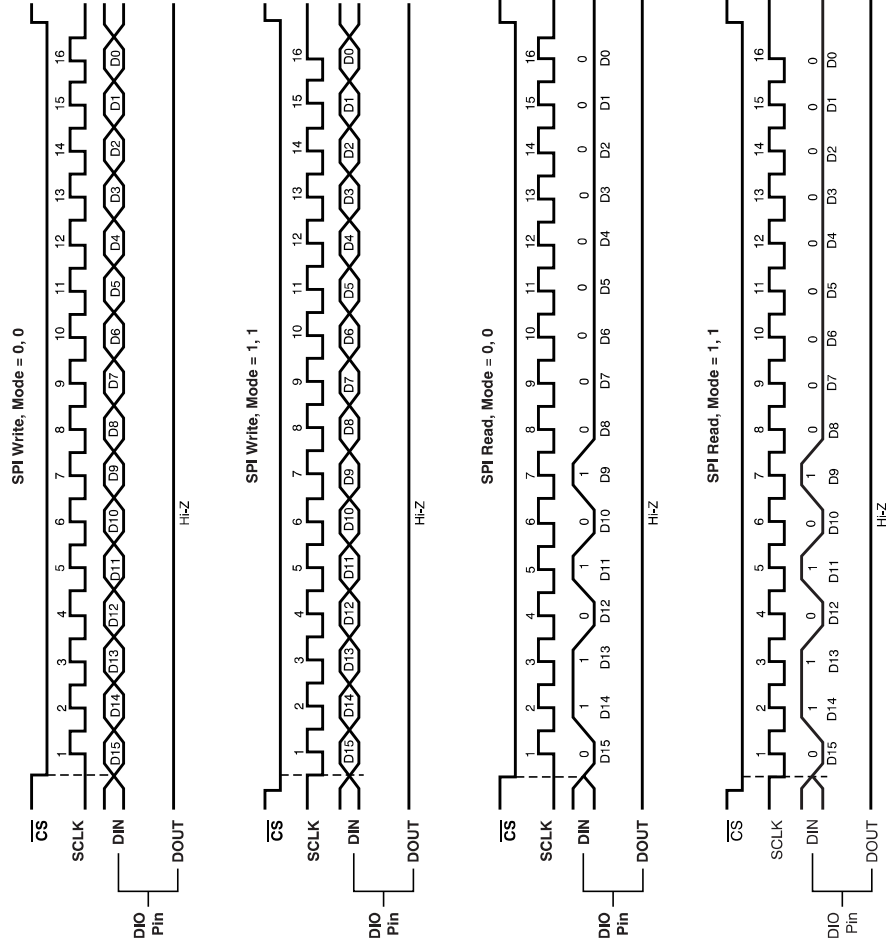
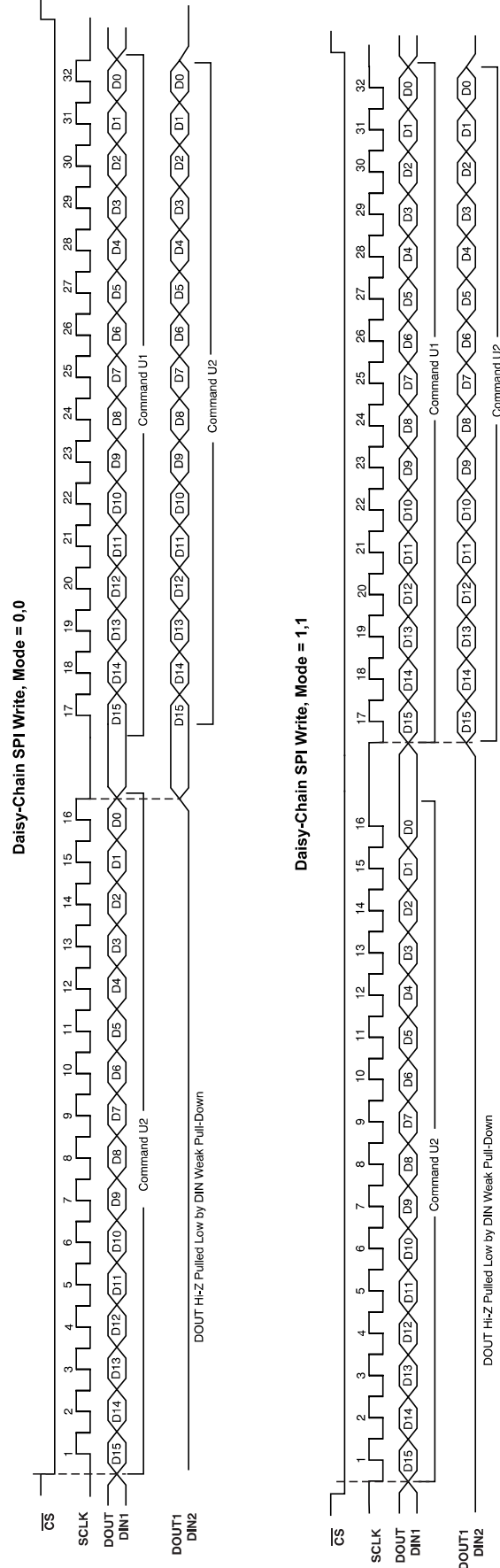
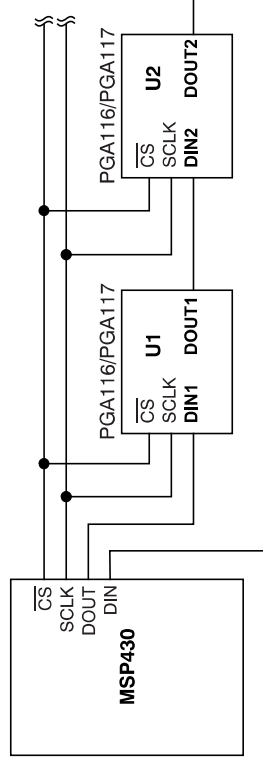
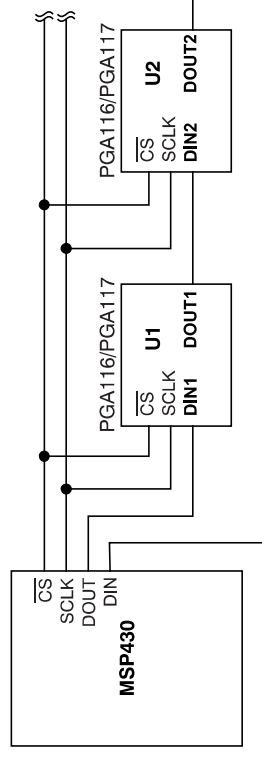


Figure 62. SPI Serial Interface Timing Diagrams



**Figure 63. SPI Daisy-Chain Write Timing Diagrams**



Daisy-Chain SPI Read, Mode = 0,0

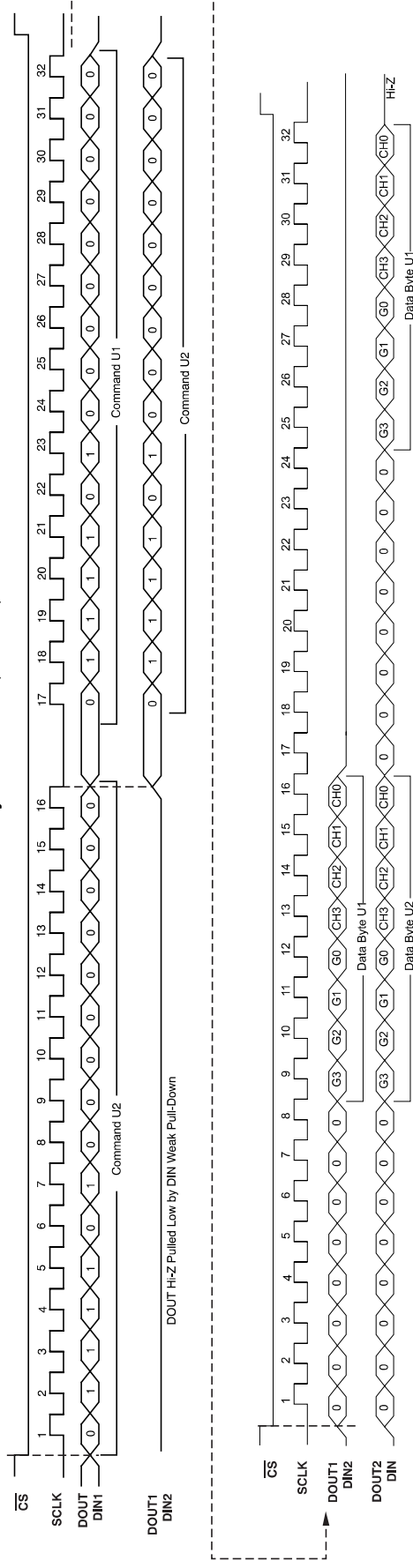
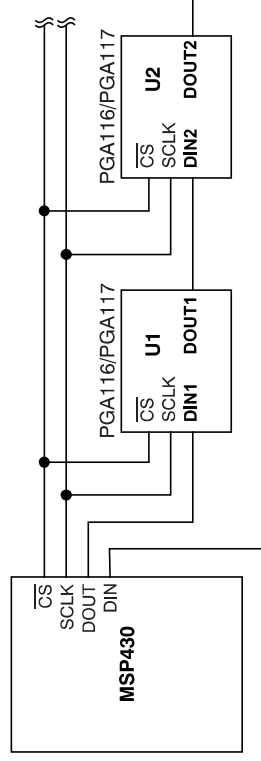


Figure 64. SPI Daisy-Chain Read Timing Diagram (Mode 0,0)



Daisy-Chain SPI Read, Mode = 1,1

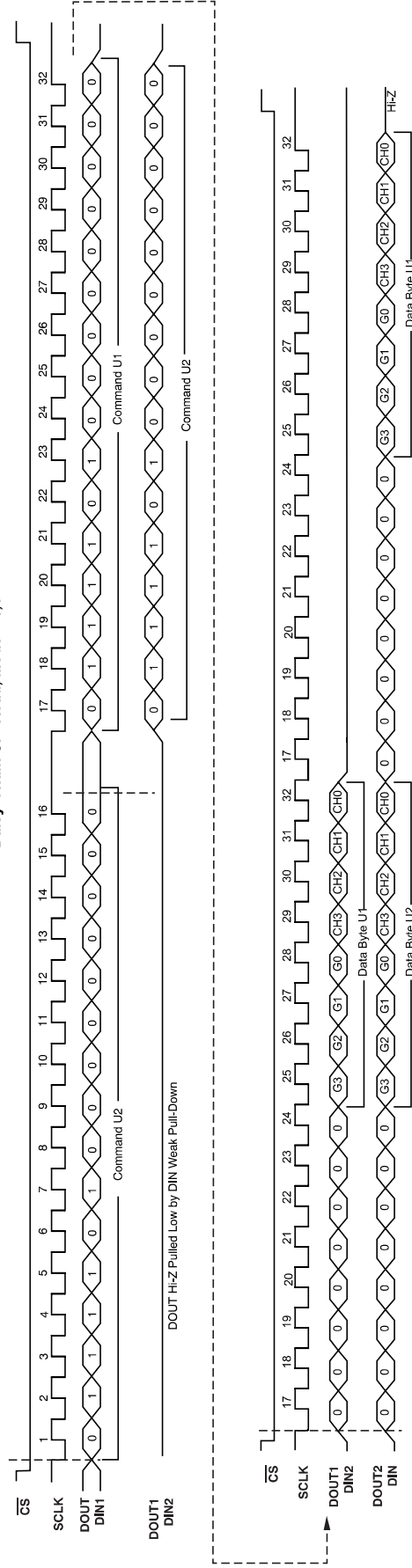


Figure 65. SPI Daisy-Chain Read Timing Diagram (Mode 1,1)

## 8.6.4 SPI Commands

**Table 3. SPI Commands (PGA112 and PGA113)<sup>(1)(2)</sup>**

| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 | D7 | D6 | D5 | D4 | D3  | D2  | D1  | D0  | THREE-WIRE<br>SPI COMMAND |
|-----|-----|-----|-----|-----|-----|----|----|----|----|----|----|-----|-----|-----|-----|---------------------------|
| 0   | 1   | 1   | 0   | 1   | 0   | 1  | 0  | 0  | 0  | 0  | 0  | 0   | 0   | 0   | 0   | READ                      |
| 0   | 0   | 1   | 0   | 1   | 0   | 1  | 0  | G3 | G2 | G1 | G0 | CH3 | CH2 | CH1 | CH0 | WRITE                     |
| 0   | 0   | 0   | 0   | 0   | 0   | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 0   | 0   | 0   | NOP WRITE                 |
| 1   | 1   | 1   | 0   | 0   | 0   | 0  | 1  | 0  | 0  | 0  | 0  | 0   | 0   | 0   | 0   | SDN_DIS<br>WRITE          |
| 1   | 1   | 1   | 0   | 0   | 0   | 0  | 1  | 1  | 1  | 1  | 1  | 0   | 0   | 0   | 1   | SDN_EN WRITE              |

(1) SDN = Shutdown mode. Enter Shutdown mode by issuing an SDN\_EN command. Shutdown mode is cleared (returned to the last valid write configuration) by a SDN\_DIS command or by any valid Write command.

(2) POR (Power-on-Reset) value of internal Gain/Channel Select Register is all 0s; this value sets Gain = 1, and Channel = V<sub>CAL</sub>/CH0.

**Table 4. SPI Daisy-Chain Commands<sup>(1)(2)</sup>**

| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 | D7 | D6 | D5 | D4 | D3  | D2  | D1  | D0  | DAISY-CHAIN<br>COMMAND |
|-----|-----|-----|-----|-----|-----|----|----|----|----|----|----|-----|-----|-----|-----|------------------------|
| 0   | 0   | 0   | 1   | 0   | 0   | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 0   | 0   | 0   | NOP                    |
| 1   | 1   | 1   | 1   | 0   | 0   | 0  | 1  | 0  | 0  | 0  | 0  | 0   | 0   | 0   | 0   | SDN_DIS                |
| 1   | 1   | 1   | 1   | 0   | 0   | 0  | 1  | 1  | 1  | 1  | 1  | 0   | 0   | 0   | 1   | SDN_EN                 |
| 0   | 1   | 1   | 1   | 1   | 0   | 1  | 0  | 0  | 0  | 0  | 0  | 0   | 0   | 0   | 0   | READ                   |
| 0   | 0   | 1   | 1   | 1   | 0   | 1  | 0  | G3 | G2 | G1 | G0 | CH3 | CH2 | CH1 | CH0 | WRITE                  |

(1) SDN = Shutdown Mode. Shutdown Mode is entered by an SDN\_EN command. Shutdown Mode is cleared (returned to the last valid write configuration) by a SDN\_DIS command or by any valid Write command.

(2) POR (Power-on-Reset) value of internal Gain/Channel Register is all 0s; this value sets Gain = 1, V<sub>CAL</sub>/CH0 selected.

**Table 5. Gain Selection Bits (PGA112 and PGA113)**

| G3 | G2 | G1 | G0 | BINARY GAIN | SCOPE GAIN |
|----|----|----|----|-------------|------------|
| 0  | 0  | 0  | 0  | 1           | 1          |
| 0  | 0  | 0  | 1  | 2           | 2          |
| 0  | 0  | 1  | 0  | 4           | 5          |
| 0  | 0  | 1  | 1  | 8           | 10         |
| 0  | 1  | 0  | 0  | 16          | 20         |
| 0  | 1  | 0  | 1  | 32          | 50         |
| 0  | 1  | 1  | 0  | 64          | 100        |
| 0  | 1  | 1  | 1  | 128         | 200        |

**PGA112, PGA113, PGA116, PGA117**

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**Table 6. MUX Channel Selection Bits**

| CH3 | CH2 | CH1 | CH0 | PGA112, PGA113      | PGA116, PGA117      |
|-----|-----|-----|-----|---------------------|---------------------|
| 0   | 0   | 0   | 0   | VCAL/CH0            | VCAL/CH0            |
| 0   | 0   | 0   | 1   | CH1                 | CH1                 |
| 0   | 0   | 1   | 0   | X <sup>(1)</sup>    | CH2                 |
| 0   | 0   | 1   | 1   | X                   | CH3                 |
| 0   | 1   | 0   | 0   | X                   | CH4                 |
| 0   | 1   | 0   | 1   | X                   | CH5                 |
| 0   | 1   | 1   | 0   | X                   | CH6                 |
| 0   | 1   | 1   | 1   | X                   | CH7                 |
| 1   | 0   | 0   | 0   | X                   | CH8                 |
| 1   | 0   | 0   | 1   | X                   | CH9                 |
| 1   | 0   | 1   | 0   | X                   | X <sup>(1)</sup>    |
| 1   | 0   | 1   | 1   | Factory Reserved    | Factory Reserved    |
| 1   | 1   | 0   | 0   | CAL1 <sup>(2)</sup> | CAL1 <sup>(2)</sup> |
| 1   | 1   | 0   | 1   | CAL2 <sup>(3)</sup> | CAL2 <sup>(3)</sup> |
| 1   | 1   | 1   | 0   | CAL3 <sup>(4)</sup> | CAL3 <sup>(4)</sup> |
| 1   | 1   | 1   | 1   | CAL4 <sup>(5)</sup> | CAL4 <sup>(5)</sup> |

- (1) X = channel is not used.  
 (2) CAL1: connects to GND.  
 (3) CAL2: connects to 0.9 V<sub>CAL</sub>.  
 (4) CAL3: connects to 0.1 V<sub>CAL</sub>.  
 (5) CAL4: connects to V<sub>REF</sub>.

## 9 Applications and Implementation

### NOTE

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

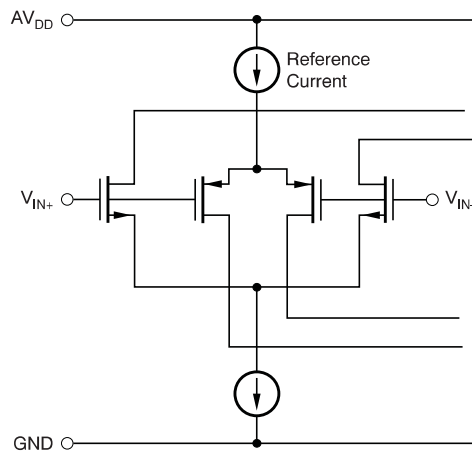
### 9.1 Application Information

The PGA11x family of devices are single-ended input, single-supply, programmable gain amplifiers (PGAs) with an input multiplexer. Multiplexer channel selection and gain selection are done through a standard SPI interface. The PGA112 and PGA113 have a two-channel input MUX and the PGA116 and PGA117 devices have a 10-channel input MUX. The PGA112 and PGA116 devices provide binary gain selections (1, 2, 4, 8, 16, 32, 64, 128) and the PGA113 and PGA117 devices provide scope gain selections (1, 2, 5, 10, 20, 50, 100, 200). All models use a split-supply architecture with an analog supply,  $AV_{DD}$ , and a digital supply,  $DV_{DD}$ . This split-supply architecture allows for ease of interface to analog-to-digital converters (ADCs) and microcontrollers in mixed-supply voltage systems, such as where the analog supply is 5 V and the digital supply is 3 V. Four internal calibration channels are provided for system-level calibration. The channels are tied to GND,  $0.9 V_{CAL}$ ,  $0.1 V_{CAL}$ , and  $V_{REF}$ , respectively.  $V_{CAL}$ , an external voltage connected to  $V_{CAL}/CH0$ , acts as the system calibration reference. If  $V_{CAL}$  is the system ADC reference, then gain and offset calibration on the ADC are easily accomplished through the PGA11x devices using only one MUX input. If calibration is not used, then  $V_{CAL}/CH0$  can be used as a standard MUX input. All four versions provide a  $V_{REF}$  pin that can be tied to ground or, for ease of scaling, to midsupply in single-supply systems where midsupply is used as a virtual ground. The PGA112 and PGA113 devices offer a software-controlled shutdown feature for low standby power. The PGA116 and PGA117 devices offer both hardware- and software-controlled shutdown for low standby power. The PGA112 and PGA113 devices have a 3-wire SPI digital interface; the PGA116 and PGA117 devices have a four-wire SPI digital interface. The PGA116 and PGA117 devices also have daisy-chain capability.

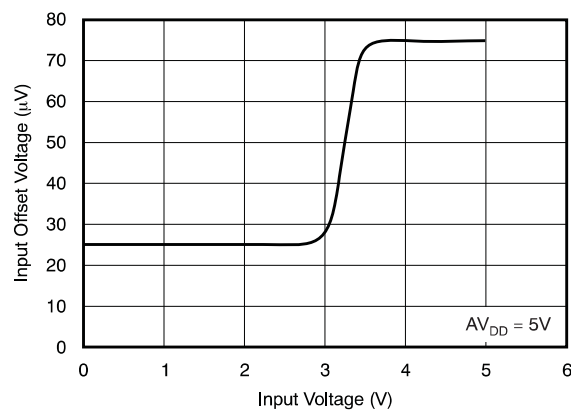
#### 9.1.1 Op Amp: Input Stage

The PGA op amp is a rail-to-rail input and output (RRIO) single-supply op amp. The input topology uses two separate input stages in parallel to achieve rail-to-rail input. As [Figure 66](#) shows, there is a PMOS transistor on each input for operation down to ground; there is also an NMOS transistor on each input in parallel for operation to the positive supply rail. When the common-mode input voltage (that is, the single-ended input, because this PGA is configured internally for noninverting gain) crosses a level that is typically about 1.5 V less than the positive supply, there is a transition between the NMOS and PMOS transistors. The result of this transition appears as a small input offset voltage transition that is reflected to the output by the selected PGA gain. This transition may be either increasing or decreasing, and differs from part to part as described in [Figure 67](#) and [Figure 68](#). These figures illustrate possible differences in input offset voltage between two different devices when used with  $AV_{DD} = 5$  V. Because the exact transition region varies from device to device, the [Electrical Characteristics:  \$V\_S = AV\_{DD} = DV\_{DD} = 5\$  V](#) table specifies an input offset voltage above and below this input transition region.

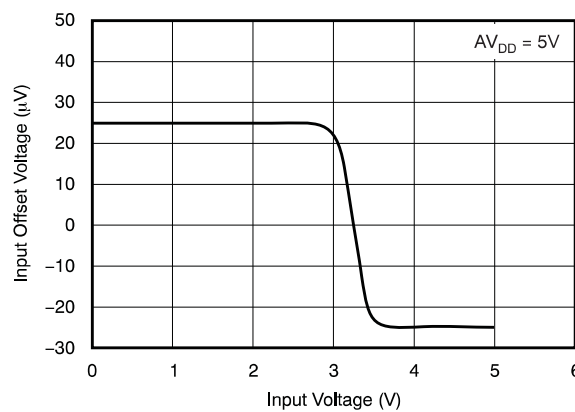
## Application Information (continued)



**Figure 66. PGA Rail-To-Rail Input Stage**



**Figure 67.  $V_{OS}$  Versus Input Voltage—Case 1**



**Figure 68.  $V_{OS}$  Versus Input Voltage—Case 2**

### 9.1.2 Op Amp: General Gain Equations

Figure 69 shows the basic configuration for using the PGA as a gain block.  $V_{OUT} / V_{IN}$  is the selected noninverting gain, depending on the model selected, for either binary or scope gains.

## Application Information (continued)

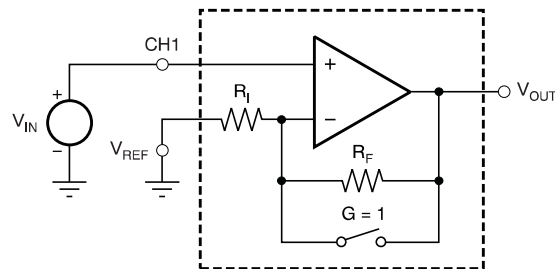


Figure 69. PGA Used as a Gain Block

$$V_{OUT} = G \times V_{IN}$$

where

- $G = 1, 2, 4, 8, 16, 32, 64,$  and  $128$  (binary gains)
- $G = 1, 2, 5, 10, 20, 50, 100,$  and  $200$  (scope gains)

Figure 70 shows the PGA configuration and gain equations for  $V_{REF} = AV_{DD}/2$ .  $V_{OUT0}$  is  $V_{OUT}$  when CH0 is selected and  $V_{OUT1}$  is  $V_{OUT}$  when CH1 is selected. Notice the  $V_{REF}$  pin has no effect for  $G = 1$  because the internal feedback resistor,  $R_F$ , is shorted out. This configuration allows for positive and negative voltage excursions around a midsupply virtual ground.

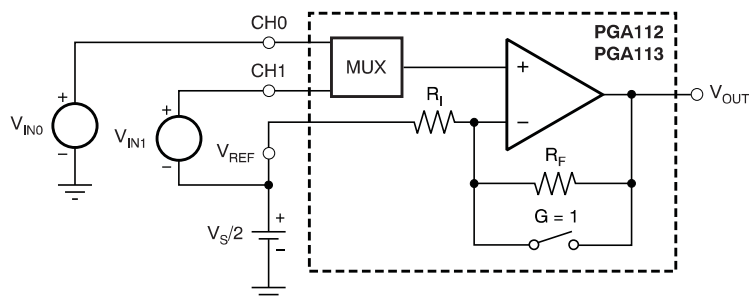


Figure 70. PGA112 and PGA113 Configuration for Positive and Negative Excursions Around Midsupply Virtual Ground

$$V_{OUT0} = G \times V_{IN0} - AV_{DD}/2 \times (G - 1) \quad (2)$$

When:  $G = 1$

Then:  $V_{OUT0} = G \times V_{IN0}$

$$V_{OUT1} = G \times (V_{IN1} + AV_{DD}/2) - AV_{DD}/2 \times (G - 1)$$

$$V_{OUT1} = G \times V_{IN1} + AV_{DD}/2, \text{ where: } -AV_{DD}/2 < G \times V_{IN1} < +AV_{DD}/2$$

where

- $G = 1, 2, 4, 8, 16, 32, 64,$  and  $128$  (binary gains)
- $G = 1, 2, 5, 10, 20, 50, 100,$  and  $200$  (scope gains)

Table 7 details the internal typical values for the op amp internal feedback resistor ( $R_F$ ) and op amp internal input resistor ( $R_I$ ) for both binary and scope gains.

## Application Information (continued)

**Table 7. Typical  $R_F$  and  $R_I$  Versus Gain**

| BINARY GAIN (V/V) | $R_F$ ( $\Omega$ ) | $R_I$ ( $\Omega$ ) | SCOPE GAIN (V/V) | $R_F$ ( $\Omega$ ) | $R_I$ ( $\Omega$ ) |
|-------------------|--------------------|--------------------|------------------|--------------------|--------------------|
| 1                 | 0                  | 3.25 k             | 1                | 0                  | 3.25 k             |
| 2                 | 3.25 k             | 3.25 k             | 2                | 3.25 k             | 3.25 k             |
| 4                 | 9.75 k             | 3.25 k             | 5                | 13 k               | 3.25 k             |
| 8                 | 22.75 k            | 3.25 k             | 10               | 29.25 k            | 3.25 k             |
| 16                | 48.75 k            | 3.25 k             | 20               | 61.75 k            | 3.25 k             |
| 32                | 100.75 k           | 3.25 k             | 50               | 159.25 k           | 3.25 k             |
| 64                | 204.75 k           | 3.25 k             | 100              | 321.75 k           | 3.25 k             |
| 128               | 412.75 k           | 3.25 k             | 200              | 646.75 k           | 3.25 k             |

### 9.1.3 Op Amp: Frequency Response Versus Gain

[Table 8](#) documents how small-signal bandwidth and slew rate change correspond to changes in PGA gain.

Full power bandwidth (that is, the highest frequency that a sine wave can pass through the PGA for a given gain) is related to slew rate by [Equation 4](#):

$$SR \text{ (V/}\mu\text{s)} = 2\pi f \times V_{OP} (1 \times 10^{-6})$$

where

- SR = Slew rate in V/ $\mu$ s
- f = Frequency in Hz
- $V_{OP}$  = Output peak voltage in volts

(4)

#### 9.1.3.1 Example:

For  $G = 8$ , then  $SR = 10.6 \text{ V/}\mu\text{s}$  (slew rate rise is minimum slew rate).

For a 5-V system, choose  $0.1 \text{ V} < V_{OUT} < 4.9 \text{ V}$  or  $V_{OUTPP} = 4.8 \text{ V}$  or  $V_{OUTP} = 2.4 \text{ V}$ .

$$SR \text{ (V/}\mu\text{s)} = 2\pi f \times V_{OP} (1 \times 10^{-6}).$$

$$10.6 = 2\pi f (2.4) (1 \times 10^{-6}) \rightarrow f = 702.9 \text{ kHz}$$

This example shows that a  $G = 8$  configuration can produce a 4.8- $V_{PP}$  sine wave with frequency up to 702.9 kHz. This computation only shows the theoretical upper limit of frequency for this example, but does not indicate the distortion of the sine wave. The acceptable distortion depends on the specific application. As a general guideline, maintain two to three times the calculated slew rate to minimize distortion on the sine wave. For this example, the application should only use  $G = 8$ , 4.8  $V_{PP}$ , up to a frequency range of 234 kHz to 351 kHz, depending upon the acceptable distortion. For a given gain and slew rate requirement, check for adequate small-signal bandwidth (typical –3-dB frequency) to assure that the frequency of the signal can be passed without attenuation.

### 9.1.4 Analog MUX

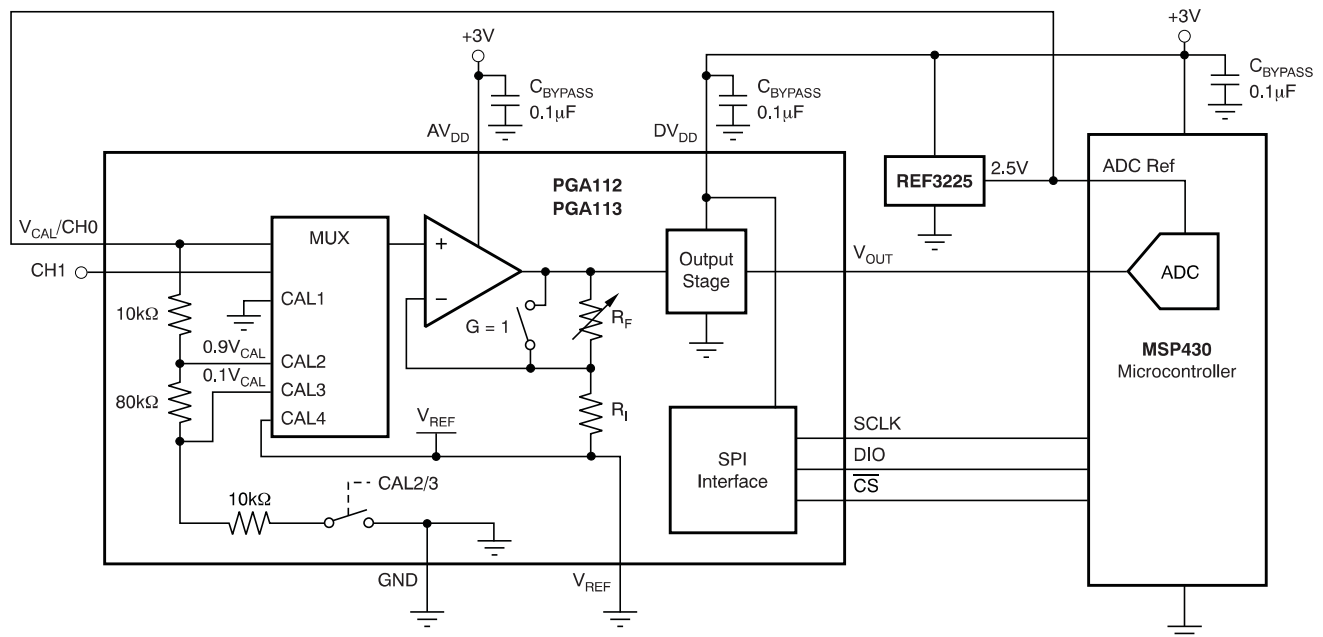
The analog input MUX provides two input channels for the PGA112 and PGA113 devices and 10 input channels for the PGA116 and PGA117 devices. The MUX switches are designed to be break-before-make and thereby eliminate any concerns about shorting the two input signal sources together.

Four internal MUX CAL channels are included in the analog MUX for ease of system calibration. These CAL channels allow ADC gain and offset errors to be calibrated out. This calibration does not remove the offset and gain errors of the PGA for gains greater than 1, but most systems should see a significant increase in the ADC accuracy. In addition, these CAL channels can be used by the ADC to read the minimum and maximum possible voltages from the PGA. With these minimum and maximum levels known, the system architecture can be designed to indicate an out-of-range condition on the measured analog input signals if these levels are ever measured.

To use the CAL channels,  $V_{\text{CAL}}/\text{CH0}$  must be permanently connected to the system ADC reference. There is a typical 100-k $\Omega$  load from  $V_{\text{CAL}}/\text{CH0}$  to ground. Table 9 illustrates how to use the CAL channels with  $V_{\text{REF}} = \text{ground}$ . Table 10 describes how to use the CAL channels with  $V_{\text{REF}} = \text{AV}_{\text{DD}}/2$ . The  $V_{\text{REF}}$  pin must be connected to a source that is low-impedance for both DC and AC to maintain gain and nonlinearity accuracy. Worst-case current demand on the  $V_{\text{REF}}$  pin occurs when  $G = 1$  because there is a 3.25-k $\Omega$  resistor between  $V_{\text{OUT}}$  and  $V_{\text{REF}}$ . For a 5-V system with  $\text{AV}_{\text{DD}}/2 = 2.5 \text{ V}$ , the  $V_{\text{REF}}$  pin buffer must source and sink  $2.5 \text{ V}/3.25 \text{ k}\Omega = 0.7 \text{ mA}$  minimum for a  $V_{\text{OUT}}$  that can swing from ground to 5 V.

**Table 8. Frequency Response versus Gain ( $C_L = 100 \text{ pf}$ ,  $R_L = 10 \text{ k}\Omega$ )**

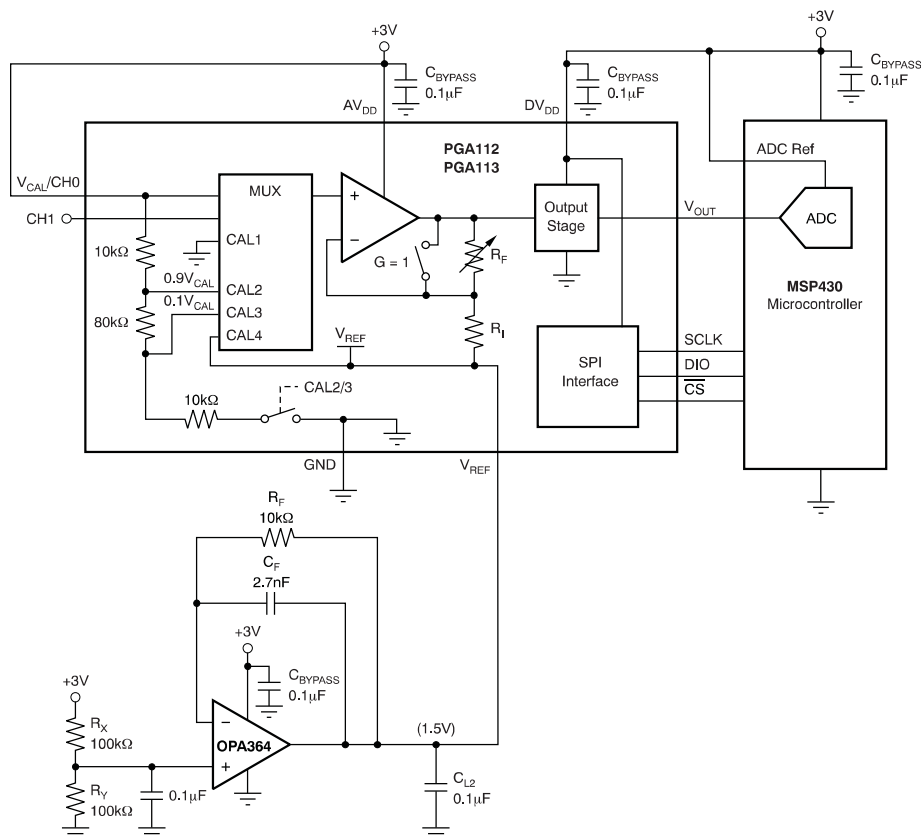
| BINARY GAIN (V/V) | TYPICAL –3dB FREQUENCY (MHz) | SLEW RATE-FALL (V/ $\mu\text{s}$ ) | SLEW RATE-RISE (V/ $\mu\text{s}$ ) | 0.1% SETTling TIME: 4V <sub>PP</sub> ( $\mu\text{s}$ ) | 0.01% SETTling TIME: 4V <sub>PP</sub> ( $\mu\text{s}$ ) | SCOPE GAIN (V/V) | TYPICAL –3dB FREQUENCY (MHz) | SLEW RATE-FALL (V/ $\mu\text{s}$ ) | SLEW RATE-RISE (V/ $\mu\text{s}$ ) | 0.1% SETTling TIME: 4V <sub>PP</sub> ( $\mu\text{s}$ ) | 0.01% SETTling TIME: 4V <sub>PP</sub> ( $\mu\text{s}$ ) |
|-------------------|------------------------------|------------------------------------|------------------------------------|--------------------------------------------------------|---------------------------------------------------------|------------------|------------------------------|------------------------------------|------------------------------------|--------------------------------------------------------|---------------------------------------------------------|
| 1                 | 10                           | 8                                  | 3                                  | 2                                                      | 2.55                                                    | 1                | 10                           | 8                                  | 3                                  | 2                                                      | 2.55                                                    |
| 2                 | 3.8                          | 9                                  | 6.4                                | 2                                                      | 2.6                                                     | 2                | 3.8                          | 9                                  | 6.4                                | 2                                                      | 2.6                                                     |
| 4                 | 2                            | 12.8                               | 10.6                               | 2                                                      | 2.6                                                     | 5                | 1.8                          | 12.8                               | 10.6                               | 2                                                      | 2.6                                                     |
| 8                 | 1.8                          | 12.8                               | 10.6                               | 2                                                      | 2.6                                                     | 10               | 1.8                          | 12.8                               | 10.6                               | 2.2                                                    | 2.6                                                     |
| 16                | 1.6                          | 12.8                               | 12.8                               | 2.3                                                    | 2.6                                                     | 20               | 1.3                          | 12.8                               | 9.1                                | 2.3                                                    | 2.8                                                     |
| 32                | 1.8                          | 12.8                               | 13.3                               | 2.3                                                    | 3                                                       | 50               | 0.9                          | 9.1                                | 7.1                                | 2.4                                                    | 3.8                                                     |
| 64                | 0.6                          | 4                                  | 3.5                                | 3                                                      | 6                                                       | 100              | 0.38                         | 4                                  | 3.5                                | 4.4                                                    | 7                                                       |
| 128               | 0.35                         | 2.5                                | 2.5                                | 4.8                                                    | 8                                                       | 200              | 0.23                         | 2.3                                | 2                                  | 6.9                                                    | 10                                                      |



**Figure 71. Using CAL Channels With  $V_{\text{REF}} = \text{Ground}$**

**Table 9. Using the MUX CAL Channels With  $V_{REF} = GND$**   
**( $AV_{DD} = 3\text{ V}$ ,  $DV_{DD} = 3\text{ V}$ ,  $ADC\text{ Ref} = 2.5\text{ V}$ , and  $V_{REF} = GND$ )**

| FUNCTION           | MUX SELECT | GAIN SELECT | MUX INPUT                  | OP AMP (+In) | OP AMP ( $V_{OUT}$ ) | DESCRIPTION                                                                                                                                                                     |
|--------------------|------------|-------------|----------------------------|--------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Minimum Signal     | CAL1       | 1           | GND                        | GND          | 50 mV                | Minimum signal level that the MUX, op amp, and ADC can read. Op amp $V_{OUT}$ is limited by negative saturation.                                                                |
| Gain Calibration   | CAL2       | 1           | $0.9 \times (V_{CAL}/CH0)$ | 2.25 V       | 2.25 V               | 90% ADC Ref for system full-scale or gain calibration of the ADC.                                                                                                               |
| Maximum Signal     | CAL2       | 2           | $0.9 \times (V_{CAL}/CH0)$ | 2.25 V       | 2.95 V               | Maximum signal level that the MUX, op amp, and ADC can read. Op amp $V_{OUT}$ is limited by positive saturation. System is limited by ADC max input of 2.5 V (ADC Ref = 2.5 V). |
| Offset Calibration | CAL3       | 1           | $0.1 \times (V_{CAL}/CH0)$ | 0.25 V       | 0.25 V               | 10% ADC Ref for system offset calibration of the ADC.                                                                                                                           |
| Minimum Signal     | CAL4       | 1           | $V_{REF}$                  | GND          | 50 mV                | Minimum signal level that the MUX, op amp, and ADC can read. Op amp $V_{OUT}$ is limited by negative saturation.                                                                |



**Figure 72. Using CAL Channels With  $V_{REF} = AV_{DD}/2$**

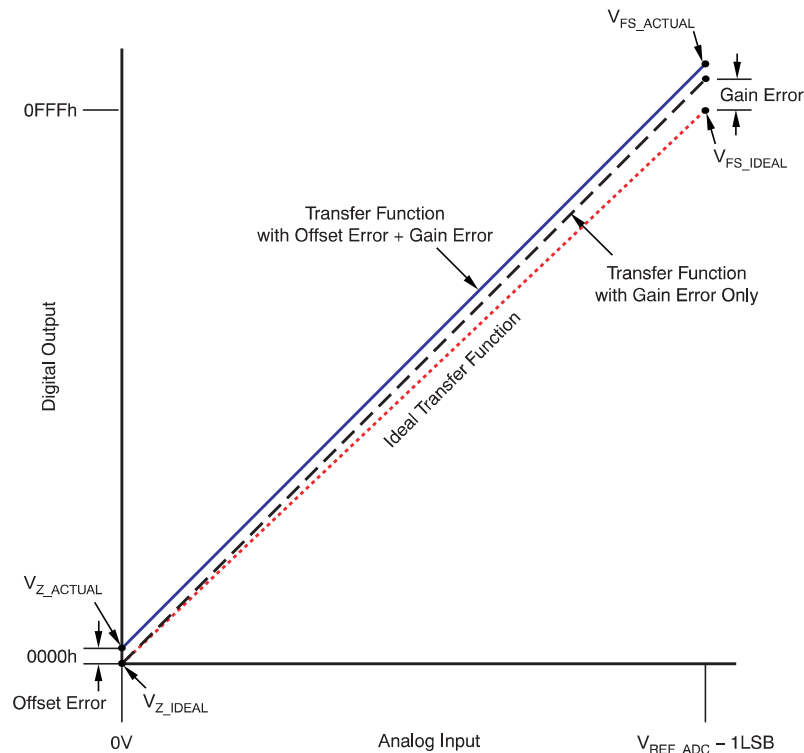
**Table 10. Using the MUX CAL Channels With  $V_{REF} = AV_{DD}/2$**

**Table 10. Using the MUX CAL Channels With  $V_{REF} = AV_{DD}/2$   
( $AV_{DD} = 3\text{ V}$ ,  $DV_{DD} = 3\text{ V}$ , ADC Ref = 3 V, and  $V_{REF} = 1.5\text{ V}$ ) (continued)  
( $AV_{DD} = 3\text{ V}$ ,  $DV_{DD} = 3\text{ V}$ , ADC Ref = 3 V, and  $V_{REF} = 1.5\text{ V}$ )**

| FUNCTION               | MUX SELECT | GAIN SELECT | MUX INPUT                  | OP AMP (+In) | OP AMP (V <sub>OUT</sub> ) | DESCRIPTION                                                                                                             |
|------------------------|------------|-------------|----------------------------|--------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Minimum Signal         | CAL1       | 1           | GND                        | GND          | 50 mV                      | Minimum signal level that the MUX, op amp, and ADC can read. Op amp V <sub>OUT</sub> is limited by negative saturation. |
| Gain Calibration       | CAL2       | 1           | $0.9 \times (V_{CAL}/CH0)$ | 2.7 V        | 2.7 V                      | 90% ADC Ref for system full-scale or gain calibration of the ADC.                                                       |
| Maximum Signal         | CAL2       | 4 or 5      | $0.9 \times (V_{CAL}/CH0)$ | 2.25 V       | 2.95 V                     | Maximum signal level that the MUX, op amp, and ADC can read. Op amp V <sub>OUT</sub> is limited by positive saturation. |
| Offset Calibration     | CAL3       | 1           | $0.1 \times (V_{CAL}/CH0)$ | 0.3 V        | 0.3 V                      | 10% ADC Ref for system offset calibration of the ADC.                                                                   |
| V <sub>REF</sub> Check | CAL4       | 1           | V <sub>REF</sub>           | 1.5 V        | 1.5 V                      | Midsupply voltage used as V <sub>REF</sub> .                                                                            |

### 9.1.5 System Calibration Using The PGA

Analog-to-digital converters (ADCs) contain two major errors that can be easily removed by calibration at a system level. These errors are gain error and offset error, as shown in Figure 73. Figure 73 shows a typical transfer function for a 12-bit ADC. The analog input is on the x-axis with a range from 0 V to ( $V_{REF\_ADC} - 1\text{LSB}$ ), where  $V_{REF\_ADC}$  is the ADC reference voltage. The y-axis is the hexadecimal equivalent of the digital codes that result from ADC conversions. The dotted red line represents an ideal transfer function with 0000h representing 0 V analog input and 0FFFh representing an analog input of ( $V_{REF\_ADC} - 1\text{LSB}$ ). The solid blue line illustrates the offset error. Although the solid blue line includes both offset error and gain error, at an analog input of 0 V the offset error voltage,  $V_{Z\_ACTUAL}$ , can be measured. The dashed black line represents the transfer function with gain error. The dashed black line is equivalent to the solid blue line without the offset error, and can be measured and computed using  $V_{Z\_ACTUAL}$  and  $V_{Z\_IDEAL}$ . The difference between the dashed black line and the dotted red line is the gain error. Gain and offset error can be computed by taking zero input and full-scale input readings. Using these error calculations, compute a calibrated ADC reading to remove the ADC gain and offset error.



**Figure 73. ADC Offset and Gain Error**

In practice, the zero input (0 V) or full-scale input ( $V_{REF\_ADC} - 1LSB$ ) of ADCs cannot always be measured because of internal offset error and gain error. However, if measurements are made very close to the full-scale input and the zero input, both zero and full-scale can be calibrated very accurately with the assumption of linearity from the calibration points to the desired end points of the ADC ideal transfer function. For the zero calibration, choose  $10\%V_{REF\_ADC}$ ; this value should be above the internal offset error and sufficiently out of the noise floor range of the ADC. For the gain calibration, choose  $90\%V_{REF\_ADC}$ ; this value should be less than the internal gain error and sufficiently below the tolerance of  $V_{REF}$ . These key points can be summarized in this way:

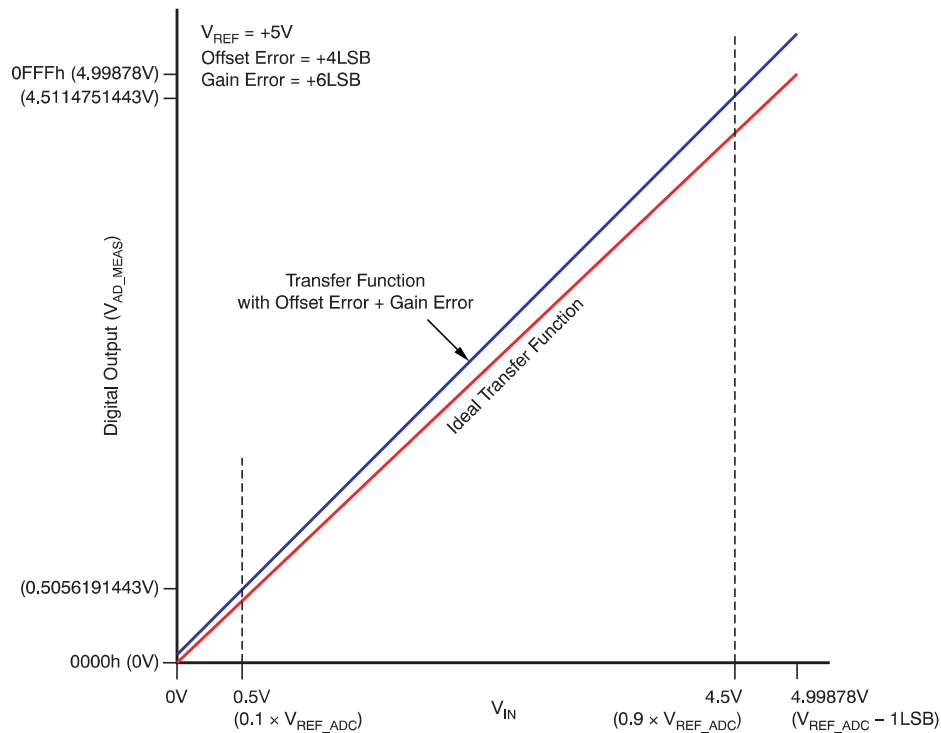
For zero calibration:

- The ADC cannot read the ideal zero because of offset error
- Must be far enough above ground to be above noise floor and ADC offset error
- Therefore, choose  $10\%V_{REF\_ADC}$  for zero calibration

For gain calibration:

- The ADC cannot read the ideal full-scale because of gain error
- Must be far enough below full-scale to be below the  $V_{REF}$  tolerance and ADC gain error
- Therefore, choose  $90\%V_{REF\_ADC}$  for gain calibration

The 12-bit ADC example in [Figure 74](#) illustrates the technique for calibrating an ADC using a  $10\%V_{REF\_ADC}$  and  $90\%V_{REF\_ADC}$  reading where  $V_{REF\_ADC}$  is the ADC reference voltage. The  $10\%V_{REF}$  reading also contains a gain error because it is not a  $V_{IN} = 0$  calibration point. First, use the  $90\%V_{REF}$  and  $10\%V_{REF}$  points to compute the measured gain error. The measured gain error is then used to remove the gain error from the  $10\%V_{REF}$  reading, giving a measured  $10\%V_{REF}$  number. The measured  $10\%V_{REF}$  number is used to compute the measured offset error.



**Figure 74. 12-Bit Example of ADC Calibration for Gain and Offset Error**

The gain error and offset error in ADC readings can be calibrated by using  $10\%V_{REF\_ADC}$  and  $90\%V_{REF\_ADC}$  calibration points. Because the calibration is ratiometric to  $V_{REF\_ADC}$ , the exact value of  $V_{REF\_ADC}$  does not need to be known in the end application.

Follow these steps to compute a calibrated ADC reading:

1. Take the ADC reading at  $V_{IN} = 90\% \times V_{REF}$  and  $V_{IN} = 10\% \times V_{REF}$ . The ADC readings for  $10\%V_{REF}$  and  $90\%V_{REF}$  are taken.

$$V_{REF90} = 0.9(V_{REF\_ADC}) \quad (5)$$

$$V_{REF10} = 0.1(V_{REF\_ADC}) \quad (6)$$

$$V_{MEAS90} = ADC_{MEASUREMENT} \text{ at } V_{REF90} \quad (7)$$

$$V_{MEAS10} = ADC_{MEASUREMENT} \text{ at } V_{REF10} \quad (8)$$

2. Compute the ADC measured gain. The slope of the curve connecting the measured  $10\%V_{REF}$  and measured  $90\%V_{REF}$  point is computed and compared to the slope between the ideal  $10\%V_{REF}$  and ideal  $90\%V_{REF}$ . This result is the measured gain.

$$G_{MEAS} = \frac{V_{MEAS90} - V_{MEAS10}}{V_{REF90} - V_{REF10}} \quad (9)$$

3. Compute the ADC measured offset. The measured offset is computed by taking the difference between the measured  $10\%V_{REF}$  and the (ideal  $10\%V_{REF}$ )  $\times$  (measured gain).

$$O_{MEAS} = V_{MEAS10} - (V_{REF10} \times G_{MEAS}) \quad (10)$$

4. Compute the calibrated ADC readings.

$$V_{AD\_MEAS} = \text{Any } V_{IN} \text{ ADC}_{MEASUREMENT} \quad (11)$$

$$V_{ADC\_CAL} = \frac{V_{AD\_MEAS} - O_{MEAS}}{G_{MEAS}} \quad (12)$$

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Any ADC reading can therefore be calibrated by removing the gain error and offset error. The measured offset is subtracted from the ADC reading and then divided by the measured gain to give a corrected reading. If this calibration is performed on a timed basis, relative to the specific application, gain and offset error over temperature are also removed from the ADC reading by calibration.

For example; given:

- 12-Bit ADC
- ADC Gain Error = 6 LSB
- ADC Offset Error = 4 LSB
- ADC Reference ( $V_{REF\_ADC}$ ) = 5 V
- Temperature = 25°C

Table 11 shows the resulting system accuracy.

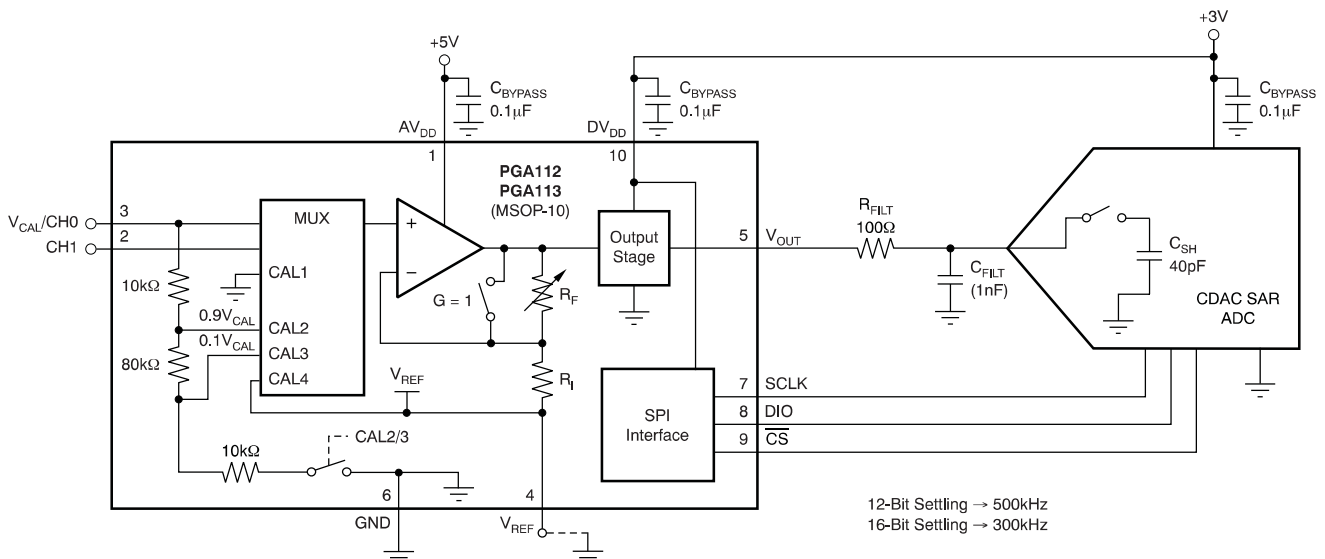
**Table 11. Bits of System Accuracy<sup>(1)</sup> (To 0.5 LSB)**

| $V_{IN}$           | ADC ACCURACY WITHOUT CALIBRATION | ADC ACCURACY WITH PGA112 CALIBRATION |
|--------------------|----------------------------------|--------------------------------------|
| 10% $V_{REF\_ADC}$ | 8.80 Bits                        | 12.80 Bits                           |
| 90% $V_{REF\_ADC}$ | 7.77 Bits                        | 11.06 Bits                           |

(1) Difference in maximum input offset voltage for  $V_{IN} = 10\%V_{REF\_ADC}$  and  $V_{IN} = 90\%V_{REF\_ADC}$  is the reason for different accuracies.

### 9.1.6 Driving and Interfacing to ADCs

CDAC SAR ADCs contain an input sampling capacitor,  $C_{SH}$ , to sample the input signal during a sample period as shown in Figure 75. After the sample period,  $C_{SH}$  is removed from the input signal. Subsequent comparisons of the charge stored on  $C_{SH}$  are performed during the ADC conversion process. To achieve optimal op amp stability, input signal settling, and the demands for charge from the input signal conditioning circuitry, most ADC applications are optimized by the use of a resistor ( $R_{FILT}$ ) and capacitor ( $C_{FILT}$ ) filter placed between the op amp output and ADC input. For the PGA112 and PGA113 devices, or the PGA116 and PGA117 devices, setting  $C_{FILT} = 1$  nF and  $R_{FILT} = 100$   $\Omega$  yields optimum system performance for sampling converters operating at speeds up to 500 kHz, depending upon the application settling time and accuracy requirements.



**Figure 75. Driving and Interfacing to ADCs**

### 9.1.7 Power Supplies

Figure 76 shows a typical mixed-supply voltage system where the analog supply,  $AV_{DD}$ , is 5 V and the digital supply voltage,  $DV_{DD}$ , is 3 V. The analog output stage of the PGA and the SPI interface digital circuitry are both powered from  $DV_{DD}$ . When considering the power required for  $DV_{DD}$ , use the *Electrical Characteristics:  $V_S = AV_{DD} = DV_{DD} = 5\text{ V}$*  table and add any load current anticipated on  $V_{OUT}$ ; this load current must be provided by  $DV_{DD}$ . This split-supply architecture ensures compatible logic levels with the microcontroller. It also ensures that the PGA output cannot run the input for the onboard ADC into an overvoltage condition; this condition could cause device latch-up and system lock-up, and require power-supply sequencing. Each supply pin should be individually bypassed with a 0.1  $\mu\text{F}$  ceramic capacitor directly at the device to ground. If there is only one power supply in the system,  $AV_{DD}$  and  $DV_{DD}$  can both be connected to the same supply; however, TI recommends using individual bypass capacitors directly at each respective supply pin to a single point ground.  $V_{OUT}$  is diode-clamped to  $AV_{DD}$  (as shown in Figure 76); therefore, set  $DV_{DD}$  less than or equal to  $AV_{DD} + 0.3\text{ V}$ .  $DV_{DD}$  and  $AV_{DD}$  must be within the operating voltage range of 2.2 V to 5.5 V.

At initial power-on, the state of the PGA is  $G = 1$  and Channel 0 active.

#### NOTE

For most applications, set  $AV_{DD} \geq DV_{DD}$  to prevent  $V_{OUT}$  from driving current into  $AV_{DD}$  and raising the voltage level of  $AV_{DD}$ .

### 9.1.8 Shutdown and Power-On-Reset (POR)

The PGA112 and PGA113 devices have a software shutdown mode, and the PGA116 and PGA117 devices offer both a hardware and software shutdown mode. When the PGA11x is shut down, it goes into a low-power standby mode. The *Electrical Characteristics:  $V_S = AV_{DD} = DV_{DD} = 5\text{ V}$*  table details the current draw in shutdown mode with and without the SPI interface being clocked. In shutdown mode,  $R_F$  and  $R_I$  remain connected between  $V_{OUT}$  and  $V_{REF}$ .

When  $DV_{DD}$  is less than 1.6 V, the digital interface is disabled and the channel and gain selections are held to the respective POR states of Gain = 1 and Channel =  $V_{CAL}/CH0$ . When  $DV_{DD}$  is above 1.8 V, the digital interface is enabled and the POR gain and channel states remain unchanged until a valid SPI communication is received.

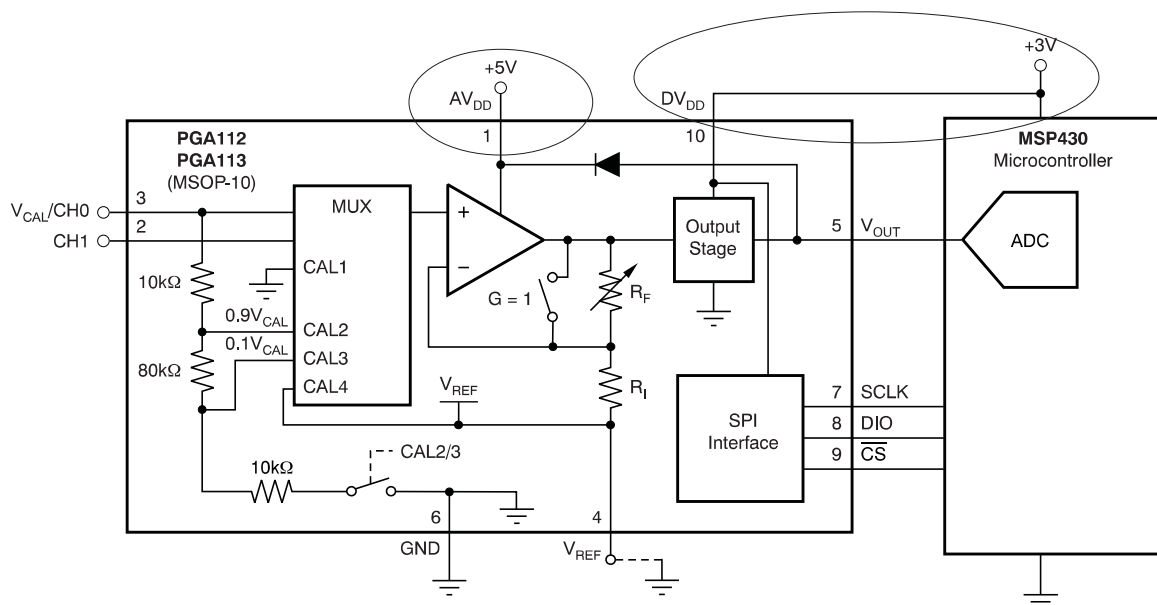
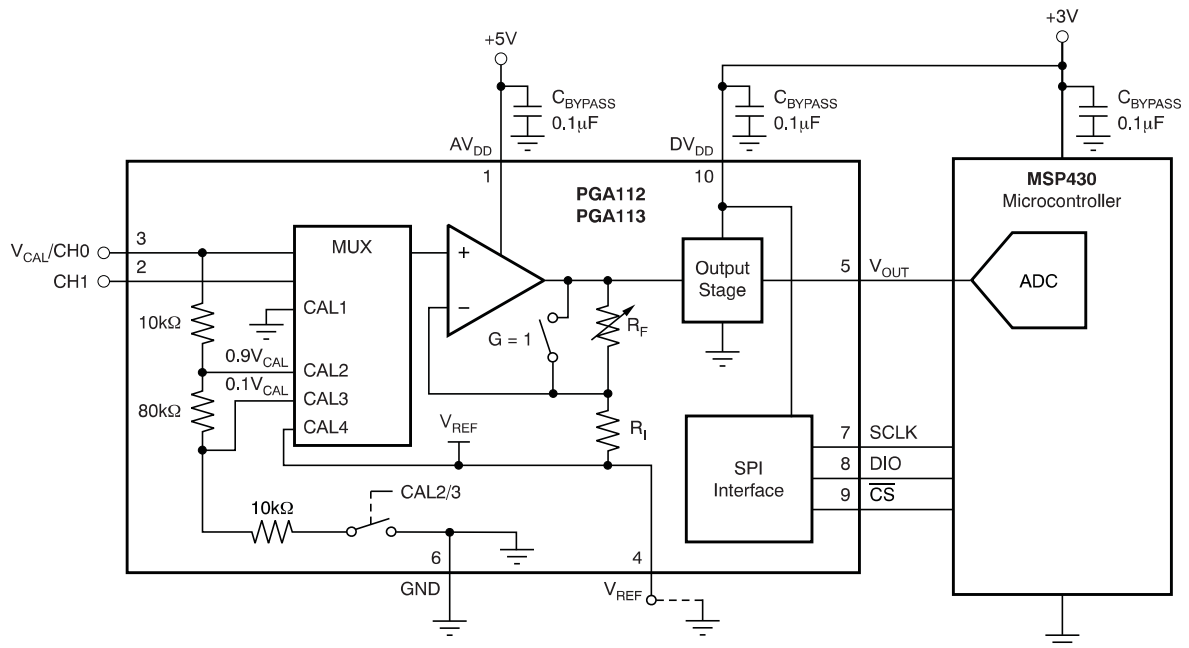
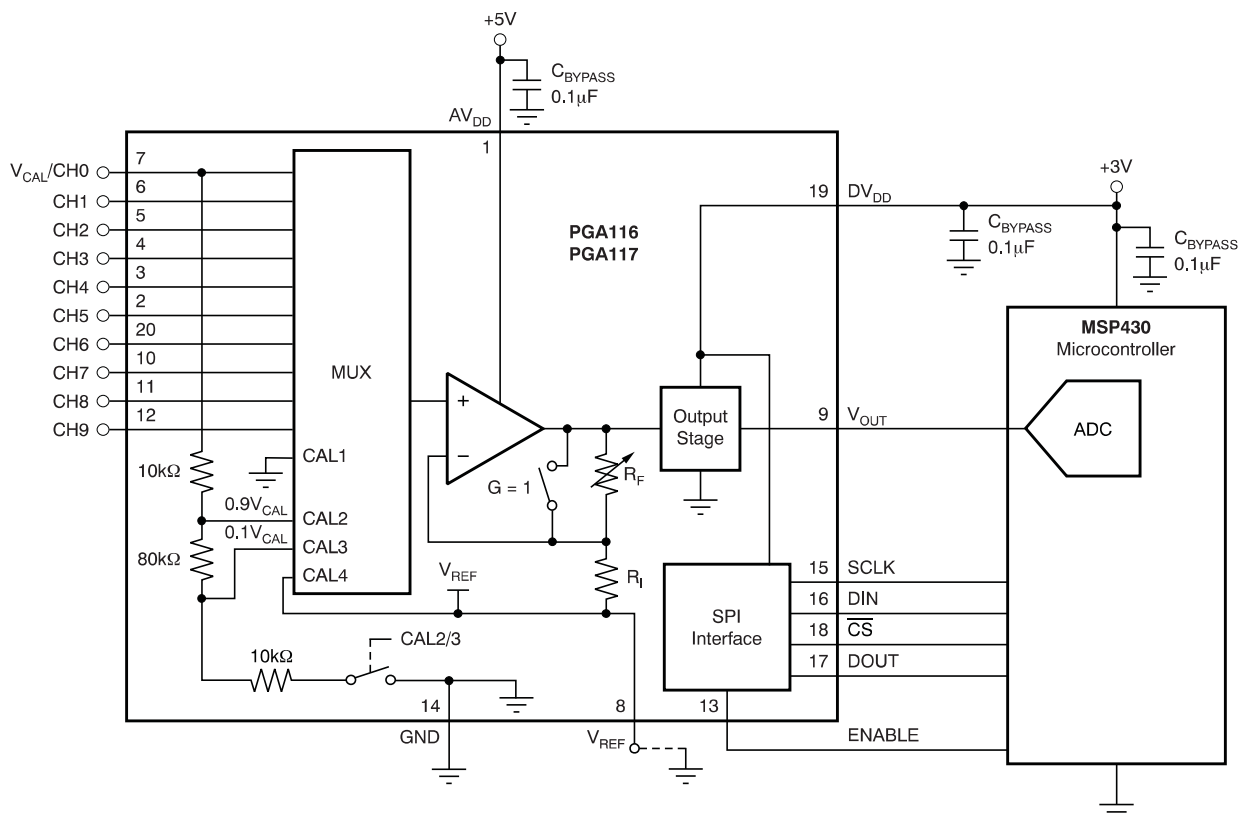


Figure 76. Split Power-Supply Architecture:  $AV_{DD} \neq DV_{DD}$

**PGA112, PGA113, PGA116, PGA117**

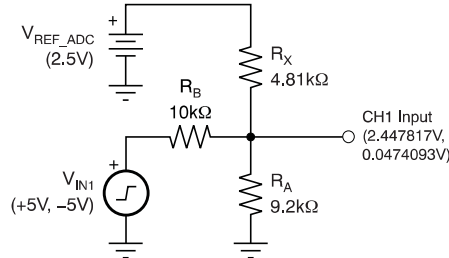
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**Figure 77. PGA112, PGA113 (VSSOP-10) Typical Application Schematic**
**9.1.9 Typical Connections: PGA116, PGA117 (TSSOP-20)**

**Figure 78. PGA116, PGA117 (TSSOP-20)**

## 9.2 Typical Applications

### 9.2.1 Bipolar Input to Single-Supply Scaling



**Figure 79. Bipolar to Single-Ended Input Algorithm**

#### 9.2.1.1 Design Requirements

**Table 12. Bipolar to Single-Ended Input Scaling<sup>(1)(2)</sup>**

| $V_{REF\_ADC}$ (V) | $V_{IN1}$ (V) | CH1 INPUT | $R_A$ (kΩ) | $R_X$ (Ω) | $R_B$ (kΩ) |
|--------------------|---------------|-----------|------------|-----------|------------|
| 2.5                | –5            | 0.047613  | 9.2        | 4.81 k    | 10         |
|                    | 0             | 1.247613  |            |           |            |
|                    | 5             | 2.447613  |            |           |            |
| 2.5                | –10           | 0.050317  | 3.16       | 2.4 k     | 10         |
|                    | 0             | 1.250317  |            |           |            |
|                    | 10            | 2.450317  |            |           |            |
| 3                  | –5            | 0.058003  | 13.5       | 5.76 k    | 10         |
|                    | 0             | 1.498003  |            |           |            |
|                    | 5             | 2.938003  |            |           |            |
| 3                  | –10           | 0.059303  | 4.02       | 2.87 k    | 10         |
|                    | 0             | 1.499303  |            |           |            |
|                    | 10            | 2.939303  |            |           |            |
| 4.096              | –5            | 0.082224  | 37         | 7.87 k    | 10         |
|                    | 0             | 2.048304  |            |           |            |
|                    | 5             | 4.014384  |            |           |            |
| 4.096              | –10           | 0.086018  | 6.49       | 3.92 k    | 10         |
|                    | 0             | 2.052098  |            |           |            |
|                    | 10            | 4.018178  |            |           |            |
| 5                  | –5            | 0.093506  | 24         | 965       | 10         |
|                    | 0             | 2.493506  |            |           |            |
|                    | 5             | 4.893506  |            |           |            |
| 5                  | –10           | 0.095227  | 9.2        | 4.81 k    | 10         |
|                    | 0             | 2.495227  |            |           |            |
|                    | 10            | 4.895227  |            |           |            |

(1) Scaling is based on  $0.02(V_{REF\_ADC})$  to  $0.98(V_{REF\_ADC})$ , using standard 0.1% resistor values.

(2) Assumes symmetrical  $V_{IN}$  and symmetrical scaling for CH1 input minimum and maximum.

### 9.2.1.2 Detailed Design Procedure

This process assumes a symmetrical  $V_{IN1}$  and that symmetrical scaling is used for CH1 input minimum and maximum values. The following steps give the algorithm to compute resistor values for references not listed in [Table 12](#).

Step 1: Choose the following:

- $V_{REF\_ADC} = 2.5 \text{ V}$  (ADC reference voltage)
- $|V_{IN1}| = 5$   
(magnitude of  $V_{IN}$ , assuming scaling is for  $\pm V_{IN1}$ )
- Choose  $R_B$  as a standard resistor value. The input on-channel current multiplied by  $R_B$  should be less than the input offset voltage, such that  $R_B$  is not a major source of inaccuracy.  
 $R_B = 10 \text{ k}\Omega$  (select as a starting value for resistors)
- For the most negative  $V_{IN1}$ , choose the percentage (in decimal format) of  $V_{REF\_ADC}$  desired at the ADC input.  
 $k_{VO-} = 0.02$   
(CH1 input =  $k_{VO-} \times V_{REF\_ADC}$  when  $V_{IN1} = -V_{IN1}$ )
- For the most positive  $V_{IN1}$ , choose the percentage (in decimal format) of  $V_{REF\_ADC}$  desired at the ADC input. Because this scaling is based on symmetry,  $k_{VO+}$  must be the same percentage away from  $V_{REF\_ADC}$  at the upper limit as at the lower limit where  $k_{VO-}$  is computed.  
 $k_{VO+} = 1 - k_{VO-}$   
 $k_{VO+} = 1 - 0.02 = 0.98$   
(CH1 input =  $k_{VO+} \times V_{REF\_ADC}$  when  $V_{IN1} = +V_{IN1}$ )

Step 2: Compute the following:

- To simplify analysis, create one constant called  $k_{VO}$ .  
 $k_{VO} = k_{VO+} - k_{VO-}$   
 $0.96 = 0.98 - 0.02$  (13)
- A constant,  $g$ , is created to simplify resistor value computations.

$$g = \frac{k_{VO} \times V_{REF\_ADC}}{2 \times |V_{IN1}| - k_{VO} \times V_{REF\_ADC}}$$

$$0.315789474 = \frac{0.96 \times 2.5}{2 \times 5 - 0.96 \times 2.5}$$
 (14)

- $R_A$  is now selected from the starting value of  $R_B$  and the  $g$  constant.

$$R_A = \frac{2 \times R_B \times g}{1 - g}$$

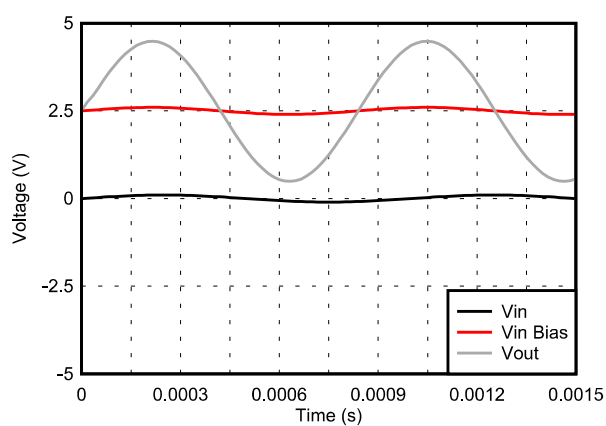
$$9.23077 \text{ k}\Omega = \frac{2 \times 10 \text{ k}\Omega \times 0.315789474}{1 - 0.315789474}$$
 (15)

- $R_X$  can now be computed from the starting value of  $R_B$  and the computed value for  $R_A$ .

$$R_X = \frac{R_B \times R_A}{R_B + R_A}$$

$$4.81 \text{ k}\Omega = \frac{10 \text{ k}\Omega \times 9.23077 \text{ k}\Omega}{10 \text{ k}\Omega + 9.23077 \text{ k}\Omega}$$
 (16)

### 9.2.1.3 Application Curve

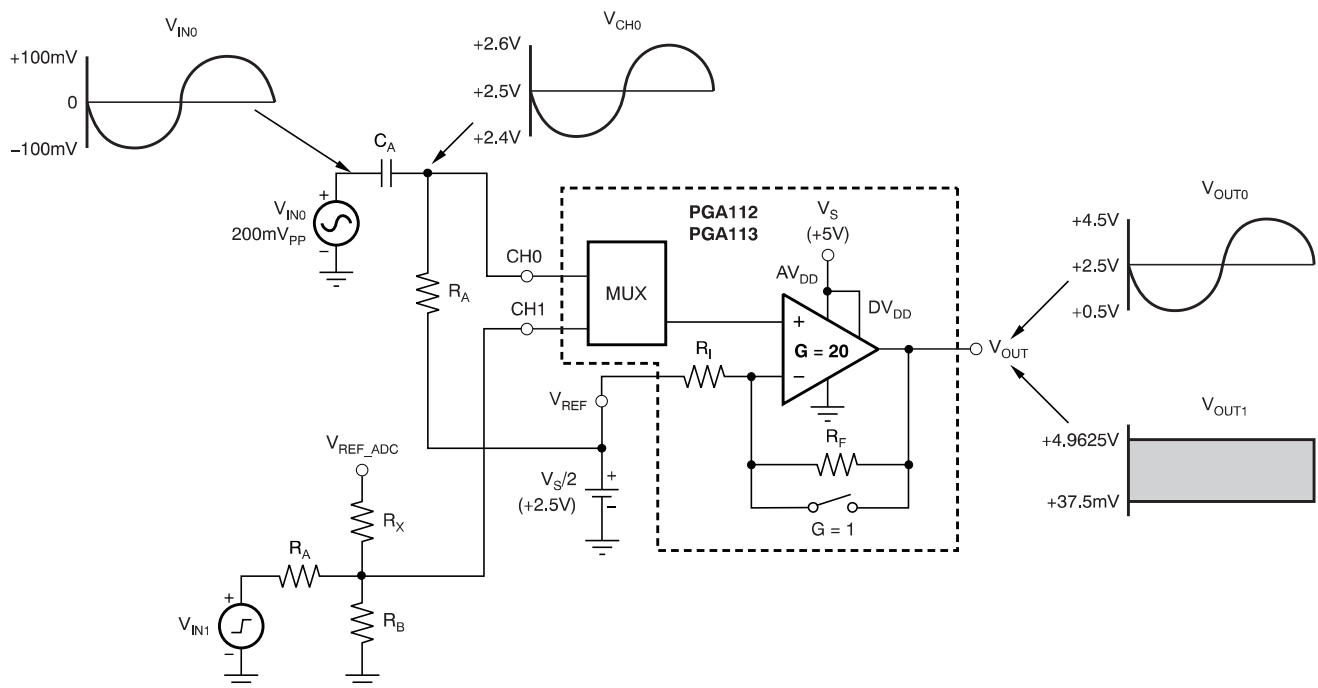


**Figure 80. Voltage (V) vs Time (s)**

## 9.2.2 Typical Application: General-Purpose Input Scaling

Figure 81 is an example application that demonstrates the flexibility of the PGA for general-purpose input scaling.  $V_{IN0}$  is a  $\pm 100$ -mV input that is ac-coupled into CH0. The PGA112 and PGA113 are powered from a 5-V supply voltage,  $V_S$ , and configured with the  $V_{REF}$  pin connected to  $V_S/2$  (2.5 V).  $V_{CH0}$  is the  $\pm 100$ -mV input, level-shifted and centered on  $V_S/2$  (2.5 V). A gain of 20 is applied to CH0, and because of the PGA113 configuration, the output voltage at  $V_{OUT}$  is  $\pm 2$  V centered on  $V_S/2$  (2.5 V).

CH1 is set to  $G = 1$ ; through a resistive divider and scalar network, we can read  $\pm 5$  V or 0 V. This setting provides bipolar to single-ended input scaling. Table 12 summarizes the scaling resistor values for  $R_A$ ,  $R_X$ , and  $R_B$  for different ADC Ref voltages.  $V_{REF\_ADC}$  is the reference voltage used for the ADC connected to the PGA112 and PGA113 output. It is assumed the ADC input range is 0 V to  $V_{REF\_ADC}$ . The Table 12 section gives the algorithm to compute resistor values for references not listed in Table 12. As a general guideline,  $R_B$  should be chosen such that the input on-channel current multiplied by  $R_B$  is less than or equal to the input offset voltage. This value ensures that the scaling network contributes no more error than the input offset voltage. Individual applications may require other design trade-offs.



**Figure 81. General-Purpose Input Scaling**

## 10 Power Supply Recommendations

**Power-supply bypass:** Bypass each power-supply pin separately. Use a ceramic capacitor connected directly from the power-supply pin to the ground pin of the IC on the same PCB plane. Vias can then be used to connect to ground and voltage planes. This configuration keeps parasitic inductive paths out of the local bypass for the PGA. Good analog design practice dictates the use of a large value tantalum bypass capacitor on the PCB for each respective voltage.

## 11 Layout

### 11.1 Layout Guidelines

#### 11.1.1 High Gain and Wide Bandwidth Considerations

As a result of the combination of wide bandwidth and high gain capability of the PGA112 and PGA113 devices and PGA116 and PGA117 devices, there are several printed-circuit-board (PCB) design and system recommendations to consider for optimum application performance.

1. **Power-supply bypass:** Refer to [Power Supply Recommendations](#).
2. **Signal trace routing:** Keep  $V_{OUT}$  and other low impedance traces away from MUX channel inputs that are high impedance. Poor signal routing can cause positive feedback, unwanted oscillations, or excessive overshoot and ringing on step-changing signals. If the input signals are particularly noisy, separate MUX input channels with guard traces on either side of the signal traces. Connect the guard traces to ground near the PGA and at the signal entry point into the PCB. On multilayer PCBs, ensure that there are no parallel traces near MUX input traces on adjacent layers; capacitive coupling from other layers can be a problem. Use ground planes to isolate MUX input signal traces from signal traces on other layers.

Additionally, group and route the digital signals into the PGA as far away as possible from the analog MUX input signals. Most digital signals are fast rise and fall time signals with low-impedance drive capability that can easily couple into the high-impedance inputs of the input MUX channels. This coupling can create unwanted noise that gains up to  $V_{OUT}$ .

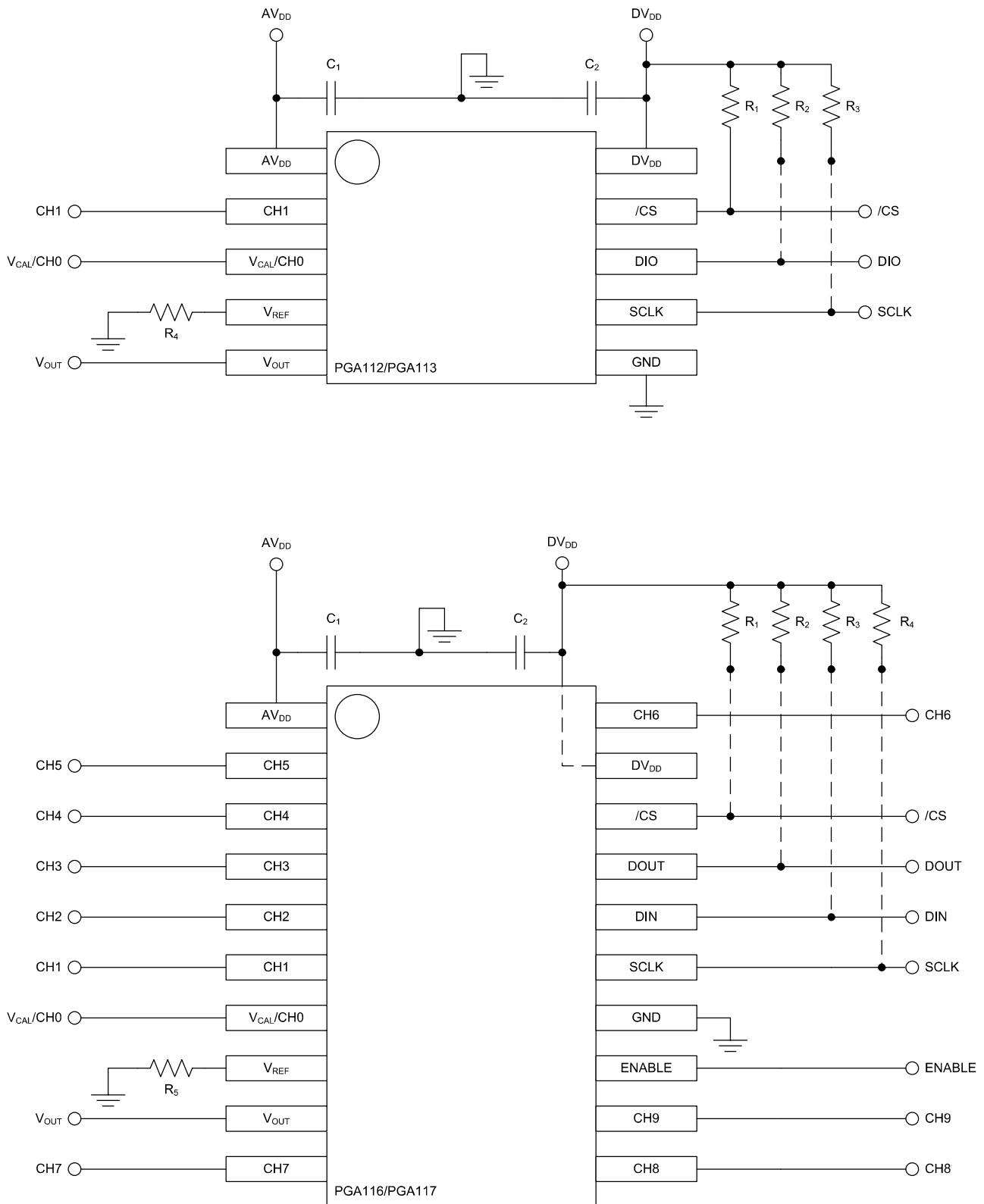
3. **Input MUX channels and source impedance:** Input MUX channels are high-impedance; when combined with high gain, the channels can pick up unwanted noise. Keep the input signal sources low-impedance ( $< 10\text{ k}\Omega$ ). Also, consider bypassing input MUX channels with a ceramic bypass capacitor directly at the MUX input pin. Bypass capacitors greater than 100 pF are recommended. Lower impedances and a bypass capacitor placed directly at the input MUX channels keep crosstalk between channels to a minimum as a result of parasitic capacitive coupling from adjacent PCB traces and pin-to-pin capacitance.

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### 11.2 Layout Example



**Figure 82. PGA11x Layout Example**

## 12 Device and Documentation Support

### 12.1 Documentation Support

#### 12.1.1 Related Documentation

For related documentation see the following:

- *Shelf-Life Evaluation of Lead-Free Component Finishes*, [SZZA046](#).
- *PGA112/113EVM Users Guide*, [SBOU073](#).

### 12.2 Related Links

The table below lists quick access links. Categories include technical documents, support and community resources, tools and software, and quick access to sample or buy.

**Table 13. Related Links**

| PARTS  | PRODUCT FOLDER             | SAMPLE & BUY               | TECHNICAL DOCUMENTS        | TOOLS & SOFTWARE           | SUPPORT & COMMUNITY        |
|--------|----------------------------|----------------------------|----------------------------|----------------------------|----------------------------|
| PGA112 | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> |
| PGA113 | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> |
| PGA116 | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> |
| PGA117 | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> | <a href="#">Click here</a> |

### 12.3 Community Resources

The following links connect to TI community resources. Linked contents are provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's [Terms of Use](#).

**TI E2E™ Online Community** *TI's Engineer-to-Engineer (E2E) Community*. Created to foster collaboration among engineers. At [e2e.ti.com](#), you can ask questions, share knowledge, explore ideas and help solve problems with fellow engineers.

**Design Support** *TI's Design Support* Quickly find helpful E2E forums along with design support tools and contact information for technical support.

### 12.4 Trademarks

E2E is a trademark of Texas Instruments.

SPI is a trademark of Motorola.

All other trademarks are the property of their respective owners.

### 12.5 Electrostatic Discharge Caution



These devices have limited built-in ESD protection. The leads should be shorted together or the device placed in conductive foam during storage or handling to prevent electrostatic damage to the MOS gates.

### 12.6 Glossary

[SLYZ022](#) — *TI Glossary*.

This glossary lists and explains terms, acronyms, and definitions.

## 13 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

## PACKAGING INFORMATION

| Orderable part number        | Status<br>(1) | Material type<br>(2) | Package   Pins   | Package qty   Carrier | RoHS<br>(3) | Lead finish/<br>Ball material<br>(4) | MSL rating/<br>Peak reflow<br>(5) | Op temp (°C) | Part marking<br>(6) |
|------------------------------|---------------|----------------------|------------------|-----------------------|-------------|--------------------------------------|-----------------------------------|--------------|---------------------|
| <a href="#">PGA112AIDGSR</a> | Active        | Production           | VSSOP (DGS)   10 | 2500   LARGE T&R      | Yes         | NIPDAU   NIPDAUAG                    | Level-2-260C-1 YEAR               | -40 to 125   | P112                |
| PGA112AIDGSR.A               | Active        | Production           | VSSOP (DGS)   10 | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-2-260C-1 YEAR               | -40 to 125   | P112                |
| <a href="#">PGA112AIDGST</a> | Active        | Production           | VSSOP (DGS)   10 | 250   SMALL T&R       | Yes         | NIPDAU   NIPDAUAG                    | Level-2-260C-1 YEAR               | -40 to 125   | P112                |
| PGA112AIDGST.A               | Active        | Production           | VSSOP (DGS)   10 | 250   SMALL T&R       | Yes         | NIPDAU                               | Level-2-260C-1 YEAR               | -40 to 125   | P112                |
| PGA112AIDGSTG4               | Active        | Production           | VSSOP (DGS)   10 | 250   SMALL T&R       | Yes         | NIPDAU                               | Level-2-260C-1 YEAR               | -40 to 125   | P112                |
| <a href="#">PGA113AIDGSR</a> | Active        | Production           | VSSOP (DGS)   10 | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-2-260C-1 YEAR               | -40 to 125   | P113                |
| PGA113AIDGSR.A               | Active        | Production           | VSSOP (DGS)   10 | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-2-260C-1 YEAR               | -40 to 125   | P113                |
| <a href="#">PGA113AIDGST</a> | Active        | Production           | VSSOP (DGS)   10 | 250   SMALL T&R       | Yes         | NIPDAU                               | Level-2-260C-1 YEAR               | -40 to 125   | P113                |
| PGA113AIDGST.A               | Active        | Production           | VSSOP (DGS)   10 | 250   SMALL T&R       | Yes         | NIPDAU                               | Level-2-260C-1 YEAR               | -40 to 125   | P113                |
| PGA113AIDGSTG4               | Active        | Production           | VSSOP (DGS)   10 | 250   SMALL T&R       | Yes         | NIPDAU                               | Level-2-260C-1 YEAR               | -40 to 125   | P113                |
| <a href="#">PGA116AIPW</a>   | Active        | Production           | TSSOP (PW)   20  | 70   TUBE             | Yes         | NIPDAU                               | Level-2-260C-1 YEAR               | -40 to 125   | PGA116              |
| PGA116AIPW.A                 | Active        | Production           | TSSOP (PW)   20  | 70   TUBE             | Yes         | NIPDAU                               | Level-2-260C-1 YEAR               | -40 to 125   | PGA116              |
| <a href="#">PGA116AIPWR</a>  | Active        | Production           | TSSOP (PW)   20  | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-2-260C-1 YEAR               | -40 to 125   | PGA116              |
| PGA116AIPWR.A                | Active        | Production           | TSSOP (PW)   20  | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-2-260C-1 YEAR               | -40 to 125   | PGA116              |
| PGA116AIPWRG4                | Active        | Production           | TSSOP (PW)   20  | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-2-260C-1 YEAR               | -40 to 125   | PGA116              |
| PGA116AIPWRG4.A              | Active        | Production           | TSSOP (PW)   20  | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-2-260C-1 YEAR               | -40 to 125   | PGA116              |
| <a href="#">PGA117AIPW</a>   | Active        | Production           | TSSOP (PW)   20  | 70   TUBE             | Yes         | NIPDAU                               | Level-2-260C-1 YEAR               | -40 to 125   | PGA117              |
| PGA117AIPW.A                 | Active        | Production           | TSSOP (PW)   20  | 70   TUBE             | Yes         | NIPDAU                               | Level-2-260C-1 YEAR               | -40 to 125   | PGA117              |
| <a href="#">PGA117AIPWR</a>  | Active        | Production           | TSSOP (PW)   20  | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-2-260C-1 YEAR               | -40 to 125   | PGA117              |
| PGA117AIPWR.A                | Active        | Production           | TSSOP (PW)   20  | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-2-260C-1 YEAR               | -40 to 125   | PGA117              |
| PGA117AIPWRG4                | Active        | Production           | TSSOP (PW)   20  | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-2-260C-1 YEAR               | -40 to 125   | PGA117              |
| PGA117AIPWRG4.A              | Active        | Production           | TSSOP (PW)   20  | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-2-260C-1 YEAR               | -40 to 125   | PGA117              |

(1) **Status:** For more details on status, see our [product life cycle](#).

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

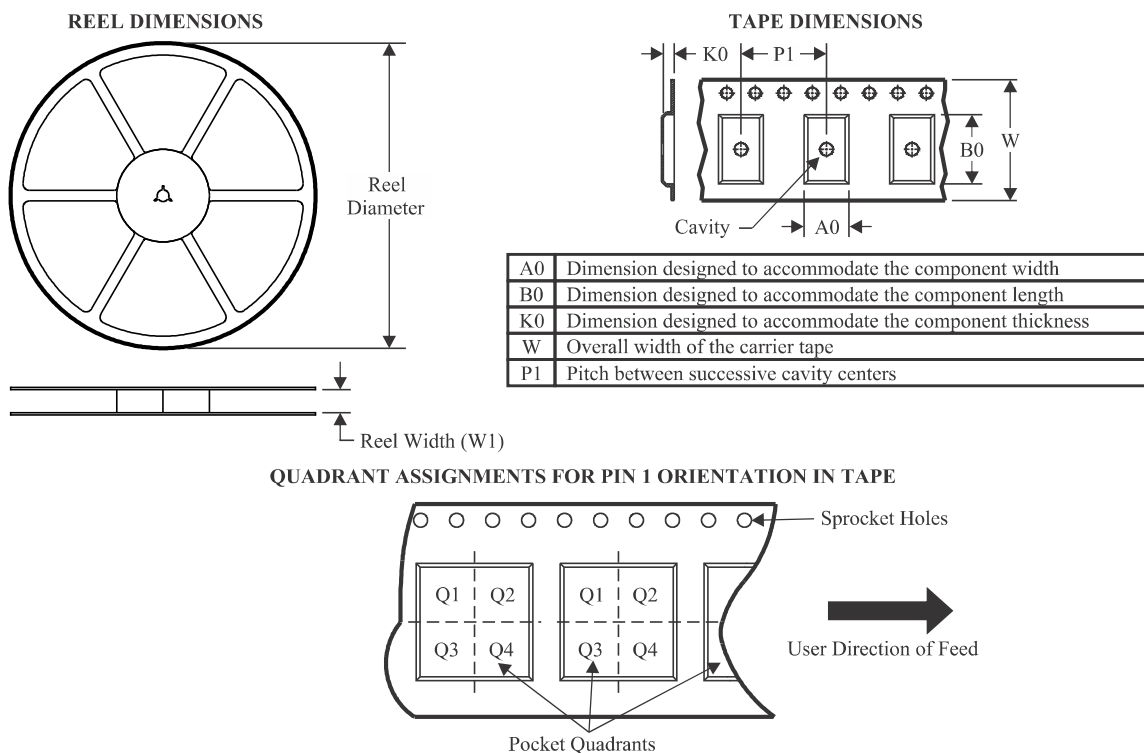
(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "-" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

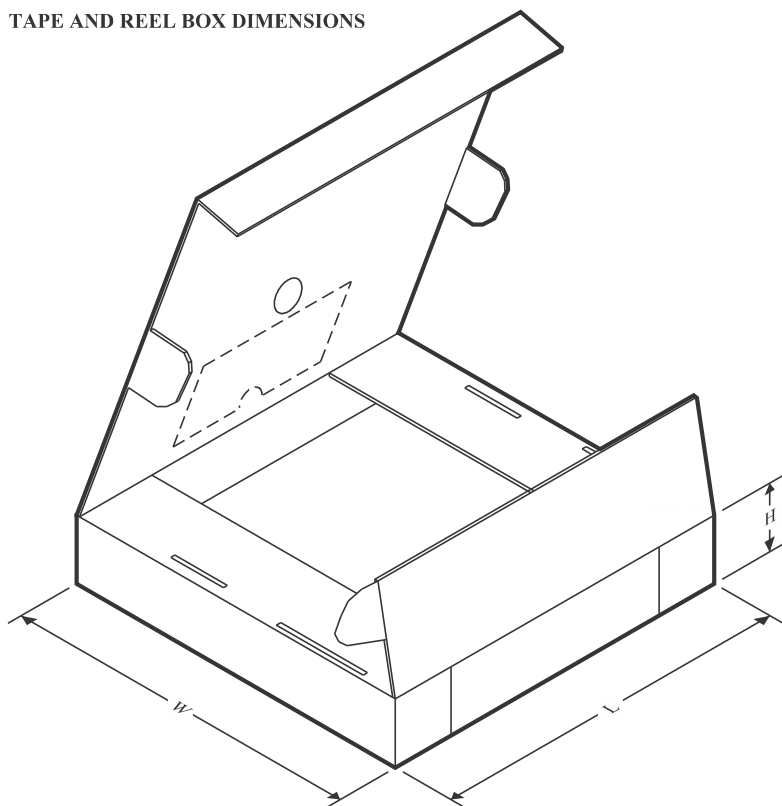
## TAPE AND REEL INFORMATION



\*All dimensions are nominal

| Device        | Package Type | Package Drawing | Pins | SPQ  | Reel Diameter (mm) | Reel Width W1 (mm) | A0 (mm) | B0 (mm) | K0 (mm) | P1 (mm) | W (mm) | Pin1 Quadrant |
|---------------|--------------|-----------------|------|------|--------------------|--------------------|---------|---------|---------|---------|--------|---------------|
| PGA112AIDGSR  | VSSOP        | DGS             | 10   | 2500 | 330.0              | 12.4               | 5.3     | 3.3     | 1.3     | 8.0     | 12.0   | Q1            |
| PGA112AIDGST  | VSSOP        | DGS             | 10   | 250  | 330.0              | 12.4               | 5.3     | 3.4     | 1.4     | 8.0     | 12.0   | Q1            |
| PGA112AIDGST  | VSSOP        | DGS             | 10   | 250  | 180.0              | 12.4               | 5.3     | 3.3     | 1.3     | 8.0     | 12.0   | Q1            |
| PGA113AIDGSR  | VSSOP        | DGS             | 10   | 2500 | 330.0              | 12.4               | 5.3     | 3.3     | 1.3     | 8.0     | 12.0   | Q1            |
| PGA113AIDGST  | VSSOP        | DGS             | 10   | 250  | 180.0              | 12.4               | 5.3     | 3.3     | 1.3     | 8.0     | 12.0   | Q1            |
| PGA116AIPWR   | TSSOP        | PW              | 20   | 2000 | 330.0              | 16.4               | 6.95    | 7.0     | 1.4     | 8.0     | 16.0   | Q1            |
| PGA116AIPWRG4 | TSSOP        | PW              | 20   | 2000 | 330.0              | 16.4               | 6.95    | 7.0     | 1.4     | 8.0     | 16.0   | Q1            |
| PGA117AIPWR   | TSSOP        | PW              | 20   | 2000 | 330.0              | 16.4               | 6.95    | 7.0     | 1.4     | 8.0     | 16.0   | Q1            |
| PGA117AIPWRG4 | TSSOP        | PW              | 20   | 2000 | 330.0              | 16.4               | 6.95    | 7.0     | 1.4     | 8.0     | 16.0   | Q1            |

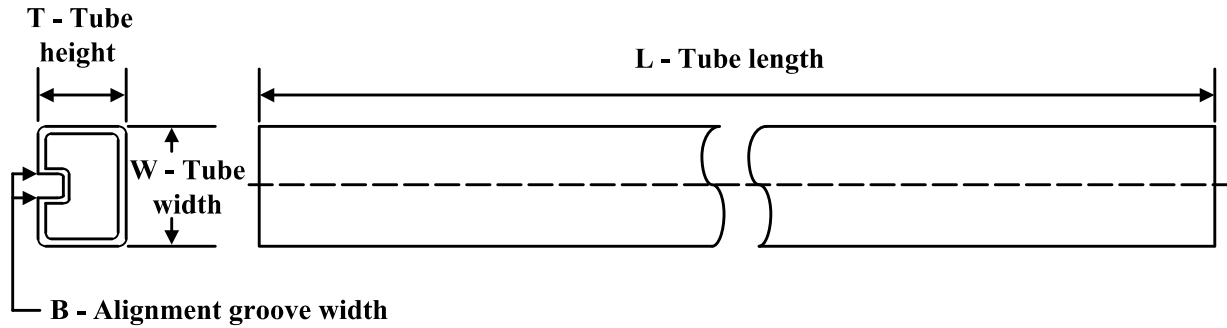
## TAPE AND REEL BOX DIMENSIONS



\*All dimensions are nominal

| Device        | Package Type | Package Drawing | Pins | SPQ  | Length (mm) | Width (mm) | Height (mm) |
|---------------|--------------|-----------------|------|------|-------------|------------|-------------|
| PGA112AIDGSR  | VSSOP        | DGS             | 10   | 2500 | 367.0       | 367.0      | 38.0        |
| PGA112AIDGST  | VSSOP        | DGS             | 10   | 250  | 366.0       | 364.0      | 50.0        |
| PGA112AIDGST  | VSSOP        | DGS             | 10   | 250  | 213.0       | 191.0      | 35.0        |
| PGA113AIDGSR  | VSSOP        | DGS             | 10   | 2500 | 367.0       | 367.0      | 38.0        |
| PGA113AIDGST  | VSSOP        | DGS             | 10   | 250  | 213.0       | 191.0      | 35.0        |
| PGA116AIPWR   | TSSOP        | PW              | 20   | 2000 | 353.0       | 353.0      | 32.0        |
| PGA116AIPWRG4 | TSSOP        | PW              | 20   | 2000 | 353.0       | 353.0      | 32.0        |
| PGA117AIPWR   | TSSOP        | PW              | 20   | 2000 | 353.0       | 353.0      | 32.0        |
| PGA117AIPWRG4 | TSSOP        | PW              | 20   | 2000 | 353.0       | 353.0      | 32.0        |

## TUBE



\*All dimensions are nominal

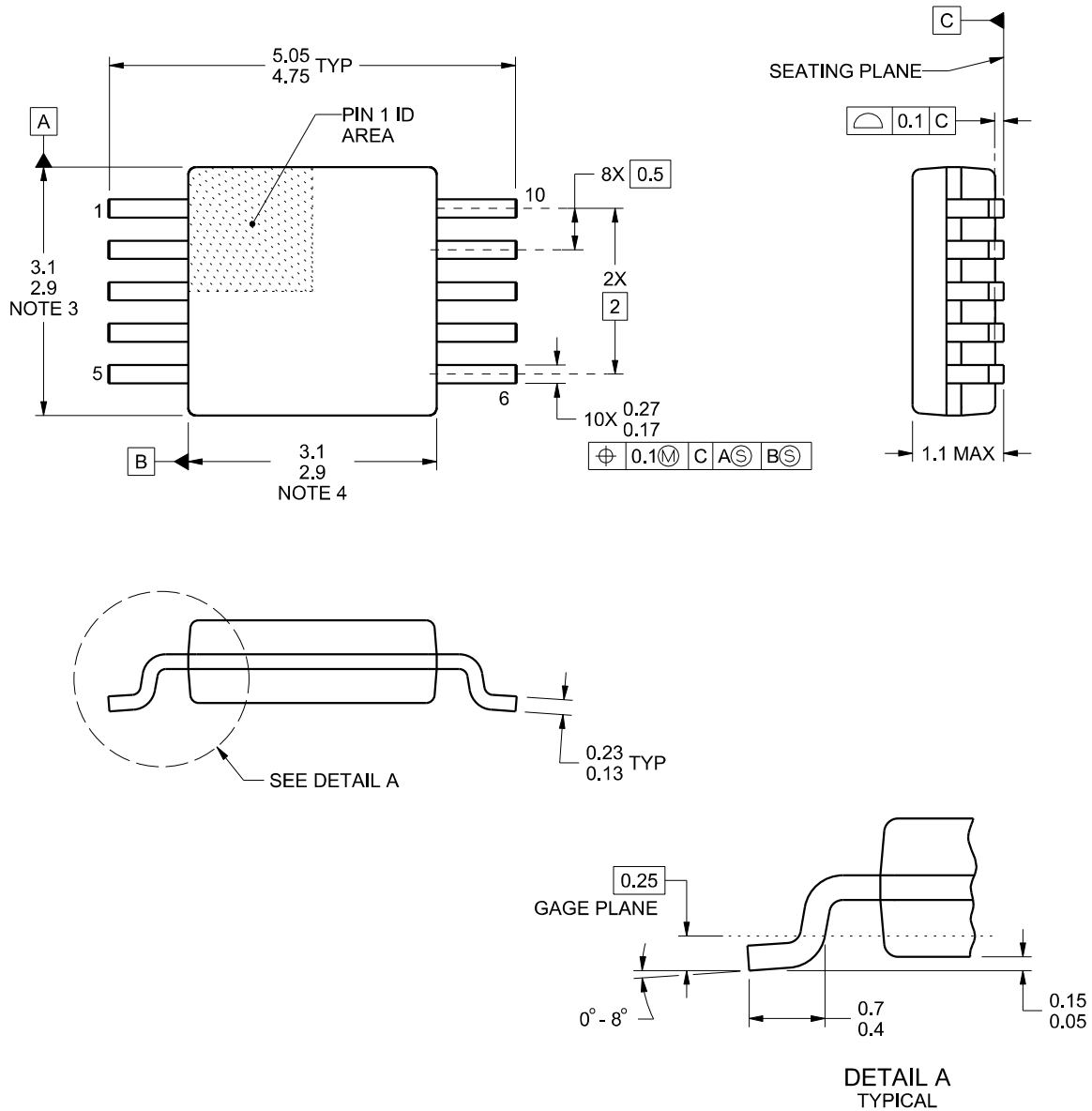
| Device       | Package Name | Package Type | Pins | SPQ | L (mm) | W (mm) | T (μm) | B (mm) |
|--------------|--------------|--------------|------|-----|--------|--------|--------|--------|
| PGA116AIPW   | PW           | TSSOP        | 20   | 70  | 530    | 10.2   | 3600   | 3.5    |
| PGA116AIPW.A | PW           | TSSOP        | 20   | 70  | 530    | 10.2   | 3600   | 3.5    |
| PGA117AIPW   | PW           | TSSOP        | 20   | 70  | 530    | 10.2   | 3600   | 3.5    |
| PGA117AIPW.A | PW           | TSSOP        | 20   | 70  | 530    | 10.2   | 3600   | 3.5    |

# DGS0010A



## PACKAGE OUTLINE VSSOP - 1.1 mm max height

SMALL OUTLINE PACKAGE



4221984/A 05/2015

### NOTES:

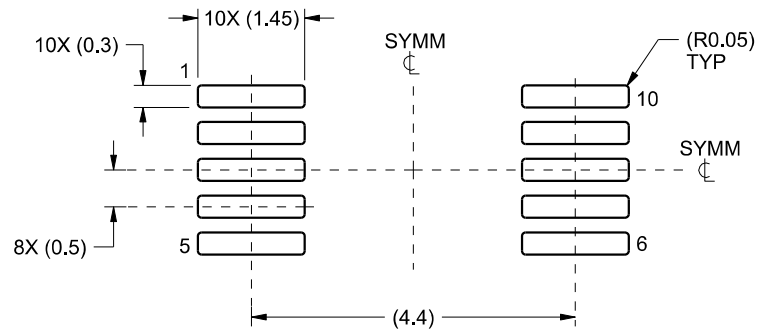
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.
5. Reference JEDEC registration MO-187, variation BA.

# EXAMPLE BOARD LAYOUT

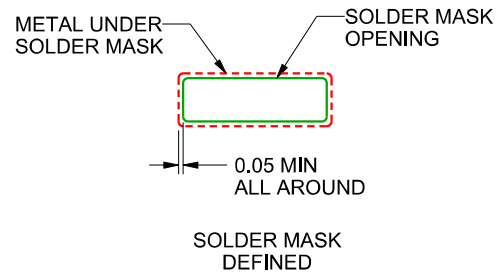
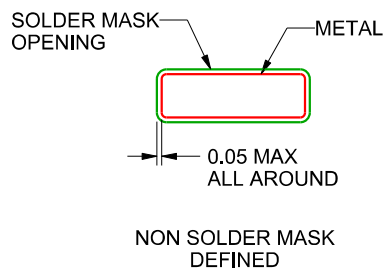
DGS0010A

VSSOP - 1.1 mm max height

SMALL OUTLINE PACKAGE



LAND PATTERN EXAMPLE  
SCALE:10X



SOLDER MASK DETAILS  
NOT TO SCALE

4221984/A 05/2015

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

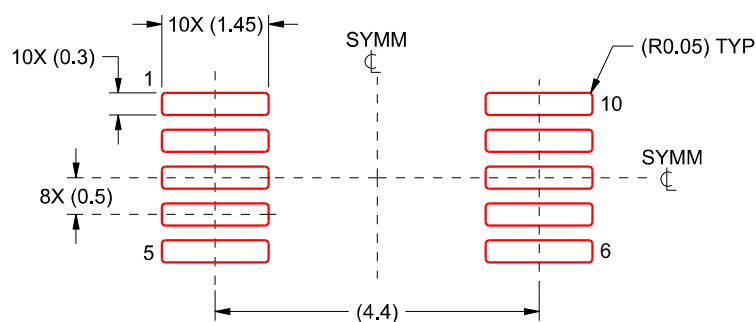
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

## EXAMPLE STENCIL DESIGN

DGS0010A

VSSOP - 1.1 mm max height

SMALL OUTLINE PACKAGE



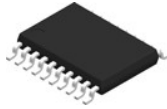
SOLDER PASTE EXAMPLE  
BASED ON 0.125 mm THICK STENCIL  
SCALE:10X

4221984/A 05/2015

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

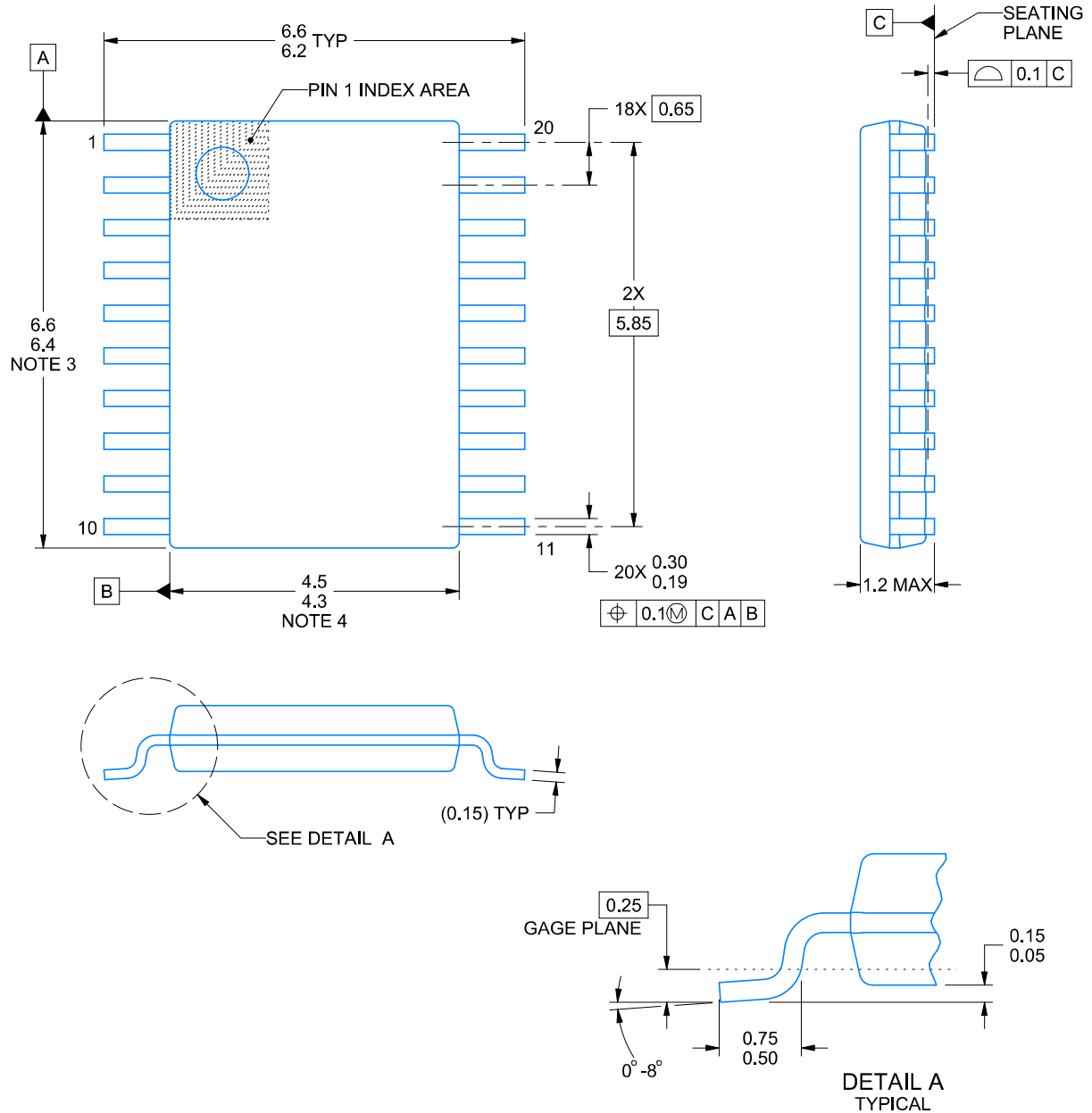
PW0020A



# PACKAGE OUTLINE

## TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



4220206/A 02/2017

### NOTES:

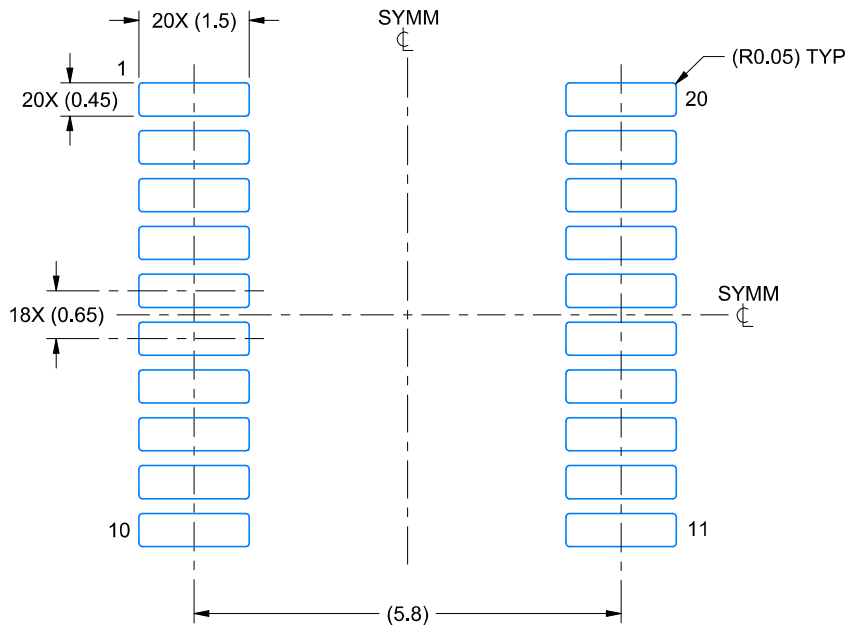
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.
5. Reference JEDEC registration MO-153.

# EXAMPLE BOARD LAYOUT

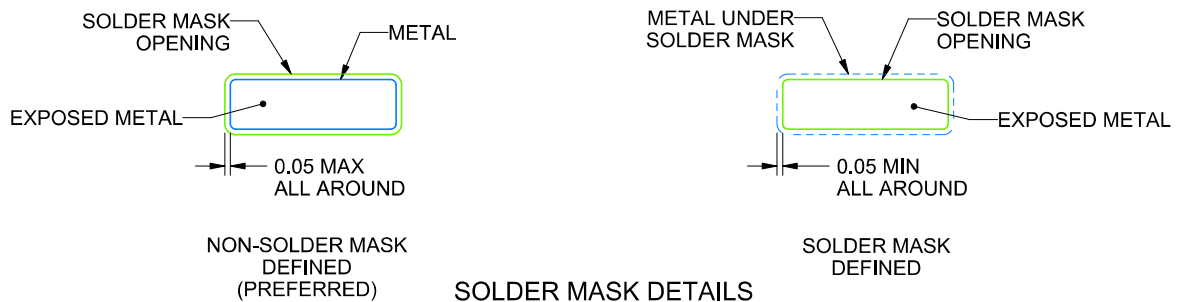
PW0020A

TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



LAND PATTERN EXAMPLE  
EXPOSED METAL SHOWN  
SCALE: 10X



SOLDER MASK DETAILS

4220206/A 02/2017

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

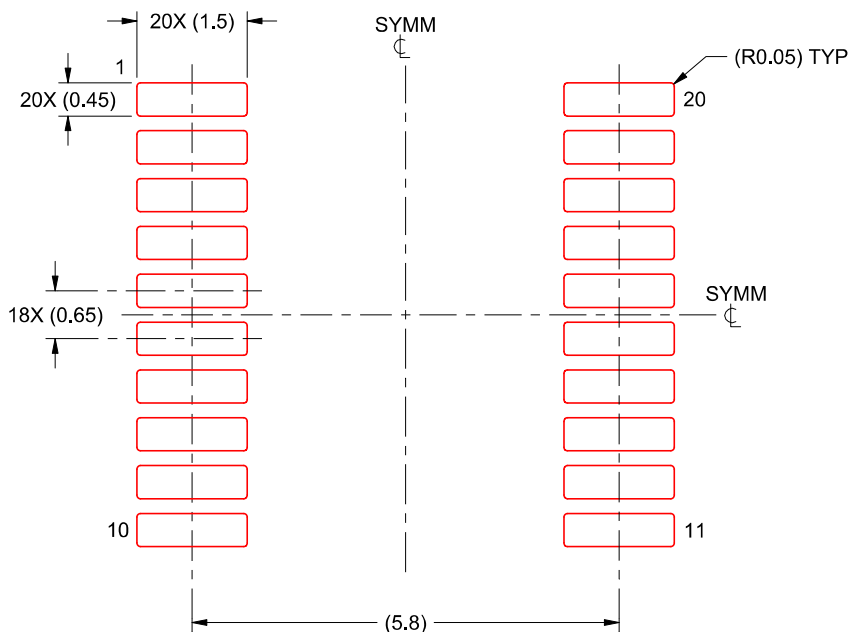
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

## EXAMPLE STENCIL DESIGN

PW0020A

TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



SOLDER PASTE EXAMPLE  
BASED ON 0.125 mm THICK STENCIL  
SCALE: 10X

4220206/A 02/2017

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

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